

**PERFORMANCE EVALUATION OF TRAFFIC SAFETY AND RL-
BASED ADAPTIVE TRAFFIC SIGNAL CONTROL UNDER
DIFFERENT CONNECTED AND AUTONOMOUS VEHICLE
PENETRATION LEVELS**

by

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**Performance Evaluation of Traffic Safety and RL-Based Adaptive
Traffic Signal Control under Different Connected and Autonomous
Vehicle Penetration Levels**

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The thesis titled “**Performance Evaluation of Traffic Safety and RL-Based Adaptive Traffic Signal Control under Different Connected and Autonomous Vehicle Penetration Levels**” submitted by **Md. Aslam Mia**, Student No.: 2004160, has been accepted as satisfactory in partial fulfillment of the requirement for the degree of **Bachelor of Science in Civil Engineering** on June 13, 2026.

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DECLARATION

I hereby declare that this thesis is my own work and to the best of my knowledge it contains no materials previously published or written by another person, or have been accepted for the award of any other degree or diploma at Bangladesh University of Engineering and Technology (BUET) or any other educational institution, except where due acknowledgement is made. I also declare that the intellectual content of this thesis is the product of my own work and any contribution made to the research by others, with whom I have worked at BUET or elsewhere, is explicitly acknowledged.

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Dedicated

to

“all my well-wishers for their encouragement, support, and inspiration”

ABSTRACT

Connected and Autonomous Vehicles (CAVs) and Reinforcement Learning (RL)-based Adaptive Traffic Signal Control (ATSC) systems have emerged as promising technologies for improving urban traffic operations. This study investigates the impact of different CAV penetration levels on traffic efficiency and traffic safety and evaluates the performance of an RL-based ATSC system under varying traffic conditions and demand patterns. The analysis was conducted using the microscopic traffic simulation software SUMO in two network scenarios: a synthetic Grid 4x4 network and a real-world Dublin City Centre (DCC) network. Three traffic conditions were considered: the HDV condition (100% HDVs), mix condition (50% HDVs and 50% CAVs), and CAV condition (100% CAVs).

Traffic efficiency was assessed using the Travel Time Rate (TTR), while traffic safety was evaluated through conflict analysis based on Post-Encroachment Time (PET) surrogate safety measures. The results show a consistent improvement in both traffic efficiency and safety with increasing CAV penetration. Across all evaluated cases, the HDV condition exhibited the highest congestion levels and conflict counts, whereas the CAV condition achieved the lowest TTR values and the greatest reduction in traffic conflicts. In the Grid 4x4 network, the CAV condition produced near-free-flow traffic conditions across most links, while in the DCC network substantial efficiency improvements were observed, particularly under free-flow conditions. In terms of safety, the CAV condition reduced traffic conflicts by up to 86.5% in the Grid 4x4 network (from 37 to 5 conflicts) and by up to 79.2% in the DCC network (from 538 to 112 conflicts) compared to the HDV condition. However, under high-demand scenarios, localized congestion and conflicts persisted even with full CAV penetration. To evaluate ATSC performance, an RL-based IDQN model was trained and tested in the Grid 4x4 network under six different traffic demand cases. The model was evaluated using average travel time and recovery time and compared with a conventional non-RL FixedTime controller. The IDQN model consistently outperformed the FixedTime controller across all traffic conditions, achieving travel time reductions of up to 30.6% in the HDV condition and 34.3% in the CAV condition. Recovery times ranged from 1 to 8 hours, demonstrating successful adaptation to varying traffic patterns and demand conditions.

Overall, the findings indicate that increasing CAV penetration can significantly enhance traffic efficiency and safety, while the integration of CAV technology with RL-based traffic signal control can further improve network performance. These results highlight the potential of intelligent transportation systems to support safer and more efficient urban mobility.

TABLE OF CONTENTS

DECLARATION.....	III
ACKNOWLEDGEMENTS	IV
ABSTRACT.....	VI
TABLE OF CONTENTS	VII
LIST OF FIGURES	XI
LIST OF TABLES	XIV
LIST OF ABBREVIATIONS	XV
CHAPTER 1 INTRODUCTION	1
1.1 Background	1
1.2 Problem statement.....	1
1.3 Objectives of the Study	4
1.4 Scope of the Study	4
1.5 Outline of the Thesis	5
CHAPTER 2 LITERATURE REVIEW.....	7
2.1 Introduction.....	7
2.2 Background	7
2.2.1 Traffic Congestion	7
2.2.2 Traffic Management.....	8
2.3 Traffic safety and Efficiency.....	9
2.4 Traffic Control Systems	14
2.4.1 Background.....	14
2.4.2 Traffic Control Systems Based on Operations.....	17

2.5 Reinforcement Learning	19
2.5.1 Background	19
2.5.2 Markov Decision Process	21
2.5.3 Reinforcement Learning Theory	23
2.6 Application of Reinforcement Learning in Traffic Control	29
2.7 Reinforcement Learning Robustness in Traffic Control Systems.....	32
2.8 Overview	33
CHAPTER 3 METHODOLOGY	34
3.1 Introduction.....	34
3.2 Vehicle Modeling	34
3.2.1 Car-Following Models	35
3.2.2 Lane-Changing Models.....	37
3.3 Traffic Patterns and Conditions	38
3.3.1 Traffic Patterns	38
3.3.2 Traffic Conditions	39
3.4 Scenarios	39
3.4.1 Grid 4x4 Scenario	39
3.4.2 DCC Scenario	47
3.5 Traffic efficiency and safety Evaluation	49
3.5.1 Traffic efficiency	49
3.5.2 Traffic safety	50
3.6 RL-based ATSC Performance Evaluation.....	51
3.6.1 IDQN (RL-based controller).....	51
3.6.2 FixedTime controller (Non-RL baseline).....	52
3.6.3 Recovery Time	52

3.6.4 Initial Training	53
3.7 Overview	53
CHAPTER 4 DATA COLLECTION AND PROCESSING	55
4.1 Introduction	55
4.2 Data Collection Process	55
4.2.1 Data Collection Process for Traffic Efficiency and Safety Evaluation.....	55
4.2.2 Data Collection Process for the Performance Evaluation of RL-Based ATSC	56
4.3 Overview	57
CHAPTER 5 RESULTS AND ANALYSIS	58
5.1 Introduction	58
5.2 Traffic Efficiency	58
5.2.1 Grid 4x4 Scenario	58
5.2.2 DCC Scenario	65
5.2.3 Results Analysis	69
5.3 Traffic Safety	70
5.3.1 Grid 4x4 Scenario	70
5.3.2 DCC Scenario	78
5.3.3 Results Analysis	82
5.4 Performance Evaluation of RL-Based ATSC.....	83
5.4.1 Initial Training	83
5.4.2 Recovery Time	84
5.4.3 Results Analysis	98
5.5 Overview	100
CHAPTER 6 CONCLUSIONS	101
6.1 Background	101

6.2 Summary of the Findings.....	101
6.3 Recommendations of the Study.....	103
6.4 Limitations of the Study.....	104
REFERENCES	106
APPENDIX.....	114

LIST OF FIGURES

Figure 2.1: An Intersection Between a Major Arterial Road	14
Figure 2.2: Reinforcement Learning Framework	20
Figure 3.1 : Grid 4x4 Scenario Network with 16 traffic lights	40
Figure 3.2: Morning traffic trip generation for the Grid 4×4 scenario	42
Figure 3.3: Blocked Roads for Grid 4x4 Scenario.....	43
Figure 3.4: Evening traffic trip generation for the Grid 4×4 scenario	44
Figure 3.5: Special event traffic trip generation for the Grid 4×4 scenario	45
Figure 3.6: DCC Scenario Network with 435 traffic lights	47
Figure 5.1: Spatial distribution of Travel Time Rate (min/km) in the Grid 4x4 network under Case 1: (a) HDV case 1, (b) mix case 1, and (c) CAV case 1	59
Figure 5.2: Spatial distribution of Travel Time Rate (min/km) in the Grid 4x4 network under Case 2: (a) HDV case 2, (b) mix case 2, and (c) CAV case 2	60
Figure 5.3: Spatial distribution of Travel Time Rate (min/km) in the Grid 4x4 network under Case 3: (a) HDV case 3, (b) mix case 3, and (c) CAV case 3	61
Figure 5.4: Spatial distribution of Travel Time Rate (min/km) in the Grid 4x4 network under Case 4: (a) HDV case 4, (b) mix case 4, and (c) CAV case 4.....	62
Figure 5.5: Spatial distribution of Travel Time Rate (min/km) in the Grid 4x4 network under Case 5: (a) HDV case 5, (b) mix case 5, and (c) CAV case 5	64
Figure 5.6: Spatial distribution of Travel Time Rate (min/km) in the Grid 4x4 network under Case 6: (a) HDV case 6, (b) mix case 6, and (c) CAV case 6	65
Figure 5.7: Spatial distribution of Travel Time Rate (min/km) in the DCC network under Case 1: (a) HDV case 1, (b) mix case 1, and (c) CAV case 1	66
Figure 5.8: Spatial distribution of Travel Time Rate (min/km) in the DCC network under Case 2: (a) HDV case 2, (b) mix case 2, and (c) CAV case 2	68
Figure 5.9: Spatial distribution of Travel Time Rate (min/km) in the DCC network under Case 3: (a) HDV case 3, (b) mix case 3, and (c) CAV case	69
Figure 5.10: Spatial distribution of the total number of conflicts at intersections in the Grid 4x4 network under Case 1: (a) HDV Case 1, (b) Mix Case 1, and (c) CAV Case 1	71

Figure 5.11: Spatial distribution of the total number of conflicts at intersections in the Grid 4x4 network under Case 2: (a) HDV Case 2, (b) Mix Case 2, and (c) CAV Case 2.....	72
Figure 5.12: Spatial distribution of the total number of conflicts at intersections in the Grid 4x4 network under Case 3: (a) HDV Case 3, (b) Mix Case 3, and (c) CAV Case 3.....	73
Figure 5.13: Spatial distribution of the total number of conflicts at intersections in the Grid 4x4 network under Case 4: (a) HDV Case 4, (b) Mix Case 4, and (c) CAV Case 4.....	74
Figure 5.14: Spatial distribution of the total number of conflicts at intersections in the Grid 4x4 network under Case 5: (a) HDV Case 5, (b) Mix Case 5, and (c) CAV Case 5.....	76
Figure 5.15: Spatial distribution of the total number of conflicts at intersections in the Grid 4x4 network under Case 6: (a) HDV Case 6, (b) Mix Case 6, and (c) CAV Case 6.....	77
Figure 5.16: Spatial distribution of the total number of conflicts at intersections in the DCC network under Case 1: (a) HDV Case 1, (b) Mix Case 1, and (c) CAV Case 1.....	78
Figure 5.17: Spatial distribution of the total number of conflicts at intersections in the DCC network under Case 2: (a) HDV Case 2, (b) Mix Case 2, and (c) CAV Case 2.....	79
Figure 5.18: Spatial distribution of the total number of conflicts at intersections in the DCC network under Case 3: (a) HDV Case 3, (b) Mix Case 3, and (c) CAV Case 3.....	81
Figure 5.19: Initial training results of the IDQN model for HDV case 1, showing Average Travel Time (sec) versus Episodes. Each episode represents one hour of simulation.....	84
Figure 5.20: Case-wise recovery times of the IDQN model under different traffic conditions in the Grid 4×4 network scenario.....	85
Figure 5.21: Average travel time versus episodes for the IDQN model and FixedTime model under the HDV condition, mix condition, and CAV condition in Case 1 of the Grid 4x4 network	87
Figure 5.22: Average travel time versus episodes for the IDQN model and FixedTime model under the HDV condition, mix condition, and CAV condition in Case 2 of the Grid 4x4 network	88
Figure 5.23: Average travel time versus episodes for the IDQN model and FixedTime model under the HDV condition, mix condition, and CAV condition in Case 3 of the Grid 4x4 network	90
Figure 5.24: Average travel time versus episodes for the IDQN model and FixedTime model under the HDV condition, mix condition, and CAV condition in Case 4 of the Grid 4x4 network	92

Figure 5.25: Average travel time versus episodes for the IDQN model and FixedTime model under the HDV condition, mix condition, and CAV condition in Case 5 of the Grid 4x4 network	93
Figure 5.26: Average travel time versus episodes for the IDQN model and FixedTime model under the HDV condition, mix condition, and CAV condition in Case 6 of the Grid 4x4 network.	95
Figure 5.27: Case-Wise Percent Decrease in Travel Times (%) of the IDQN Model for the Grid 4x4 Network under HDV, Mix, And CAV Conditions	97

LIST OF TABLES

Table 2.1: Summary of Existing Studies on the Impact of CAVs on Traffic Efficiency and Safety	13
Table 3.1: Car-Following Model Parameters for Simulated CAVs.....	35
Table 3.2: Car-Following Model Parameters for Simulated HDVs.....	36
Table 3.3: Driver State Device Parameters for Simulating Driver's Imperfection.....	37
Table 3.4 : HDVs and CAVs Penetration at HDV Condition, Mixed Condition, and CAV Condition in Grid 4x4 Scenario	41
Table 3.5: HDVs and CAVs Penetration at HDV Condition, Mixed Condition, and CAV Condition.....	48
Table 5.1: Traffic safety analysis summary for the six cases of the Grid 4x4 scenarios ..	77
Table 5.2: Traffic safety analysis summary for the three cases of the DCC scenarios	81
Table 5.3: Recovery Time (Hours) of IDQN RL Models Across Different Cases for the Grid 4x4 Scenario.....	85
Table 5.4: Case-wise Initial and Optimum Average Travel Times (s) of the IDQN Model for the Grid 4x4 Network under HDV, Mix, and CAV Conditions.....	96
Table 5.5: Case-Wise Percent Decrease in Travel Times (%) of the IDQN Model for the Grid 4x4 Network under HDV, Mix, And CAV Conditions	97
Table 5.6: Case-Wise Fixed Travel Times (sec) of the FixedTime Model for the Grid 4x4 Network under HDV, Mix, And CAV Conditions.....	98

LIST OF ABBREVIATIONS

HDVs	Human Driven Vehicles
CAVs	Connected and Autonomous Vehicles
RL	Reinforcement Learning
ATSC	Adaptive Traffic Signal Control
CACC	Cooperative Adaptive Cruise Control
ACC	Adaptive Cruise Control
IDM	Intelligent Driver Model
ACACC	Advanced Cooperative Adaptive Cruise Control
SSM	Surrogate Safety Measure
TTR	Travel Time Rate
PET	Post Encroachment Time
TTC	Time-to-Collision
SUMO	Simulation of Urban MObility

CHAPTER 1

INTRODUCTION

1.1 Background

Recently, Connected and Autonomous vehicles (CAVs) have been developing rapidly. Traffic efficiency and safety can be greatly improved through the use of vehicle-to-vehicle and vehicle-to-infrastructure wireless communication technologies in CAVs (Shladover, 2018). However, the transition from a fully Human-Driven Vehicle (HDV) system to a fully Connected and Autonomous vehicle (CAV) system may take many years. During this period, both HDVs and CAVs are likely to coexist on the same roads (Carrone et al., 2021). Uncertainties in human driving behavior primarily cause most road accidents and traffic congestion (Di Vaio et al., 2019). CAVs can improve traffic safety and efficiency by adjusting their driving behavior based on data from onboard sensors and communication with other vehicles (Guanetti et al., 2018). On the other hand, Traffic congestion is a major urban transportation problem around the world (Chakraborty, n.d.; Faheem et al., 2024a). One of the most effective approaches to solving this problem is efficient traffic flow management. Adaptive Traffic Signal Control (ATSC) Systems are effective in optimizing urban traffic flow, and reinforcement learning (RL) has emerged as a promising approach for traffic signal control in recent years. Compared to other techniques, RL-based traffic control achieves better performance and significantly reduces traffic congestion (Miletić et al., 2022a). Thus, the application of RL in ATSC offers a promising approach to improving traffic flow in urban contexts.

1.2 Problem statement

Urban transportation networks across the globe are facing escalating traffic congestion, which has serious economic and environmental consequences. With rising metropolitan populations and increased reliance on automobiles, congestion continues to intensify. According to the TomTom Traffic Index, London (UK), Bengaluru (India), Dublin

(Ireland) and Sapporo (Japan) were the most congested cities in 2022 worldwide, with an average speed of no more than 20 km/h during rush hour and more than 240 hours of traffic congestion in a year, a situation that is worsening every year. In China, data shows that the average peak speeds are around 30 km/h in 2022 in its major congested cities (Xu et al., 2024). It is estimated that traffic congestion in the United States caused approximately 8.3 billion hours of travel delay in 2020, leading to nearly \$192 billion in fuel-related costs, and each year, about 1.35 million people lose their lives in road accidents worldwide (Garg & Bouroche, 2023a). According to the 2023 INRIX Global Traffic Scorecard, traffic congestion caused economic losses of about USD 70.4 billion in the United States in 2023, while the United Kingdom suffered approximately GBP 7.5 billion in losses. In developing countries such as Bangladesh, congestion remains a major barrier to economic development, accounting for nearly 7% of GDP (Chakraborty, n.d.).

Traffic congestion can be addressed through two main approaches: supply-side and demand-side strategies. Supply-side management focuses on enhancing the capacity and performance of transportation infrastructure, for example, by constructing new roads, expanding existing networks, or improving traffic flow through advanced signal control systems. These measures aim to accommodate increasing traffic demand and improve overall operational efficiency.

On the other hand, demand-side management seeks to reduce traffic demand by influencing travel behavior. This can be achieved through strategies such as land-use planning, congestion pricing, and encouraging the use of public transportation. Such approaches are often considered more sustainable because they target the underlying causes of congestion rather than merely increasing infrastructure capacity.

In practice, both approaches are implemented based on policy objectives and societal needs. Although demand-side strategies are gaining greater emphasis for long-term sustainability, supply-side solutions continue to play a vital role in addressing immediate congestion issues and maintaining efficient transportation systems.

CAVs primarily fall under supply-side strategies for addressing traffic congestion. HDVs often create stop-and-go waves due to slow reaction times and perception errors, causing

large speed fluctuations (Di Vaio et al., 2019). This behavior is a major source of traffic congestion and leads to lower safety and efficiency. CAVs can reduce stop-and-go waves and improve traffic efficiency without affecting safety by using proper car-following strategies. Two common approaches used in CAVs for longitudinal control are adaptive cruise control (ACC) and cooperative adaptive cruise control (CACC). ACC uses the faster reaction time of autonomous vehicles to reduce traffic waves, while CACC allows vehicles to follow more closely by receiving information such as position, speed, and acceleration from the vehicle ahead through wireless communication (Garg et al., 2021).

CAVs enhance the efficiency of existing road infrastructure without needing physical expansions. They have the potential to significantly enhance traffic safety and efficiency, although it is projected that CAVs will account for at most 24.8% of vehicles by 2045 (Bansal & Kockelman, 2017). CAVs offer a wide range of benefits, including significant reductions in crash rates and traffic congestion. CAVs create data-rich environments that support a wide range of applications aimed at improving road safety, reducing congestion, and enhancing environmental sustainability, while also making travel more efficient and comfortable. These applications include connected in-vehicle infotainment systems, real-time navigation and traffic information, safety alerts, collision avoidance systems, advanced driver-assistance systems (ADAS), automated driving systems (ADS), remote diagnostics and maintenance, teleoperation in case of system failures, data monetization, and fleet management (Abdelkader et al., 2021).

On the other hand, Traffic signal control is categorized as a supply-side traffic management approach, as it regulates the movement of vehicles and pedestrians at roads and intersections to ensure safe and efficient traffic flow. These systems include automated mechanisms such as traffic signals, ramp metering, and variable speed limits, as well as static elements like road signs and marking (Rafi, n.d.). In this part, the primary focus is on traffic signal control by using RL and evaluating the improvements achieved through the integration of CAVs. RL, particularly deep RL, benefits from powerful computing and the availability of data from intelligent transport systems. This allows RL-based methods to achieve better results in traffic control compared to other methods. RL-based traffic

controllers learn through interaction with their environment, but in real-world conditions, unexpected events like accidents can change system behavior and reduce performance. In addition, issues such as unreliable sensor data make their operation more challenging. Despite its importance, this problem has not been widely studied.

1.3 Objectives of the Study

This study is primarily guided by two key objectives:

- 1) To evaluate the improvement in traffic efficiency and safety under varying traffic conditions and patterns at different penetration levels of CAVs.
- 2) To analyze the performance of RL-based ATSC under varying traffic conditions and patterns at different penetration levels of CAVs.

With the rapid advancement of intelligent transportation systems, the integration of CAVs and RL-based traffic control strategies has the potential to significantly transform urban traffic management. However, their performance may vary depending on traffic dynamics, demand patterns, and the proportion of CAVs in the traffic stream. Therefore, this study aims to systematically investigate how different levels of CAV penetration influence both traffic safety, measured through surrogate safety indicators such as Post Encroachment Time (PET), and operational efficiency. Furthermore, it examines whether RL-based ATSC can maintain consistent and reliable performance under mixed traffic environments, where interactions between HDVs and CAVs introduce additional uncertainty.

1.4 Scope of the Study

This study focuses on evaluating the impact of CAVs and RL-based ATSC on urban traffic performance under mixed traffic environments. The scope is limited to simulation-based analysis, where different traffic scenarios are modeled to represent varying traffic conditions, demand patterns, and levels of CAV penetration.

The research primarily considers the interaction between HDVs and CAVs across three penetration scenarios: 0% CAV (fully HDV), 50% CAV (mixed traffic), and 100% CAV

(fully autonomous), reflecting a realistic transition phase where both vehicle types coexist on the same network. Different penetration levels of CAVs are examined to understand their influence on traffic efficiency and safety. Traffic performance is evaluated using key indicators such as travel time rate and surrogate safety measures, particularly Post Encroachment Time (PET).

Furthermore, the study evaluates the performance of RL-based ATSC in traffic signal control across three CAV penetration scenarios: 0% CAV (fully HDV), 50% CAV (mixed traffic), and 100% CAV (fully autonomous). The analysis examines its adaptability under varying traffic patterns, with particular focus on uncertainties in mixed traffic conditions.

However, this study does not include real-world field implementation or hardware-level validation. The findings are based on simulation environments and predefined models, and factors such as communication delays, sensor inaccuracies, and real-world deployment constraints are considered only conceptually.

1.5 Outline of the Thesis

This thesis is structured into six chapters as follows:

Chapter 1: Introduction. This chapter provides an overview of the research problem, objectives, and scope of the study. It also describes the overall structure of the thesis.

Chapter 2: Literature Review. This chapter presents the necessary background knowledge for the study and reviews existing research on CAVs and RL-based ATSC, along with brief discussions on related topics.

Chapter 3: Methodology. This chapter describes the research approach, including the experimental design and performance metrics used to assess performance.

Chapter 4: Data Collection and Processing. This chapter briefly describes how data is collected from the experiments and prepared for presentation.

Chapter 5: Results and Analysis. This chapter presents the results of the study, including performance evaluation and analysis of performance degradation.

Chapter 6: Conclusions. The final chapter summarizes the key findings of the study, discusses its limitations, and suggests directions for future research on the safety and efficiency of CAVs as well as RL-based ATSC.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter provides a brief overview of CAV technologies and highlights the effects of CAVs on traffic flow characteristics and traffic safety, and discusses the fundamental concepts of CAVs, various car-following and lane-changing models, and traffic flow and safety indicators.

On the other hand, it provides an overview of the performance of RL-based Adaptive Traffic Signal Control (ATSC) systems at different levels of CAV penetration under different traffic conditions and patterns. It discusses the fundamental concepts of RL. The integration of ATSC systems and RL is explored in detail, along with an explanation of the primary implementation details and design decisions.

Finally, other related topics and key considerations are discussed to complete the literature review.

2.2 Background

2.2.1 Traffic Congestion

Traffic congestion is a growing problem in urban transportation networks around the world, having major consequences for the economy and the environment. As metropolitan populations rise and more people rely on autos for mobility, traffic congestion is becoming more common and severe (Faheem et al., 2024b). Traffic congestion leads to travel delays, resulting in lost working time and reduced economic output. The 2023 INRIX Global Traffic Scorecard highlights the scale of this problem (Rafi, n.d.). According to the TomTom Traffic Index, cities such as London, Bengaluru, Dublin, and Sapporo were among the most congested in 2022, with average rush-hour speeds below 20 km/h and over 240 hours of congestion annually. This situation continues to worsen each year. In China,

major congested cities recorded average peak speeds of around 30 km/h in 2022 (Xu et al., 2024). It is estimated that traffic congestion in the United States caused approximately 8.3 billion hours of travel delay in 2020, leading to nearly \$192 billion in fuel-related costs, and each year, about 1.35 million people lose their lives in road accidents worldwide (Garg & Bouroche, 2023a). Traffic congestion is estimated to increase fuel consumption by about 13% and emissions by around 25% (Greenwood et al., 2007). According to Bharadwaj et al. (Bharadwaj et al., 2017), traffic congestion increases travel time, resulting in around 53% higher CO₂ emissions in the Mumbai Metropolitan Region, and these higher emissions negatively affect the health of city residents. Higher emissions also lead to increased health impacts for urban residents. Levy et al. (Levy et al., 2010) estimated that increased PM_{2.5} levels caused by traffic congestion resulted in economic losses of about USD 31 billion across 83 cities in 2000.

2.2.2 Traffic Management

There are two primary strategies used in traffic management to reduce traffic congestion (O’Flaherty, 2018):

1. Supply-Side Traffic Management
2. Demand-Side Traffic Management

Supply-side traffic management focuses on increasing the capacity of transportation infrastructure, such as by constructing new roads or expanding existing facilities. In contrast, demand-side traffic management aims to reduce the amount of traffic generated, for example by imposing higher tolls on a bridge to discourage vehicle use. Generally, demand-side measures are considered more effective in improving transportation efficiency.

Supply-side management includes measures such as developing new infrastructure, regulating traffic operations, and controlling freight movement. Its primary goal is to accommodate as much traffic demand as possible. Using historical traffic data, various strategies are implemented to optimize traffic flow. For example, when an intersection

begins to experience heavy traffic, a traffic signal may be installed to regulate vehicle movements and improve the efficiency of traffic flow through the intersection.

Demand-side management follows the opposite philosophy of supply-side management. Instead of increasing road capacity to accommodate traffic, it seeks to reduce traffic demand at its source. This approach includes measures such as land-use planning and zoning regulations, congestion pricing, parking fees, and the promotion of public transportation. Demand-side management is generally regarded as a more sustainable solution to traffic congestion. This is illustrated by the Braess paradox (Braess, 1968) (which states that adding a new road link to a traffic network can sometimes increase overall travel time due to the additional demand it generates). Therefore, in some situations, avoiding the construction of new roads may be the most effective way to reduce travel time.

In practice, effective traffic management relies on a combination of both supply-side and demand-side measures, with the choice of strategies often influenced by public needs and policy decisions. Although demand-side approaches offer many advantages, they cannot solve every transportation problem, making supply-side measures necessary in many cases. Within this framework, traffic control systems play a vital role in regulating traffic flow, especially in urban areas where traffic patterns are highly complex. Among these systems, efficient traffic signal control is essential for improving urban mobility. Consequently, this thesis focuses on enhancing urban traffic signal performance through the application of reinforcement learning techniques.

2.3 Traffic safety and Efficiency

Several simulation-based studies have examined the effects of Connected and Automated Vehicles (CAVs) on the safety and efficiency of mixed traffic systems in recent years (Greenwood et al., 2007; Gueriau & Dusparic, 2020). The results indicate that at low penetration rates, typically below 25%, CAVs may negatively affect traffic efficiency. However, when the penetration rate reaches medium to high levels, generally 40% or more, they can significantly improve traffic performance (Fiddian-Green & Silen, 1975; Gueriau & Dusparic, 2020) (Ehrhart et al., 1975). To overcome the reduction in efficiency at low

penetration rates, researchers have proposed equipping Human-Driven Vehicles (HDVs) with Vehicle-to-Vehicle (V2V) communication technology, thereby converting them into connected vehicles. Moreover, utilizing information from multiple preceding vehicles, rather than relying only on the immediately preceding vehicle, can further enhance traffic efficiency. Nevertheless, simulation outcomes are highly sensitive to the car-following models used to represent the driving behavior of both CAVs and HDVs(Elgsaeter et al., 1976).

While most studies have primarily focused on evaluating the effects of Connected and Automated Vehicles (CAVs) on traffic efficiency, a limited number of recent studies have explicitly examined their impact on traffic safety (Hossain et al., 2019). These studies found that even low market penetration rates of level 4 CAVs can significantly enhance safety, with reductions in safety conflicts ranging from 12% to 47% at a 25% market penetration rate, largely because many road accidents result from human perception errors and delayed reaction times (Hossain et al., 2019). However, another recent study reported that level 2 CAVs may negatively affect traffic safety, resulting in approximately a 10% increase in safety conflicts at a 25% market penetration rate (Gueriau & Dusparic, 2020). Although such conflicts can be mitigated by adopting more conservative time headway settings, this often comes at the cost of reduced traffic efficiency [26]. Equipping HDVs with communication capabilities has also been shown to enhance traffic safety further (Garg & Bouroche, 2023b). Moreover, sharing intent information, such as future speed and acceleration, in addition to the current status of surrounding vehicles, can reduce decision-making conservatism and improve safety without sacrificing efficiency (H. M. Wang et al., 2023). Nevertheless, as with traffic efficiency studies, the simulation results are highly dependent on the car-following models used to represent CAV behavior. For example, a recent study demonstrated that at low penetration rates, CAVs may negatively affect traffic safety when the Cooperative Adaptive Cruise Control (CACC) model is used, whereas safety improvements are observed when the IDM is applied (Matsuoka, 2018). Most of the previously discussed studies evaluated the effects of CAVs at different penetration rates under the assumption of perfect communication. In real-world conditions, however, communication networks are subject to delays and packet losses, which can significantly influence vehicle performance (Matsuoka, 2018). Only a limited number of studies have

examined the effects of such communication imperfections on traffic performance (Lachmayer, 1975), (Gardos & Cole, 1976a). The results indicated that communication delays adversely affect CAV performance because control decisions are based on outdated information. Similarly, a subsequent study involving a three-vehicle platoon equipped with CACC demonstrated that communication failures force vehicles to switch from CACC to ACC, thereby reducing traffic efficiency even under string-stable time headway settings (Lachmayer, 1975).

The analysis by investigating mixed traffic operations under more realistic conditions, including imperfect communication networks and large-scale road environments (Gardos & Cole, 1976b), The findings showed that although increasing CAV penetration improves traffic efficiency under reliable communication conditions, these benefits are considerably reduced when communication failures occur. To address this limitation, Navas et al. proposed an ACACC system based on Youla–Kucera (YK) parameterization (Lachmayer, 1975). Unlike conventional systems that revert to ACC when communication with the immediately preceding vehicle is lost, ACACC can continue operating using information received from other vehicles ahead. The system employs two Proportional-Derivative (PD)-based CACC controllers with different time headways: one with a shorter headway when communication with the preceding vehicle is available and another with a larger headway when such communication is unavailable but information from other upstream vehicles is accessible. YK parameterization enables a stable transition between these controllers, allowing the system to maintain string stability at shorter headways and achieve improved traffic efficiency across various traffic scenarios involving HDVs, ACC-equipped vehicles, and CAVs.

The impact of unreliable communication on traffic safety has also attracted increasing research attention (Gardos & Cole, 1976a), (Tu et al., 2019), (Yao et al., 2020). These studies found that communication failures and the resulting degradation from CACC to ACC can negatively affect safety because vehicles must maintain larger headways, leading to speed fluctuations and an increased number of safety conflicts, particularly when leading vehicles exhibit variable driving behavior. Researchers also examined the sensitivity of traffic safety assessments to different Time-to-Collision (TTC) thresholds. While some

studies concluded that TTC threshold values have little influence on safety evaluation results, others reported that safety assessments are highly sensitive to the chosen threshold and that no single TTC value is universally appropriate for surrogate safety analysis (Sinha et al., 2020). Despite these efforts, most existing studies have evaluated safety under communication failures without incorporating realistic wireless communication network models. In addition, they often neglect the uncertainty associated with human driving behavior, such as perception errors and delayed reaction times, when modeling HDVs (Yao et al., 2020).

Furthermore, a recent experimental study conducted on the Mcity test track examined a simple two-vehicle scenario in which a CAV followed a connected HDV (Beregi et al., 2023). The results revealed that although delays and packet losses in Vehicle-to-Everything (V2X) communication can negatively affect performance, CAVs can still enhance the resilience of autonomous driving systems when onboard sensors are impaired by adverse weather conditions. The study also demonstrated that appropriate tuning of controller gains can improve the robustness of CAV control systems against communication delays.

A summary of the reviewed studies concerning the impacts of CAVs on traffic performance and road safety is presented in Table I. The literature review reveals that the findings of existing studies are often influenced by the assumptions, simulation settings, and modeling approaches adopted. One of the major limitations identified is the lack of realistic traffic scenarios, as most studies consider only a simplified form of traffic heterogeneity, typically involving HDVs and a single level of vehicle automation.

To address these limitations, the present study adopts a more comprehensive approach by considering mixed traffic environments, different vehicle classes Vehicles, diverse road network configurations, and varying traffic demand patterns. Furthermore, the study examines the effects of CAV deployment under different market penetration rates and evaluates their influence on both traffic efficiency and road safety simultaneously. The traffic demand characteristics, including traffic volumes and vehicle compositions, are calibrated using real-world data, while the adoption scenarios are designed to represent both short-term and long-term transitions toward increased CAV penetration.

Table 2.1: Summary of Existing Studies on the Impact of CAVs on Traffic Efficiency and Safety

Reference	Levels of Automation	Type of Vehicles	Penetration Rates (%)	Network Type	Network Scale	Traffic Level(s)	Impact on Traffic	Impact on Safety
Makridis et al. (Makridis et al., 2018)	Level 2 (C-ACC)	Cars	0, 25, 50, 75, 100	Highway (ring road)	119 km	1 h real data $\pm 20\%$	Fuel consumption, speeds	X
Guériau et al. (Guériau et al., 2016)	Level 2 (C-ACC)	Cars	0, 20, 40, 60, 80, 100	Highway	10 km	Saturated (2000 veh/h/lane)	Flow, speeds, density	X
Stanek et al.	Level 4 CAV	Cars	0, 10, 30, 50, 70, 90, 100	Two freeway sections	20 miles, 10 miles	AM peak period (calibrated)	Network delay, speeds	X
Talebpour and Mahmassani (Talebpour & Mahmassani, 2016a)	CV, AV Level 4	Cars	0, 10, 25, 50, 75, 90 (mix of HDV, CV and AV)	One-lane highway	Infinite length	Platoons with 100, 80, 60, 40, and 20 vehicles	Flow, density	X
Ye and Yamamoto (Ye & Yamamoto, 2019)	Level 2 (C-ACC)	Cars	0, 25, 50, 75, 100	Two-lane road	10 km	50 veh/km/lane, 100 veh/km/lane	X	TTC
Morando et al. (Morando et al., 2018a)	Level 4 (2 models)	Cars, HGVs (95/5%)	0, 25, 50, 75, 100	Signalized intersection, roundabout	1 km	1 h (8–9 AM)	X	Conflicts from TTC and PET
Xie et al. (Xie et al., 2019)	Levels 1, 2, 3, 4	Cars	0, 20, 40, 60, 80, 100	Freeway, CBD, Campus	30 × 30 km, unknown, unknown	3 uniform fixed loads	Travel times	Conflicts from TTC

2.4 Traffic Control Systems

2.4.1 Background

Traffic control systems are designed to regulate the movement of vehicles and pedestrians on roads and at intersections in order to ensure safe and efficient traffic operations. These systems include both dynamic control measures, such as traffic signals, ramp metering systems, and variable speed limits, as well as static control devices, including traffic signs, pavement markings, and similar roadway elements. In the context of this thesis, the term traffic control system specifically refers to traffic signals installed at road intersections, while other forms of traffic control are beyond the scope of this study. Figure 2.1 illustrates an example of a traffic signal located at the intersection of a major arterial road and a minor side road.

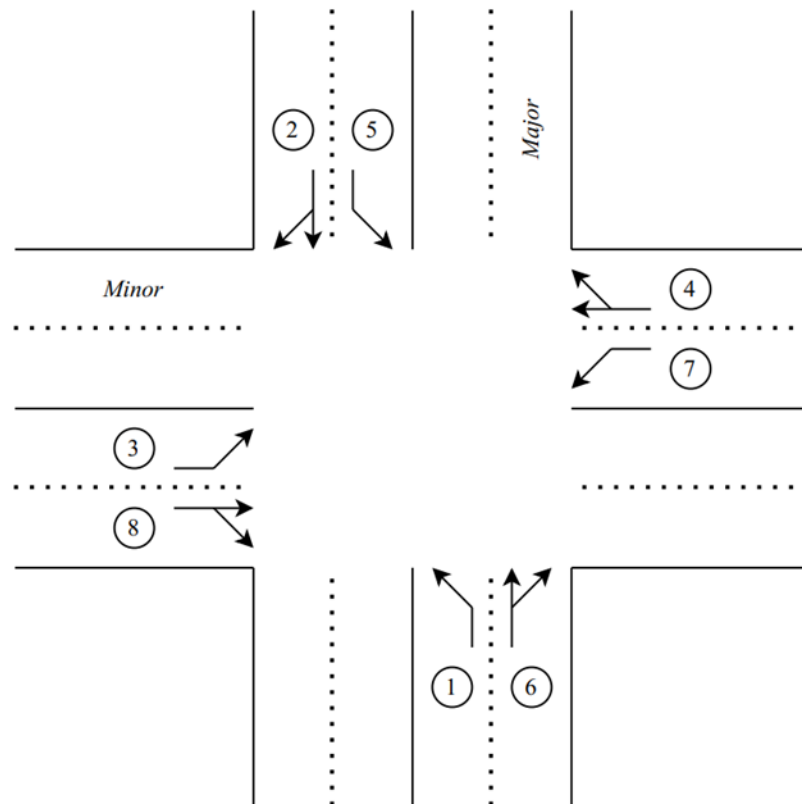


Figure 2.1: An Intersection Between a Major Arterial Road

In Figure 2.1, both roads are designed for right-hand driving. Here, the numbers 1 to 8 denote phases according to the National Electrical Manufacturers Association (NEMA) (Kurniawan et al., 2025). Each phase indicates a specific traffic movement in the intersection. For example, phase 1 indicates left-turn movement on the major arterial road. Traffic control signals manage the flow by permitting specific phases to proceed through the intersection while restricting others (Rafi, n.d.).

Lane: A lane is a designated portion of the road used by a single line of vehicles. Lanes are often marked by road markings and may be used for specific purposes, such as turning, merging, or through traffic.

Movement: A movement refers to the specific direction or path that vehicles or pedestrians take when navigating an intersection. Movements are typically categorized based on their direction, such as through movement (continuing straight), left turn, right turn, or pedestrian crossing. This thesis will mainly consider vehicular movements.

Phase: A phase ϕ in traffic signal control refers to a specific period during which a set of traffic movements (such as vehicles traveling in one direction or pedestrians crossing) are given right-of-way or allowed to move through the intersection. Each phase ϕ indicates a specific traffic movement in the intersection and is controlled by a specific set of traffic signal configurations. According to the National Electrical Manufacturers Association (NEMA), these are the phases for a major road with higher priority and a minor road with lower priority (Kurniawan et al., 2025):

a) **Phase 1:** A major road left-turn phase.

b) **Phase 2:** A major road through phase.

c) **Phase 3:** A minor road left-turn phase.

d) **Phase 4:** A minor road through phase.

e) **Phase 5:** A major road left-turn phase.

f) **Phase 6:** A major road through phase.

g) **Phase 7:** A minor road left-turn phase.

h) **Phase 8:** A minor road through phase.

i) **Phase 9:** A dedicated pedestrian phase.

Phase Groups: In traffic control systems, phase groups refer to sets of traffic signal phases that can operate simultaneously or are grouped together based on their compatibility. Compatibility is determined by whether the movements controlled by the phases can proceed without conflict. For example, in Figure 2.1, phases 1 and 2 are compatible and can be grouped together. These are also known as concurrency groups.

Phase Sequence: A phase sequence refers to the specific order in which traffic signal phases are activated at an intersection. Phases in the same phase group can run at the same time. In most traffic control systems, the phase sequence is fixed. However, it is technically possible to allow the sequence to be changed.

Cycle Time: Cycle time C is the total duration required to complete one full sequence of all traffic signal phases at an intersection. It includes the time allocated to all phases, such as green, yellow, and red times, and is typically measured in seconds. It is worth noting that it is possible to design traffic signals without cycle times.

Offset: Offset refers to the time difference between the start of the green phase at one intersection and the start of the corresponding green phase at another intersection. Some traffic control systems synchronize traffic lights with offsets.

Splits: Splits represent the distribution of the cycle time among the various phases in a signal cycle. They specify the proportion of time allocated to each phase, ensuring that different movements receive appropriate green times based on traffic demand. Splits are typically expressed as a percentage of the total cycle time.

Green Time: The duration t_g during which a traffic signal displays a green light, allowing vehicles or pedestrians to proceed through an intersection.

Red Time: The duration t_r during which a traffic signal displays a red light, indicating that vehicles or pedestrians must stop and wait.

Yellow Time: The duration t_y during which a traffic signal displays a yellow light, transitioning between green and red lights. Yellow time alerts drivers to prepare to stop or clear the intersection if already within the dilemma zone.

Minimum Green Time: The shortest duration for which the green light is displayed during a signal phase. This time ensures that vehicles waiting at the stop line have sufficient time to start moving and clear the intersection.

Maximum Green Time: The longest duration for which the green light can remain active during a signal phase. This time prevents excessive delays for conflicting movements and ensures fair distribution of green time across approaches.

In summary, a traffic signal operates by regulating the movement of vehicles and pedestrians at intersections through a structured sequence of phases, each controlling specific movements. These phases are organized into a phase sequence that repeats over a cycle time, with each phase allocated a proportion of the cycle time. For each phase, movement occurs during green time and stops during red time.

2.4.2 Traffic Control Systems Based on Operations

Over the years, a wide range of traffic control systems has been developed to improve traffic operations at intersections. Based on their operating principles, these systems can generally be classified into three broad categories (Miletić et al., 2022b):

1. Fixed-Time Control
2. Actuated Control
3. Adaptive Control

Fixed-Time Control: Fixed-time traffic signals operate based on a predetermined cycle length, which represents the total time required to complete one full sequence of all signal phases. For example, a cycle length of 90 seconds indicates that the entire signal sequence repeats every 90 seconds. Within each cycle, specific amounts of green time (or split time) and red time are allocated to different traffic movements. Vehicles are allowed to proceed

during the green phase and must stop during the red phase, while the yellow phase provides a transition period between the two.

Various methods have been developed to determine the optimal allocation of green time among competing traffic movements. One of the earliest and most widely adopted approaches is the Webster method (Webster, 1958), which uses saturation flow rates to distribute green time efficiently and minimize average vehicle delay based on theoretical principles. In addition, several other signal timing methodologies have been introduced by transportation agencies and research organizations worldwide to address different operational objectives and traffic conditions (Koonce, 2008a). Coordination between multiple traffic signals can also be achieved through established tools and techniques such as TRANSYT (Robertson, 1969), MAXBAND, and other signal optimization methods.

Actuated Control: As traffic signal technology advanced, actuated traffic control systems were developed to overcome the limitations of fixed-time signal operation (Hamilton et al., 2013). Unlike fixed-time systems, actuated signals can respond dynamically to real-time traffic conditions. This adaptability is achieved through the use of vehicle detectors that provide traffic information directly to the signal controller.

Because actuated control relies on vehicle detection, several operational parameters are required for its design. One important parameter is the minimum green time, which represents the shortest period for which a green signal must remain active. This ensures that at least one vehicle has sufficient time to pass through the intersection (Koonce, 2008b). When vehicles continue to be detected on an approach, the green phase can be extended beyond the minimum duration. However, this extension is restricted by the maximum green time, which defines the longest period a green signal can remain active. Consequently, unlike fixed-time traffic signals, actuated control systems do not operate with a predetermined cycle length, as signal timings continuously adjust according to prevailing traffic conditions.

Adaptive Control: With the development of advanced computing technologies, ATSC systems were introduced to provide a more intelligent approach to traffic signal

management (Z. Wang et al., 2023). These systems, also commonly referred to as Urban Traffic Control (UTC) systems (Hamilton et al., 2013), represent an advanced form of actuated traffic control. Similar to actuated systems, ATSC system relies on sensors to collect real-time traffic data and regulate signal operations. However, unlike conventional actuated control, ATSC system continuously analyzes changing traffic conditions and adjusts signal control strategies accordingly (Cai et al., 2009). Rather than depending on fixed or pre-defined parameters, these systems utilize field data to update and optimize signal timing parameters over time, allowing them to respond more effectively to fluctuating traffic demands.

It is best to consider these three types of systems as tools rather than viewing one system as superior to the others. Depending on location, budget constraints, and other requirements, different systems will emerge as the most optimal. As this thesis focuses on ATSC, it will limit its scope to urban environments, as ATSC is most suitable for this setting.

2.5 Reinforcement Learning

2.5.1 Background

Reinforcement Learning (RL) is a branch of machine learning in which an agent learns how to make decisions through interactions with its environment (Sutton & Barto, 2014). The learning process is based on a trial-and-error mechanism, where the agent performs actions and receives feedback in the form of rewards or penalties. By continuously evaluating the outcomes of its actions, the agent gradually learns a strategy that maximizes the cumulative reward over time.

In the RL framework, two main components are considered: the agent and the environment. The environment represents the external system in which interactions occur, while the agent is an autonomous decision-making entity that observes the environment and takes actions. At any given time, the environment is characterized by a state that reflects its current condition. The agent gathers information about the environment through

observations and uses this information to estimate the current state. Based on the observed state, the agent selects an action, which may alter the state of the environment. Following the action, the environment transitions to a new state, which the agent observes before making its next decision. During this process, the agent may receive a reward that reflects how well the chosen action contributes to achieving its objective. This continuous cycle of observation, action, state transition, and reward forms the fundamental learning mechanism of reinforcement learning. An overview of this framework (Oucherif & Baghdadi, 2025) is illustrated in Figure 2.2.

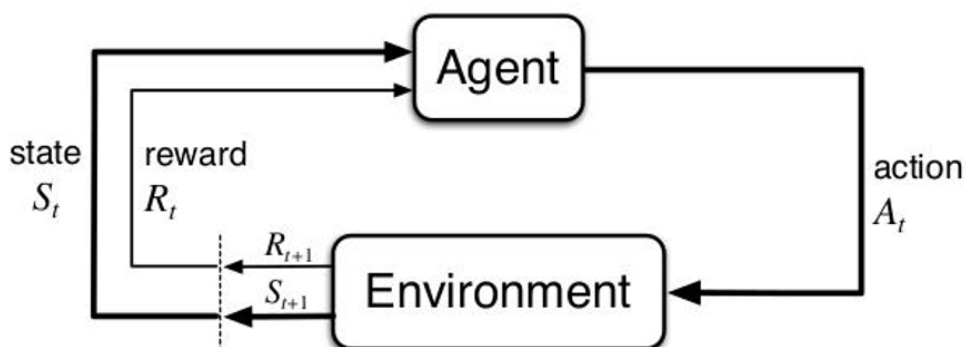


Figure 2.2: Reinforcement Learning Framework

The interaction between the agent and the environment occurs through a closed-loop process. At each time step, the agent observes the current state of the environment and selects an action according to its policy. This action affects the environment, causing it to transition to a new state and generate a reward signal. The reward provides feedback on the effectiveness of the chosen action in achieving the desired objective. Based on this feedback, the agent updates its policy to improve future decisions and maximize cumulative rewards over time. Through repeated interactions and continuous learning from experience, the agent gradually develops more effective decision-making strategies.

In the context of traffic signal control at an intersection, the environment can be viewed as the traffic network, comprising vehicles, pedestrians, and various traffic-related parameters. The agent corresponds to the traffic signal controller, which monitors the current traffic conditions, including factors such as vehicle density, traffic speed, and queue

lengths. These conditions are typically measured using sensors, allowing the agent to infer the current state of the environment. Based on the observed state, the agent selects an appropriate action, such as modifying the duration of the green signal or changing the signal phase. The selected action influences traffic operations and causes the environment to transition to a new state.

After executing an action, the agent receives a reward that reflects the effectiveness of its decision in achieving the desired traffic management objectives. The reward function is usually designed to encourage outcomes such as minimizing vehicle delays, reducing queue lengths, or maximizing traffic throughput. Using this feedback, the agent evaluates its performance and updates its decision-making policy accordingly. Through repeated interactions with the traffic environment, the agent progressively learns more effective control strategies and improves its ability to optimize traffic signal operations.

2.5.2 Markov Decision Process

Reinforcement Learning problems are typically formulated as a Markov Decision Process (MDP), which provides a mathematical framework for decision-making in stochastic environments. Before discussing MDP, it is helpful to understand Markov chains first.

A Markov chain is a mathematical model that describes a system that transitions from one state to another state with a certain probability. The key thing to note here is the probability of each state depends only on the previous state. In simpler terms, all information about the future can be known from the present, and the past can be ignored. In more formal terms, if the current state s_t in time t is known, the next state s_{t+1} can be predicted with the state transition probability $P(s_{t+1}|s_t)$. If the Markov property of the future only being dependent on the present is satisfied, the system can be described as a Markov chain and obeys the following property:

$$P(s_{t+1}|s_t) = P(s_{t+1}|s_t, s_{t-1}, s_{t-2}, \dots) \quad 2.1$$

There is one neat consequence of this fact. A Markov chain can be easily represented by a diagram listing all the known states along with the transition probabilities between each

state. Nodes can represent the states, and arrows between nodes can represent the probabilities of transitioning to the next state.

Using the traffic control example again, the states in the Markov chain represent traffic flow variables, such as vehicle density, traffic speed, or queue length at an intersection. These states describe the current state of the traffic system, and the transitions between states represent the changes in traffic flow as time progresses. The transition probabilities can be determined by factors such as traffic arrival rates, traffic light cycles, and road conditions.

For example, the states could be defined as:

- a) **State 1:** Low traffic density (few cars on the road, high speed),
- b) **State 2:** Medium traffic density (moderate number of cars, moderate speed),
- c) **State 3:** High traffic density (many cars, low speed).

The transitions between these states depend on the current traffic conditions. If the traffic density is low (State 1), there may be a high probability of staying in the same state or transitioning to medium density (State 2) as more vehicles enter the system. Conversely, if the traffic density is high (State 3), the probability of transitioning to lower densities (State 2 or State 1) could be influenced by factors such as the duration of green lights or vehicles exiting the intersection.

The Markov chain only describes the interaction within the environment. To incorporate autonomous agents into this model, the idea of a Markov Decision Process (MDP) can be used. This model simply extends the Markov chain to include the actions taken by the agent and the rewards received from the environment. In this case, the agent can observe the state of the environment s_t at time t and choose an action a_t . This action may have an impact on the environment, resulting in a new state s_{t+1} , and the agent receives a reward r_t . The goal of the agent is to maximize the cumulative reward over time.

An MDP can be modeled by the tuple (S, A, P, R) , where:

- S : The set of all possible states that represent the environment. At any given time t , the agent observes a state $s_t \in S$.

- A : The set of all possible actions that the agent can take. At time t , the agent chooses an action $a_t \in A$.
- $P(s_{t+1}|s_t, a_t)$: The state transition probability, which defines the probability of transitioning to a new state s_{t+1} given the current state s_t and action a_t .
- $R(s_t, a_t)$: The reward function, which provides a scalar reward signal r_t to the agent for taking action a_t in state s_t . This reward quantifies the immediate benefit of the action.

There is one more idea that will make this model complete. The actions the agent can take to achieve its goal are known as a policy. More formally, if at each time step t , the agent observes the current state s_t and selects an action a_t , this decision-making process is governed by a policy π . The probability of taking an action a_t by an agent having a policy π depends on the current state s_t and is given by the policy function $\pi(a|s)$. The policy is a mapping from states to actions, and it defines the agent's behavior in the environment.

To summarize the whole MDP again, at each time step t , the agent observes the current state s_t , selects an action a_t based on the output of the policy function $\pi(a_t|s_t)$. After taking the action a_t , the environment transitions to a new state s_{t+1} according to the transition probability $P(s_{t+1}|s_t, a_t)$, and the agent receives a reward $r_t = R(s_t, a_t)$. In fact, the diagram shown in Figure 2.2 is a visual representation of the MDP described here.

2.5.3 Reinforcement Learning Theory

RL is a learning process, and the learning process involves learning an optimal policy that achieves the desired objective. This is captured in the reward r_t . However, simply taking actions that maximize the immediate reward r_t may not be the optimal strategy in the long term. It would be better to consider the average rewards received over time for optimization. The returns function encapsulates this idea well. If an agent follows a policy π and receives a reward r_t at time t , then the expected return over time is given by:

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k} \quad (2.2)$$

Here, γ is the discount factor where $0 \leq \gamma \leq 1$, and k is the time horizon. To understand this equation, assume $\gamma = 1$. In other words, there is no discount occurring. In this case, the return function becomes:

$$G_t = \sum_{k=0}^{\infty} r_{t+k} \quad (2.3)$$

Here, the return function is simply the cumulative reward the agent receives over time. As the agent interacts with the environment and keeps receiving the rewards, then the return function simply sums up all the rewards. As the returns function considers rewards received into the future, it is a method of incorporating long-term goals into the agent. Here in equation 2.3, it is assumed that the MDP never terminates and goes on forever. In case the MDP terminates at time T , the return function can be modified to include a terminal state:

$$G_t = \sum_{k=0}^T r_{t+k} \quad (2.4)$$

Referring back to equation 2.2, when $\gamma = 1$, as it is here, the returns function gives both the immediate reward and the future rewards equal weights. However, when $\gamma < 1$, the returns function gives more weight to the immediate reward and less weight to the future rewards. There are two main reasons for introducing the discount factor γ . Firstly, it is infeasible to optimize agents for infinite returns in most cases. Secondly, for most practical purposes, it is better to give more weight to immediate concerns. In fact, there is even a biological basis for this, as neurons in the brain respond more to immediate signals. The discount factor solves all these problems and introduces a parameter that can be tuned to control the balance between immediate rewards and future rewards. When $\gamma = 0$, the agent will only consider immediate rewards and will follow a greedy strategy. When $\gamma = 1$, the agent will consider everything into the future for calculating rewards. Setting $0 < \gamma < 1$ provides a middle ground between the two extremes.

One final question remains: how are the returns calculated? It helps to structure the interaction between the agent and the environment as episodes. An episode refers to a complete sequence of interactions between the agent and the environment, starting from an initial state and ending at a terminal state. It consists of a series of states, actions, and rewards that the agent experiences. In the case of finite episodes, the calculation is straightforward. The return is calculated by summing up the rewards (with the corresponding discount factors) received over the episode. In the case of infinite episodes, the summation is calculated up to a specified number of steps into the future. The discount factor ensures that rewards far into the future are less relevant for the agent, and as a result, it makes it feasible for the agent to learn in non-terminating environments. With the significance of the return function G established, now the learning problem can be formally stated. In reinforcement learning, the agent aims to maximize the expected return G_t from a policy π by learning an optimal policy π^* such that:

$$\pi^* = \arg \max_{\pi} E [G_t | \pi] \quad (2.5)$$

In other words, the agent learns the best policy that will give it the maximum average returns. Note the use of the expectation operator E in equation 2.5. As there is a stochastic or random component in the MDP, it is more practical to work with the average. Using the traffic control example again, if the reward function is defined as the negative of the average vehicle delay at an intersection, then the agent will try to learn a policy that minimizes delays for all vehicles during its operations.

Another thing to note in RL is that the agent does not have access to the transition probability $P(s_{t+1}|s_t, a_t)$. This is also more prevalent in real-world problems. For example, in the traffic control example, there is no realistic way of knowing the exact transition probabilities from a high traffic state to a low traffic state. In RL, this problem is solved by taking an empirical approach. The agent learns the environment dynamics directly or indirectly by interacting with the environment.

Now, the core optimization goal has been defined. The next step is to devise a method for learning the optimal policy. This involves systematically improving the agent's decisions based on a measurable performance criterion. It is not clear how the agent will learn the

optimal policy by looking at equation 2.5. It would be helpful to discuss value functions and Bellman equations before moving forward.

A value function is a way to estimate how good a particular state or state-action pair is. It provides a numerical representation of expected future rewards, allowing the agent to quantify the long-term benefit of decisions. Higher values from the value function indicate better states or state-action pairs. The agent does not know the optimal policy initially, so it needs to estimate the value of being in different states or taking different actions to guide its learning process. Assuming the value function is accurate, optimization also becomes a simple process. Preferring the state-action pairs with the higher values will guide the agent to the optimal policy. There are two main types of value functions:

1. State Value: The state value function represents the expected return from state s under a given policy π . It tells us how good it is to be in state s and follow policy π :

$$V^\pi(s) = E_\pi[G_t \mid S_t = s] \quad (2.6)$$

This is the expected return starting from state s and following the policy π . Sometimes, the state value function is referred to as just the value function.

2. Action Value: The action value function represents the expected return from state s after taking action a , then following policy π :

$$Q^\pi(s, a) = E_\pi[G_t \mid S_t = s, A_t = a] \quad (2.7)$$

This tells us how good it is to take a particular action a in state s and then continue following policy π . The action value function is also known as the Q-value function.

Note that the value function is simply the expected return starting from a particular state or state-action pair. The stochastic nature of the MDP means that these equations also deal with averages. So, the value functions indicate the average reward that an agent can expect from being in a state or taking an action in that state. By picking the actions or states that have the highest value according to the value function, the agent will learn to maximize its rewards with higher probability, and thus learn the optimal policy following equation 2.5.

Using the traffic control example again, an action value function will tell the agent which state is likely to result in the reduction of vehicle delay, given a particular traffic level on an approach.

Related to value functions, the Bellman equations provide a recursive relationship for computing the value of a state or action. The significance of this equation comes from the fact that it gives an iterative method for calculating the value functions. This equation is central to RL and makes it feasible to calculate the value functions, and so, solve the optimization problem shown in 2.5.

The structure of these equations is simple. These equations are derived from the idea that the value of being in a particular state is equal to the immediate reward plus the expected future return from the next state, under a particular policy. For any state s , the value function $V^\pi(s)$ can be written as:

$$V^\pi(s) = E_\pi[r_{t+1} + \gamma V^\pi(S_{t+1}) \mid S_t = s] \quad (2.8)$$

Here, S_{t+1} is the successor state to state $S_t = s$, and r_{t+1} is the reward received when going from S_t to S_{t+1} . In other words, the value of a particular state s is simply the average of the values of all successor states.

Similarly, for any state s and action a , the action value function $Q^\pi(s, a)$ can be written as:

$$Q^\pi(s, a) = E_\pi[r_{t+1} + \gamma Q^\pi(S_{t+1}, A_{t+1}) \mid S_t = s, A_t = a] \quad (2.9)$$

Here, A_{t+1} is the action taken in state S_{t+1} following policy π . In other words, the action value of a particular state-action pair (s, a) is simply the average of the action values of all successor state-action pairs.

Both equations 2.8 and 2.9 are known as Bellman equations. They simplify the process of learning the value functions by establishing a relationship between the current states and successor states. Using this relationship, another set of equations for computing the value functions can be derived, known as the Bellman optimality equations. These equations

define the value of a state or a state-action pair under the optimal policy π^* , which maximizes the expected return.

The optimal state value function $V^*(s)$ represents the maximum expected return that can be achieved starting from state s and following the optimal policy π^* . It satisfies the Bellman optimality equation:

$$V^*(s) = \max_a E [r_{t+1} + \gamma V^*(S_{t+1}) \mid S_t = s, A_t = a] \quad (2.10)$$

This equation states that the value of a state s under the optimal policy π^* is equal to the maximum over all possible actions of the expected reward plus the discounted value of the next state. Here, a is the action that maximizes the reward plus the discounted value of the next state.

Similarly, the optimal action value function $Q^*(s, a)$ represents the maximum expected return achievable starting from state s , taking action a , and following the optimal policy π^* thereafter. It satisfies the Bellman optimality equation:

$$Q^*(s, a) = E \left[r_{t+1} + \gamma \max_{A_{t+1}} Q^*(S_{t+1}, A_{t+1}) \mid S_t = s, A_t = a \right] \quad (2.11)$$

Here, A_{t+1} represents the action taken in the next state S_{t+1} that maximizes the action value function of that next state.

The key insight of the Bellman optimality equations is that to find the optimal policy, choosing the actions or states with higher values will give the desired results. Updating the value functions using the Bellman optimality equations will guide the agent to the optimal policy. The specifics of this process will not be discussed in this thesis. Updating the value functions is done by interacting with the environment and recording the state, action, and rewards. This is sometimes known as sampling. By sampling, an agent obtains a series of interactions with the environment or an episode. Computing value functions, sampling, and the updating process differ between algorithms, each having specific strengths and weaknesses. Some of the algorithms will be discussed in another section.

2.6 Application of Reinforcement Learning in Traffic Control

Over the years, reinforcement learning (RL) has attracted considerable attention as a promising approach for traffic control applications, leading to extensive research in this area (Miletić et al., 2022c), (Noaeen et al., 2022), (Wei et al., 2021), (Abdulhai & Kattan, 2003a). This growing interest is largely driven by the remarkable success of RL in solving complex decision-making problems, including board games such as Go (Silver et al., 2017), video games (Lim, n.d.), and autonomous driving tasks (Kiran et al., 2022). Notably, RL achieved superhuman performance in Go by defeating the world champion in 2016. Motivated by these achievements, researchers have explored the potential of RL to address the dynamic and complex challenges associated with traffic control systems.

Research on the application of RL to traffic control dates back to the 1990s. Among the earliest contributions, Mikami and Kakazu investigated the feasibility of applying RL techniques to traffic signal control (Mikami & Kakazu, 1994). Subsequently, Thorpe and Anderson were the first to propose the use of the SARSA algorithm for traffic control applications, while Wiering et al. pioneered the use of model-based RL approaches in this field (M.A., n.d.). Abdulhai et al. later introduced Q-learning as a method for improving traffic signal performance (Abdulhai et al., 2003). In addition, the survey conducted by Abdulhai and Kattan represents one of the earliest comprehensive reviews of reinforcement learning applications in traffic control systems (Abdulhai & Kattan, 2003b).

Research in reinforcement learning-based traffic control progressed steadily throughout the 2000s, with several studies introducing innovative techniques to address emerging challenges in traffic management. De Oliveira et al. proposed an RL framework incorporating context detection to account for the non-stationary nature of traffic flow (de Oliveira et al., n.d.). The approach adapts to changing traffic conditions by selecting different models according to the prevailing traffic context. Zhang et al. utilized Conditional Random Fields (CRFs) to model the joint decision-making processes of multiple traffic signals, enabling coordinated control across intersections (Zhang et al., 2007). CRFs provide a probabilistic framework for representing complex decision-making problems involving interdependent actions. Furthermore, Su and Tham combined Q-

learning with grid computing technologies to facilitate real-time traffic signal optimization, thereby improving the computational efficiency and responsiveness of RL-based traffic control systems (Su & Tham, 2007).

Around 2010, interest in applying reinforcement learning to traffic control increased substantially, largely due to improvements in computational capabilities. During this period, several studies explored new approaches to enhance the scalability, adaptability, and effectiveness of RL-based traffic management systems. LA and Bhatnagar investigated the use of function approximation techniques within RL frameworks to improve the scalability and flexibility of traffic control applications (LA & Bhatnagar, 2011). El-Tantawy and Abdulhai examined the integration of RL-based traffic signal control with other traffic management measures, including ramp metering and variable message signs, to address more complex transportation challenges (El-Tantawy & Abdulhai, 2010). Natarajan et al. combined imitation learning with reinforcement learning to improve traffic signal performance through knowledge transfer from expert-generated data (Natarajan et al., n.d.). Imitation learning aims to enhance machine learning models by leveraging high-quality demonstrations provided by domain experts. Nuli and Mathew focused on traffic environments characterized by heterogeneous vehicle compositions and weak lane discipline, conditions commonly found in developing countries (Nuli & Mathew, 2013). In addition, A et al. incorporated cumulative prospect theory into RL frameworks to better capture human decision-making behavior and risk preferences in traffic control applications (A. et al., 2015).

The emergence of Deep Reinforcement Learning (DRL) marked a major milestone in the development of intelligent traffic control systems. DRL extends traditional reinforcement learning by employing Deep Neural Networks (DNNs) as function approximators. This capability allows the efficient handling of large and complex state-action spaces that are typical in traffic management problems. One of the earliest studies in this area was conducted by Li et al., who applied DRL to optimize traffic signal timing strategies (Li et al., 2016). Subsequently, Wei et al. proposed IntelliLight, a DRL-based traffic signal control system that determines whether to maintain the current signal phase or switch to the next phase based on prevailing traffic conditions (Wei et al., 2018). Zheng et al.

introduced the FRAP model, which exploits phase competition mechanisms to improve signal control efficiency while simplifying the state representation and reducing training complexity (G. Zheng et al., 2019).

Further advancements followed with the development of coordinated and multi-agent traffic control approaches. Wei et al. proposed CoLight, which utilizes graph neural networks to coordinate traffic signal operations across a network of intersections (Wei, Xu, et al., 2019). They also introduced PressLight, a method that applies lane-based pressure concepts to optimize signal control decisions (Wei, Chen, et al., 2019). Chu et al. developed MA2C, a scalable centralized Multi-Agent Reinforcement Learning (MARL) framework based on the Advantage Actor-Critic algorithm (Chu et al., 2020). The framework leverages the advantage function, defined as the difference between the state-value function and the action-value function, to improve learning performance. Chen et al. later proposed MPLight, a decentralized MARL approach that combines the concepts of lane-based pressure and phase competition from PressLight and FRAP, respectively (C. Chen et al., 2020). Zang et al. introduced MetaLight, which employs meta-reinforcement learning techniques to enable traffic control models to adapt efficiently to different intersection configurations and traffic conditions (Zang et al., 2020). Meta-reinforcement learning enhances model generalization, allowing learned policies to perform effectively in previously unseen environments. In addition, Yen et al. proposed 2DSARSA, a centralized deep on-policy learning approach for optimizing traffic signal control (Yen et al., 2020).

Today, machine learning-based traffic control systems are widely regarded as the state of the art in intelligent transportation research. Most modern approaches rely on deep learning and reinforcement learning techniques, with DRL emerging as the dominant methodology. The success of DRL can be attributed to the availability of powerful computing resources and the large volumes of data generated by sensors deployed in Intelligent Transportation Systems (ITS) (Wei, Zheng, et al., 2019). Consequently, DRL-based methods consistently achieve superior traffic control performance compared to conventional techniques. The findings presented in this thesis further support this observation. Therefore, the primary focus of this research is on DRL-based traffic signal control methods.

Although the development of an RL-based traffic control system involves numerous design considerations, from a traffic engineering perspective the definition of actions, states, and rewards is often more important than the specific learning algorithm employed. As reinforcement learning continues to evolve, more sophisticated algorithms and new data sources are expected to emerge. However, the fundamental design principles governing actions, states, and rewards are likely to remain relatively stable. Furthermore, the design of these components directly influences the selection of sensing technologies and communication infrastructure required to support the traffic control system. For this reason, the discussion in this thesis focuses primarily on the design and formulation of actions, states, and reward functions for reinforcement learning agents in traffic signal control applications.

2.7 Reinforcement Learning Robustness in Traffic Control Systems

Having established the necessary background, the discussion can now shift to the primary focus of this thesis: robustness. In the context of adaptive traffic control, robustness refers to the capability of a control algorithm to sustain its performance when confronted with unexpected events and changing operating conditions. This characteristic is particularly important for real-world traffic management systems, where unforeseen disruptions and uncertainties are inevitable (R. Chen et al., 2022). Despite its practical significance, robustness has received relatively limited attention in research on reinforcement learning-based adaptive traffic control systems. This thesis seeks to contribute to this underexplored area.

Reinforcement learning agents acquire knowledge through interactions with their environment. These interactions are governed by the environment's dynamics, commonly represented by state transition probabilities. RL-based traffic control systems model these dynamics either implicitly in model-free approaches or explicitly in model-based approaches. However, under real-world conditions, these transition probabilities may change significantly when unexpected traffic events occur. For example, a traffic accident that blocks a roadway can alter traffic patterns throughout the network, resulting in changes to the underlying transition dynamics. Consequently, an RL-based traffic controller that

was trained under normal conditions may experience a temporary degradation in performance. In addition to traffic incidents, other sources of uncertainty, such as sensor malfunctions or inaccurate measurements, can further affect the effectiveness of RL-based traffic control systems in practical deployments (R. Chen et al., 2022).

In this thesis, such disruptions and uncertainties are collectively referred to as adverse traffic events and conditions. Any event or condition that causes a temporary decline in system performance is considered adverse. Accordingly, robustness is defined as the ability of a traffic control system to maintain a stable and acceptable level of performance despite the presence of these adverse events and conditions.

2.8 Overview

Chapter 2 reviews the literature related to CAVs, traffic efficiency, traffic safety, and RL-based ATSC. It discusses the development of CAV technology and its potential impacts on urban traffic operations. Previous studies on the effects of CAV penetration on traffic efficiency and safety are reviewed, along with research on RL-based traffic signal control systems. The chapter also examines the key concepts of RL-based ATSC, traffic performance measures, and safety assessment methods. Finally, the research gaps identified from the literature are presented, providing the motivation for this study.

CHAPTER 3

METHODOLOGY

3.1 Introduction

To achieve the specific objectives outlined in Chapter 1, it is necessary to establish a clear methodology for data collection, analysis, and evaluation. Chapter 2 presented a comprehensive literature review of existing studies on traffic efficiency and safety under varying traffic conditions and patterns, different penetration levels of CAVs, and RL-based ATSC.

This chapter describes the methodology and experimental framework adopted to evaluate and achieve the research objectives. It begins by defining the vehicle modeling approach used in this study is presented. Next, it describes the key traffic events and conditions that influence adaptive traffic signal control, as well as traffic efficiency and safety under different CAV penetration levels. The chapter then discusses the parameters employed to assess traffic efficiency and safety, along with the traffic signal control models considered, including both RL-based and conventional non-RL approaches. Finally, the experimental design is described, covering the performance metrics and evaluation procedures used to ensure a comprehensive assessment of the proposed methodology. SUMO (Simulation of Urban Mobility) was used as the simulation platform for all experiments conducted in this thesis.

3.2 Vehicle Modeling

Vehicles with different levels of automation and communication capabilities exhibit distinct driving behaviors in both longitudinal and lateral movements. Consequently, accurately evaluating their impacts on traffic performance through simulation requires the use of separate behavioral models for each vehicle type. To represent these differences, traffic simulation studies commonly employ various car-following and lane-changing models for HDVs and CAVs. This section presents the car-following and lane-changing models adopted in this study. The selected models were chosen based on a comprehensive

review of the relevant literature and their suitability for representing the characteristics of the vehicle types considered.

3.2.1 Car-Following Models

Car-following models describe how a vehicle adjusts its speed and maintains a safe distance or time headway relative to the vehicle traveling ahead. Since different vehicle types exhibit distinct driving characteristics, a variety of car-following models have been developed to represent their behavior in traffic simulation studies. Commonly used models include the Intelligent Driver Model (IDM), Enhanced Intelligent Driver Model (E-IDM), Gipps Model, Krauss Model, Wiedemann Model, and field-tested Adaptive Cruise Control (ACC) and Cooperative Adaptive Cruise Control (CACC) models. These models are widely employed to capture the driving behavior of HDVs, CAVs, and degraded CAVs under different traffic conditions (Ivanchev et al., 2019).

3.2.1.1 CAVs modeling

To simulate CAVs, this study adopts a realistic Cooperative Adaptive Cruise Control (CACC) car-following model whose parameters have been calibrated and validated using data obtained from field experiments conducted under real-world traffic conditions. The model is designed to accurately represent the behavior of CAVs and incorporates four distinct control modes: speed control, gap control, gap-closing control, and collision avoidance. These control modes enable the vehicle to adapt its driving behavior according to surrounding traffic conditions while maintaining safe and efficient vehicle operations. Table 3.1 presents the parameters used to model CAVs (Garg & Bouroche, 2023c) in the simulation.

Table 3.1: Car-Following Model Parameters for Simulated CAVs

Parameter	Value
Speed deviation	0.05
Time headway	0.6 s
Minimum gap	1.5 m
Maximum acceleration	2.9 m s^{-2}
Deceleration	7.5 m s^{-2}
Emergency deceleration	9 m s^{-2}

3.2.1.2 HDVs modeling

HDVs were modeled using the Intelligent Driver Model (IDM) Car-Following Model. The driving behavior of HDVs differs from that of CAVs because human drivers have longer reaction times, make perception errors, and maintain different safety gaps and time headways. Therefore, HDVs require a separate modeling approach. In this study, the IDM model was used together with driver reaction time and perception error parameters to represent the longitudinal driving behavior of human drivers. The IDM is widely used in traffic simulation because it can realistically reproduce stop-and-go traffic behavior. Including human reaction delays and perception errors further improves the realism of the model. Table 3.2 presents the parameters used to model HDVs (Garg & Bouroche, 2023c) in the simulation.

Table 3.2: Car-Following Model Parameters for Simulated HDVs

Parameter	Value
Maximum acceleration a	1.0 m s^{-2}
Desired deceleration b	2.0 m s^{-2}
Time headway T	1.5 s
Free flow speed v_f	33.3 m s^{-2}
Minimum gap s_0	2.0 m
Imperfection σ	0.5

Human drivers generally exhibit large reaction times and perception errors in their driving decision making. To simulate these reaction times and perception errors as the uncertainties in human's driving behavior, a driver state device is added to each HDV with parameters listed in Table 3.3 according to (Maerivoet, n.d.) and (Andreotti et al., 2020). Parameter values are chosen in such a way that perception errors are mainly due to the distance and speed difference errors between an ego vehicle and its preceding vehicle.

Table 3.3: Driver State Device Parameters for Simulating Driver's Imperfection

Parameter	Description	Value
actionStepLength	Driver reaction time in executing the decision logic	0.5 s
initialAwareness	Driver awareness	1.1
errorTimeScaleCoefficient	Time scale constant of the perception error process	100
errorNoiseIntensityCoefficient	Noise intensity constant of the perception error process	0.5
speedDifferenceErrorCoefficient	Scaling coefficient for the relative speed difference error	2
headwayErrorCoefficient	Scaling coefficient for the relative distance difference error	2
speedDifferenceChangePerceptionThreshold	Threshold value for the perception of changes in the speed difference	0.5
headwayChangePerceptionThreshold	Threshold value for the perception of changes in the distance difference	0.5

3.2.2 Lane-Changing Models

Lane-changing models describe how vehicles decide whether to move from one lane to another based on the surrounding traffic conditions. These models use information about

nearby vehicles in both the current and target lanes to determine if a lane change should be performed. Various lane-changing models have been developed using approaches such as the Gipps model, utility theory, cellular automata, Markov processes, and fuzzy logic (Treiber et al., 2006), (Z. Zheng, 2014).

A commonly used approach for lane-changing decisions is the gap acceptance model. In this model, the available gap in the target lane is compared with a critical gap, which represents the minimum safe gap required for a lane change (Kesting & Treiber, 2006). A vehicle is allowed to change lanes only when the available gap exceeds the critical gap. The size of the critical gap generally depends on the speed of the subject vehicle and the speeds of the leading and following vehicles in the target lane. Once a suitable gap is identified, the vehicle adjusts its speed to align with the gap and performs the lane-change maneuver safely.

In this study, the default lane-changing model available in SUMO, known as LC2013, is used to represent the lane-changing behavior of both HDVs and CAVs.

3.3 Traffic Patterns and Conditions

Traffic patterns and operating conditions should represent realistic situations that traffics are likely to encounter in real-world environments and respond to effectively.

3.3.1 Traffic Patterns

Traffic patterns describe the fluctuations in traffic volume and flow over time that result from both recurring and non-recurring influences. Recurring influences include predictable daily changes, such as morning and evening peak-hour traffic, whereas non-recurring influences arise from occasional events that generate unusual traffic demand, such as sporting events, concerts, or festivals. The traffic patterns considered in this thesis are as follows:

1. Daily Traffic Variations: Regular changes in traffic demand associated with commuting activities, including morning and evening peak periods.

2. Special Events: Temporary increases in traffic volume caused by events such as sports matches, concerts, festivals, and other large public gatherings.

3.3.2 Traffic Conditions

Traffic conditions refer to changes in the physical or operational characteristics of a road network that may significantly influence traffic flow and the performance of traffic control systems. They may arise from factors such as road closures, traffic accidents, construction activities, adverse weather conditions, or failures in traffic monitoring equipment. The traffic conditions considered in this thesis are listed below:

- Roadblocks: Physical obstructions caused by accidents, road maintenance activities, construction work, or debris that partially or completely block one or more lanes, thereby disrupting normal traffic flow.

3.4 Scenarios

A scenario represents a particular road network within which all simulation experiments are performed. Each scenario contains multiple cases, where a case refers to a specific configuration of the scenario defined by a unique combination of traffic patterns and traffic conditions. In this thesis, two traffic simulation scenarios were developed using the SUMO simulation platform:

1. Grid 4x4
2. Dublin City Center (DCC)

The Grid 4×4 scenario is a synthetic road network designed to facilitate controlled experimentation and analysis. In contrast, the DCC scenario represents a real-world urban traffic network covering approximately 5 km x 3.5 km of Dublin city centre and includes 435 signalized intersections.

3.4.1 Grid 4x4 Scenario

The Grid 4×4 scenario consists of a synthetic 4×4 road network containing 16 signalized intersections. This scenario is designed to facilitate controlled experimentation under

different traffic compositions. Simulations are conducted using three levels of vehicle penetration, and each penetration level includes six cases representing various traffic patterns and traffic conditions. A map of the road network and the locations of the traffic signals is presented in Figure 3.1. As illustrated in Figure 3.2, the roads running horizontally are major roads consisting of four lanes, whereas the vertically oriented roads are minor roads with two lanes. All roads within the network have a length of 200 m, except for those located at the network boundaries, which are 75 m long.

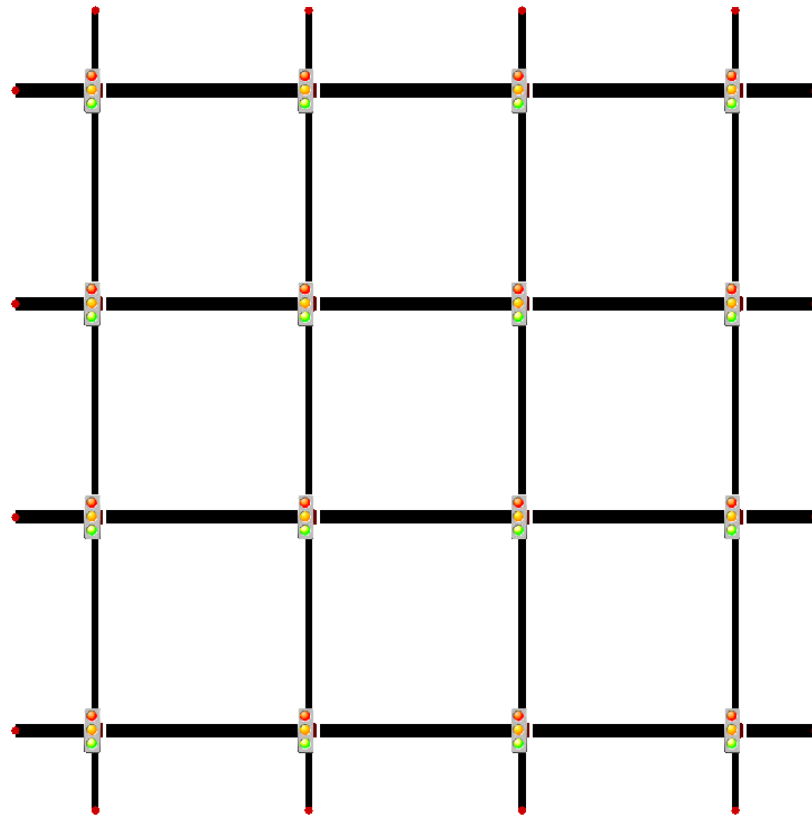


Figure 3.1: Grid 4x4 Scenario Network with 16 traffic lights

The study considers three vehicle penetration scenarios consisting of HDVs and CAVs: HDV Condition, Mixed Condition, and CAV Condition. The corresponding penetration levels of HDVs and CAVs are shown in Table 3.4. Each traffic condition consists of six

cases. Therefore, with three traffic conditions, a total of eighteen cases are evaluated in this study.

Table 3.4: HDVs and CAVs Penetration at HDV Condition, Mixed Condition, and CAV Condition in Grid 4x4 Scenario

Condition Name	HDV	CAV
HDV Condition	100 %	0 %
Mixed Condition	50 %	50 %
CAV Condition	0 %	100 %

3.4.1.1 Trip Generation

Before discussing the individual cases, it is important to describe how trips were generated for this scenario. The randomTrips.py tool in SUMO was utilized to generate trips and produce multiple *.rou.xml route files. These route files represent various traffic patterns and types, which were combined in different configurations to create the traffic demand for all the cases considered in this study. There are four base traffic types that were created for all six cases in this scenario. The traffic types are:

1. Random traffic (Period: 5 seconds)
2. Morning traffic (Period: 5 seconds)
3. Evening traffic (Period: 5 seconds)
4. Special event traffic (Period: 8 seconds)

For generating random traffic, the period was set to one vehicle every 5 seconds, with both the sources and destinations completely randomized. The random traffic type is applied to all cases. The other traffic types will be discussed along with the respective relevant cases.

3.4.1.2 Six cases for each of the three conditions

Case 1: This case represents morning peak-hour traffic and is created by combining

morning traffic demand with random traffic. The simulation is conducted for a duration of one hour. Morning traffic trips are generated by distributing trip origins throughout the network while assigning destinations to the central roads. In the Grid 4×4 scenario, these central roads represent the Central Business District (CBD). The morning traffic pattern is illustrated in Figure 3.2. This setup is intended to model commuter movements from different areas of the network toward the CBD during the morning rush hour.

HDV Case 1 was generated using the HDV condition, Mix Case 1 was generated using the Mixed condition, and CAV Case 1 was generated using the CAV condition.

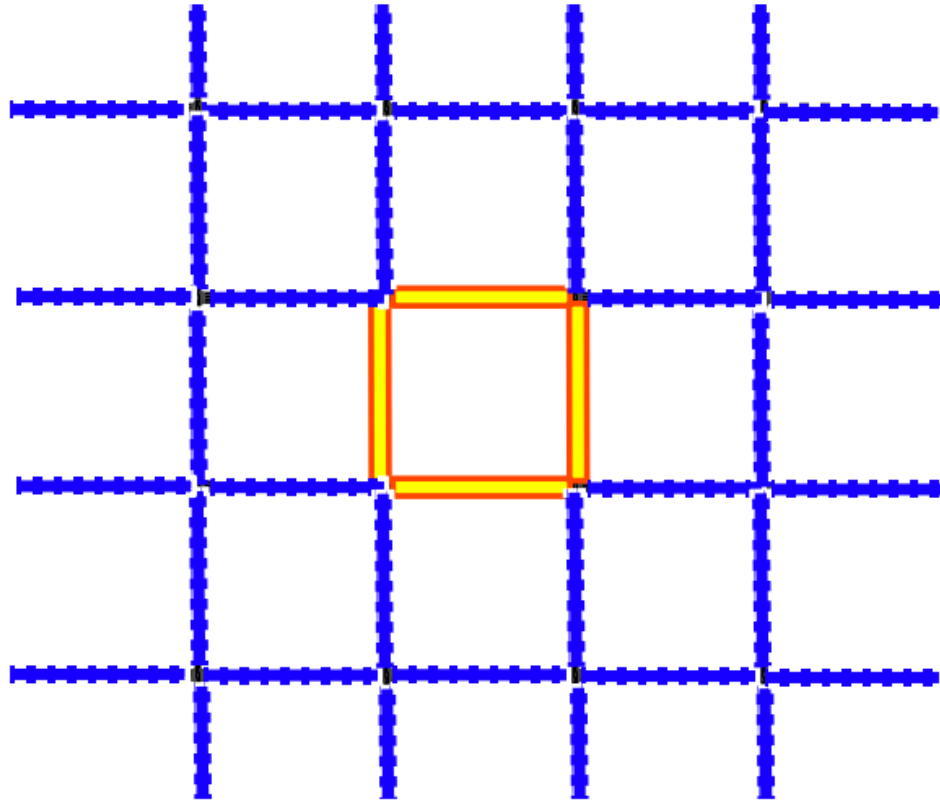


Figure 3.2: Morning traffic trip generation for the Grid 4×4 scenario

In Figure 3.2, The blue roads indicate the origins of trips, while the yellow roads with red borders denote the destinations. This traffic pattern models commuter travel from

residential areas to the central business district.

Case 2: This case also represents morning rush hour traffic. It is designed by combining morning traffic with random traffic, along with the addition of two blocked roads.

HDV Case 2 was generated using the HDV condition, Mix Case 2 was generated using the Mixed condition, and CAV Case 2 was generated using the CAV condition.

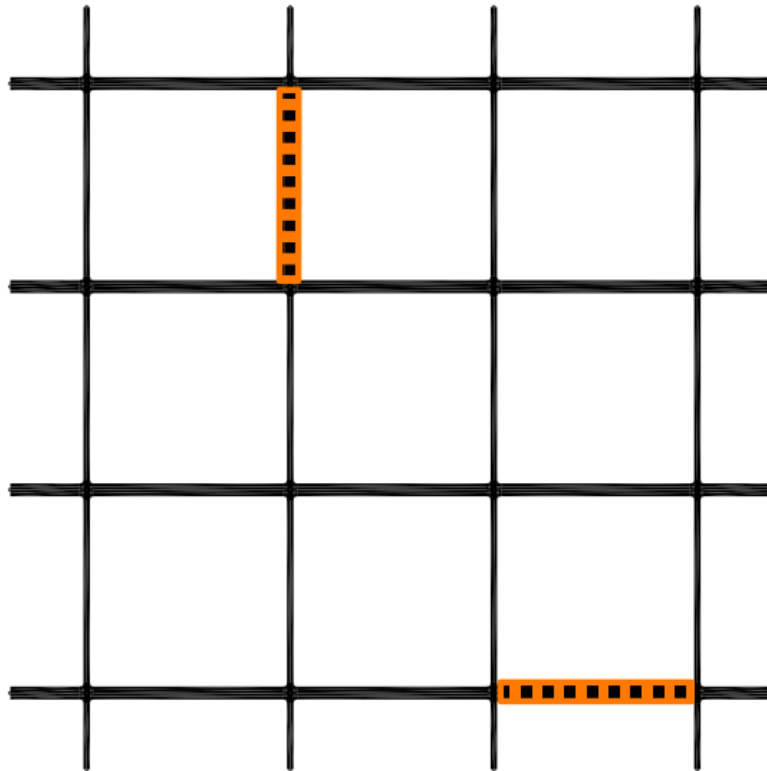


Figure 3.3: Blocked Roads for Grid 4x4 Scenario

In Figure 3.3, The orange, striped roads indicate the blocked roads. There are two blocked roads in the network.

Case 3: This case represents evening peak-hour traffic and is created by combining evening traffic demand with random traffic. Evening traffic trips are generated by assigning the central roads of the network, which represent the central business district (CBD), as trip

origins, while the destinations are located along the network edges, representing residential areas. This configuration is intended to simulate commuters traveling from the CBD to different parts of the network after work hours. The simulation is conducted for a duration of one hour. The evening traffic pattern is illustrated in Figure 3.4.

HDV Case 3 was generated using the HDV condition, Mix Case 3 was generated using the Mixed condition, and CAV Case 3 was generated using the CAV condition.

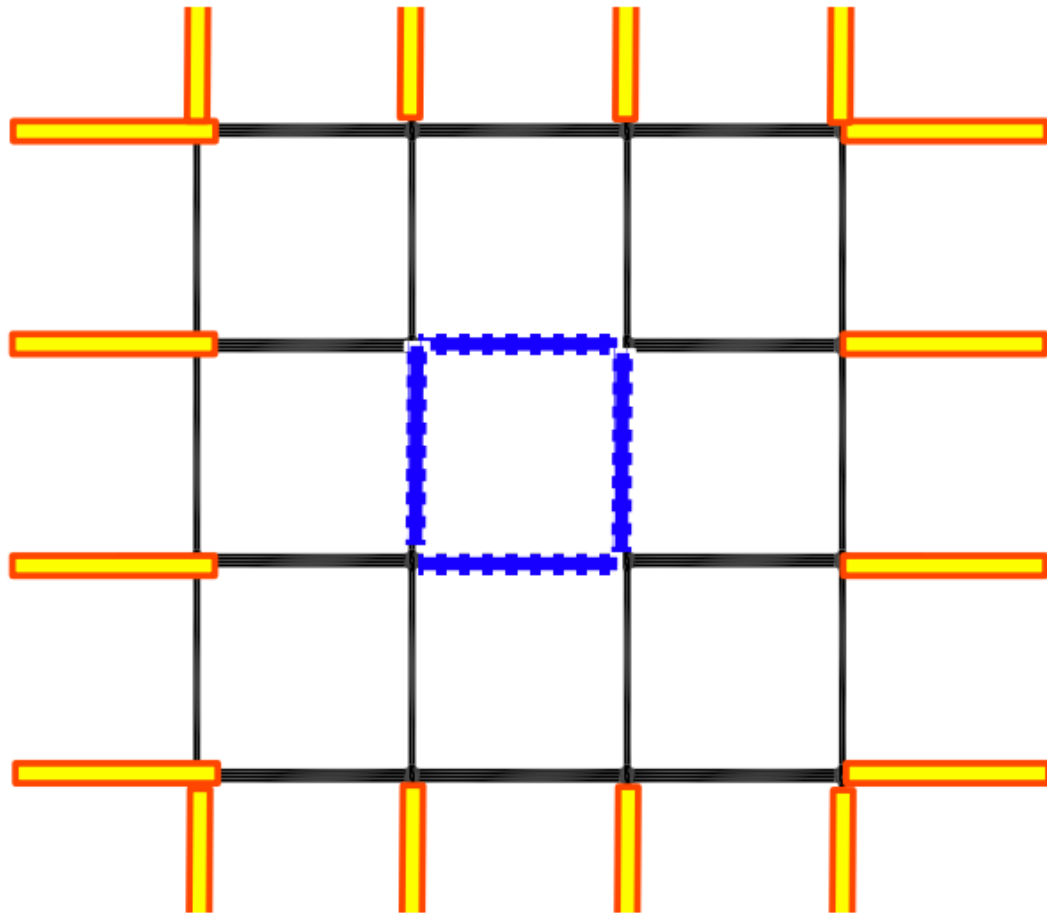


Figure 3.4: Evening traffic trip generation for the Grid 4×4 scenario

In Figure 3.4, The blue highlighted roads represent the trip origins, while the yellow roads outlined in red indicate the destinations. This traffic pattern models the movement of vehicles from the central business district to residential areas during the evening peak period.

Case 4: This case represents evening rush hour traffic with special event traffic. It is designed by combining evening traffic, random traffic, and special event traffic. The special event traffic is generated by setting the origins to be distributed across the network and the destinations to be focused at the bottom end of the network, where the event is taking place. This configuration aims to simulate the surge of vehicles traveling from various parts of the network to attend a special event at evening. The simulation is conducted for a duration of one hour. The special event traffic is shown in Figure 3.5.

HDV Case 4 was generated using the HDV condition, Mix Case 4 was generated using the Mixed condition, and CAV Case 4 was generated using the CAV condition.

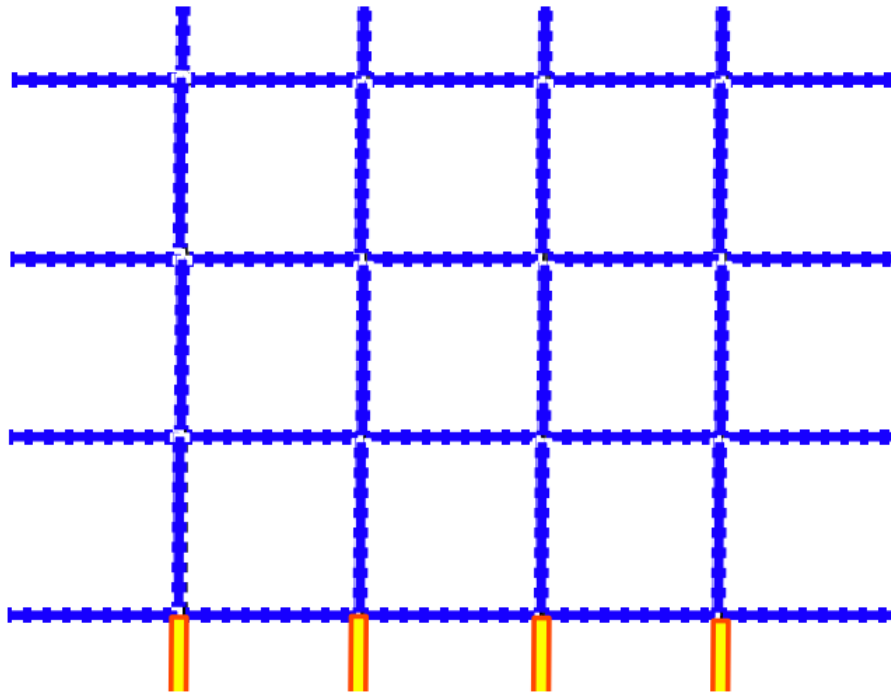


Figure 3.5: Special event traffic trip generation for the Grid 4×4 scenario

The blue highlighted roads represent the trip origins, while the yellow roads outlined in red indicate the destinations. This traffic pattern simulates the movement of vehicles from various parts of the network toward a special event located at the bottom end of the network.

Case 5: This case represents evening rush hour traffic. It is designed by combining evening traffic with random traffic. The main difference between this case and Case 3 is the use of a friction factor to represent adverse weather conditions, specifically wet road conditions. A friction factor of 0.65 (Weber et al., n.d.) was applied to simulate the wet condition. The simulation is conducted for a duration of one hour.

HDV Case 5 was generated using the HDV condition, Mix Case 5 was generated using the Mixed condition, and CAV Case 5 was generated using the CAV condition.

Case 6: This case combines all four traffic types: morning peak-hour traffic, evening peak-hour traffic, special event traffic, and random traffic. It is also categorized as a long-duration case, as the simulation runs for 14 hours. Cases 1 to 5 each have a duration of 1 hour. The traffic patterns included in this case are described below:

1. Morning Traffic (0–4 hours): During the first five hours of the simulation, morning peak-hour traffic is generated, representing vehicles traveling from residential areas to the central business district, as described in Case 1.
2. Evening Traffic (10–14 hours): From the tenth to the fourteenth hour, evening peak-hour traffic is introduced, simulating vehicles traveling from the central business district back to residential areas, as described in Case 3.
3. Special Event Traffic (6–9 hours): During the final three hours of the simulation, special event traffic is added, representing vehicles converging toward an event located at the bottom end of the network, as described in Case 4.
4. Random Traffic (0–14 hours): Random traffic is present throughout the entire simulation period. It introduces variability into the network and represents non-peak-hour travel demand. This traffic operates simultaneously with the morning, evening, and special event traffic patterns.

HDV Case 6 was generated using the HDV condition, Mix Case 6 was generated using the Mixed condition, and CAV Case 6 was generated using the CAV condition.

All six cases are used to evaluate traffic efficiency and safety at HDV condition, mix condition and CAV condition in the road network. These cases are also used to analyze the performance of the RL-based ATSC under the same traffic compositions and vehicle penetration scenarios.

3.4.2 DCC Scenario

The DCC scenario represents a real-world urban traffic network covering approximately 5 km x 3.5 km of Dublin city centre and includes 435 signalized intersections. Simulations are conducted using three levels of vehicle penetration, and each penetration level includes three cases representing various traffic patterns and traffic conditions. A map of the road network and the locations of the traffic signals is presented in Figure 3.6.



Figure 3.6: DCC Scenario Network with 435 traffic lights

The study considers three vehicle penetration scenarios consisting of HDVs and CAVs: HDV Condition, Mixed Condition, and CAV Condition. The corresponding penetration levels of HDVs and CAVs are shown in Table 3.4. Each traffic condition consists of Three cases. Therefore, with three traffic conditions, a total of nine cases are evaluated in this study.

Table 3.5: HDVs and CAVs Penetration at HDV Condition, Mixed Condition, and CAV Condition

Condition Name	HDV	CAV
HDV Condition	100 %	0 %
Mixed Condition	50 %	50 %
CAV Condition	0 %	100 %

3.4.2.1 Three cases for each of the three conditions

The demand patterns and traffic volumes were generated using real-world data. Specifically, several months of data were averaged while excluding holidays and weekends to represent a typical weekday traffic condition. Traffic demand was derived from loop detector data available in an open dataset.

Based on the data analysis, a specific time period of the day was selected for each network to represent three distinct traffic states: free-flow conditions (typically observed in the early morning), saturated conditions (where traffic volumes are high and congestion begins to form), and congested conditions (representing the peak demand level observed in both the simulation and real data). Data are collected from here, https://github.com/maxime-gueriau/ITSC2020_CAV_impact

Case 1 : This case represents the free-flow condition of a real-world urban traffic network in Dublin city centre, which is typically observed during the early morning hours between 4 and 5 a.m.

HDV Case 1 was generated using the HDV condition, Mix Case 1 was generated using the Mixed condition, and CAV Case 1 was generated using the CAV condition.

Case 2 : This case represents the saturated conditions of a real-world urban traffic network in Dublin city centre where traffic volumes are high and congestion begins to form and which is typically observed hours between 8 and 9 a.m.

HDV Case 2 was generated using the HDV condition, Mix Case 2 was generated using the Mixed condition, and CAV Case 2 was generated using the CAV condition.

Case 3 : This case represents the congested conditions of a real-world urban traffic network in Dublin city centre, which is typically observed hours between 5 and 6 p.m.

HDV Case 3 was generated using the HDV condition, Mix Case 3 was generated using the Mixed condition, and CAV Case 3 was generated using the CAV condition.

All three cases are used to evaluate traffic efficiency and safety at HDV condition, mix condition and CAV condition in the road network.

3.5 Traffic efficiency and safety Evaluation

Both the Grid 4×4 and DCC network scenarios are used to evaluate traffic efficiency and safety, and both follow the same procedure for performance assessment.

3.5.1 Traffic efficiency

Various metrics, including travel time, throughput, average speed, total travel delay, and congestion index, have been used in previous studies to assess the impact of CAVs and HDVs on traffic efficiency (Guériau & Dusparic, 2020), (Shladover et al., 2012), (Liu & Fan, 2020), (Talebpour & Mahmassani, 2016b). In this thesis, traffic efficiency is evaluated using the travel time metric, as it is considered the most suitable indicator for assessing the performance of large-scale road networks. To evaluate traffic efficiency across the entire network, the travel time of each road segment (edge) was collected. The travel time was then normalized by dividing it by the corresponding segment length, resulting in a travel time rate.

Travel time (TT) represents the time required for vehicles to traverse a specific edge of the road network. To calculate this metric, the mean speed of vehicles on each edge was

collected using edge-based network state detectors and combined with the corresponding edge length to determine the travel time. In addition, the travel time rate (TTR) was calculated by normalizing the travel time with respect to the length of each road segment.

$$TT \text{ (sec)} = \sum_i \frac{q_i}{V_i} \quad (1)$$

$$TTR \text{ (min/km)} = \frac{TT_i}{q_i} * 16.6667 \quad (2)$$

where v_i and q_i are the mean speed and length of the i th edge respectively (Garg & Bouroche, 2023c).

3.5.2 Traffic safety

To assess the impact of HDVs and CAVs on road safety, an estimate of the collision risk between vehicles was calculated within the simulation environment. This was achieved by recording surrogate microscopic traffic safety indicators, also known as Surrogate Safety Measures (SSM), which quantify interactions between closely spaced vehicles. Following the methodology, potential conflicts between vehicles were identified and analyzed to evaluate safety performance.

For the DCC network, the Post-Encroachment Time (PET) metric was selected as the safety indicator because it is particularly suitable for evaluating intersecting conflicts (Mahmud et al., 2017), such as potential side-impact collisions. This choice was motivated by the large number of signalized intersections present in the Dublin city centre network. A conflict is considered to occur when the PET value falls below a predefined threshold, a methodology that has been shown to identify conflict locations consistent with those observed in real-world networks (Guido et al., 2019). Following the approach adopted in (Morando et al., 2018b), and to account for the differing operational characteristics of CAVs and HDVs, a PET threshold of 0.75 seconds was used for all conflict types in the urban DCC network, where conflicts predominantly occur at intersections.

3.6 RL-based ATSC Performance Evaluation

To analyze the performance of the RL-based ATSC under varying traffic conditions and different penetration levels of CAVs and HDVs, three vehicle composition scenarios were considered: HDV condition, mix condition, and CAV condition. Two traffic signal control models were evaluated in this study:

1. IDQN (RL-based controller)
2. FixedTime controller (Non-RL baseline)

The performance evaluation for this objective was conducted using only the Grid 4×4 network. The DCC network was not considered for this analysis due to technical limitations associated with its large network size, which significantly increased the computational complexity and training requirements of the RL model.

3.6.1 IDQN (RL-based controller)

IDQN is a reinforcement learning (RL) model that employs a Deep Q-Network (DQN). Although the LibSignal framework does not explicitly define the acronym IDQN, it is generally understood to represent an Independent Deep Q-Network. In this approach, a separate agent is deployed at each intersection, and each traffic signal controller operates independently without exchanging information with neighboring intersections.

Reinforcement learning is a machine learning technique in which an agent learns to make decisions through interactions with its environment. The agent observes the current state, selects an action, and receives feedback in the form of rewards. Through repeated interactions, the agent learns a policy that maximizes long-term rewards. In traffic signal control applications, the objective is to learn signal timing strategies that improve traffic flow and reduce congestion.

IDQN is a value-based RL algorithm that relies on Q-learning to estimate the expected cumulative reward associated with each state-action pair. Since traffic signal control involves large and complex state and action spaces, function approximation techniques are required. To address this challenge, Deep Q-Networks use neural networks to approximate

the Q-function, enabling the model to effectively handle large-scale and dynamic traffic environments.

In essence, the Deep Q-Network evaluates the quality of different state-action pairs and estimates their expected rewards. Based on these estimates, the IDQN agent selects the traffic signal phase that is expected to yield the best traffic performance at a given time.

3.6.2 FixedTime controller (Non-RL baseline)

The Fixed Time model is a non-RL approach that allocates a fixed amount of green time to each phase. It is a simple model that does not require any learning process. This model represents the typical fixed-cycle-time traffic signal control approach, which is the most commonly used in practice in the real world.

All code related to the implemented traffic signal control models was obtained from the LibSignal repository. The repository link is provided here: <https://github.com/DaRL-LibSignal/LibSignal>

3.6.3 Recovery Time

For evaluating this objective, a evaluation metric, recovery time, is introduced in this study. When an RL model encounters a previously unseen traffic pattern or operating condition, it requires a certain amount of time to adapt and learn an effective control strategy before achieving its optimal performance. This adaptation period is referred to as the recovery time. The recovery time differs across models and traffic scenarios and serves as an indicator of how quickly a model can adjust to changes in traffic conditions.

Following the initial training phase, the RL models were evaluated across all cases within each scenario. For every case, the models were executed for a fixed number of episodes, and the average travel time for each episode was recorded. Since lower average travel times correspond to better traffic control performance, these data can be used to determine the time required for a model to regain its optimal performance level.

However, the relationship between average travel time and episode number often exhibits significant fluctuations, resulting in a spiky performance curve. These variations occur because the model may temporarily alternate between higher and lower performance levels during the learning process. To reduce the impact of these fluctuations and obtain a more reliable estimate of recovery time, a window-based analysis combined with predefined threshold values was employed in this thesis. The recovery time is calculated by using the LibSignal code directly in this study.

3.6.4 Initial Training

The initial training phase refers to training the RL models on a designated training case before evaluating their performance on previously unseen cases. This process is essential because RL-based traffic signal controllers learn through interactions with the traffic environment and therefore require training prior to deployment. Performance degradation in unseen cases is expected because the traffic characteristics in those cases differ from those encountered during training. A robust RL-based traffic controller should be capable of maintaining good performance despite such variations in traffic patterns and conditions. This principle forms the basis of the experimental design adopted in this study. In contrast, non-RL models do not require an initial training phase.

For the initial training process, the RL-based traffic control model IDQN was trained using Case 1 of the HDV condition in the Grid 4x4 network for 200 episodes. Each episode represents one hour of simulation. After the training process was completed, the trained model weights were saved and subsequently used to evaluate the performance of the model under all six cases of each traffic condition, including the HDV condition, mix condition, and CAV condition.

3.7 Overview

Chapter 3 describes the methodology used to evaluate the impact of Connected and Autonomous Vehicle (CAV) penetration on traffic efficiency, traffic safety, and the performance of Reinforcement Learning (RL)-based Adaptive Traffic Signal Control (ATSC). The microscopic traffic simulation software SUMO is used as the simulation

platform for all experiments. Two network scenarios, namely the Grid 4x4 network and the Dublin City Centre (DCC) network, are developed to assess traffic efficiency and safety under three traffic conditions: HDV condition, mix condition, and CAV condition. Traffic efficiency is evaluated using travel time-based performance measures, while traffic safety is assessed using surrogate safety measures derived from vehicle interactions. To analyze the performance of RL-based traffic signal control, the IDQN model is implemented and compared with a conventional FixedTime controller in the Grid 4x4 network. Furthermore, a recovery time metric is introduced to evaluate the adaptability of the RL model when exposed to unseen traffic conditions and demand patterns, providing insight into its ability to regain optimal performance after changes in the traffic environment.

CHAPTER 4

DATA COLLECTION AND PROCESSING

4.1 Introduction

To achieve the specific objectives presented in Chapter 1, the methodology described in Chapter 3 was implemented. The objective was to assess improvements in traffic efficiency and safety under different traffic conditions and patterns across varying penetration levels of CAVs and HDVs, as well as to evaluate the performance of RL-based ATSC under these conditions. This chapter focuses on the data collection and analysis procedures employed in the study. It begins with an overview of the key data collected, followed by a detailed description of the methods used for data acquisition. Most of the data collection details were discussed in Chapter 3. This chapter covers the remaining details and presents the simulation framework and outlines the process used for analyzing the collected data.

4.2 Data Collection Process

In this study, all scenarios were simulated using the SUMO platform. The simulation parameters were carefully calibrated to represent real-world traffic conditions as closely as possible. For the Grid 4×4 network scenario, a total of 18 simulation runs were conducted. Among these, six runs were performed under the HDV condition, six under the mix condition, and six under the CAV condition for the evaluation of all research objectives. Similarly, for the DCC network scenario, a total of nine simulation runs were conducted, consisting of three runs under the HDV condition, three under the mix condition, and three under the CAV condition for the evaluation of all objectives. All SUMO environment files can be found at https://github.com/mdaslamcee/row_data_SUMO.

4.2.1 Data Collection Process for Traffic Efficiency and Safety Evaluation

In this study, data were collected from multiple simulation runs to evaluate the ability of different traffic control models to adapt to varying traffic patterns and conditions. The same

procedure was followed for both the Grid 4×4 and DCC network scenarios to obtain traffic efficiency and safety data.

4.2.1.1 Traffic Efficiency Data Collection

Traffic efficiency data were collected from each simulation run by extracting edge-based traffic information generated by SUMO. After collecting the edge data for all simulation cases, a Python-based data processing script was used to calculate the Travel Time Rate (TTR). The script was designed to extract the required information from the edge data files, compute the travel time rate for each edge of the network, and generate corresponding color map visualizations based on the calculated TTR values. The data and code is available at https://github.com/mdaslamcee/graph_plot_data.

4.2.1.2 Traffic Safety Data Collection

Traffic safety was evaluated by measuring the total number of traffic conflicts occurring at network intersections. In this study, the Surrogate Safety Measure (SSM) approach was adopted using the Post Encroachment Time (PET) indicator. To collect the required safety data, all simulation cases were executed in SUMO with the SUMO configuration files modified accordingly. After data collection, a customized Python script was used to process the SSM output files and determine the total number of conflicts occurring within the network intersections. The data and code is available at https://github.com/mdaslamcee/graph_plot_data.

4.2.2 Data Collection Process for the Performance Evaluation of RL-Based ATSC

In this study, data were collected from multiple simulation runs to analyze the performance of the RL-based ATSC under varying traffic conditions and patterns at different penetration levels of HDVs and CAVs. As outlined in the methodology, the performance evaluation focused on two key metrics:

1. Average Travel Time
2. Recovery Time

The following sections provide a detailed description of the procedures used to collect these data.

Both the traffic simulation software SUMO and the RL traffic controller testing framework LibSignal were utilized for data collection. Specifically, a modified version of the LibSignal framework was employed to evaluate the performance of the RL-based traffic controller models under different traffic scenarios. The modified framework is available at <https://github.com/mdaslamcee/LibSignal>.

The simulation outputs generated by SUMO and LibSignal were recorded for each experimental scenario. However, the collected raw data could not be used directly for performance evaluation. Therefore, the data were further processed and organized into a suitable format to facilitate analysis and interpretation of the RL-based ATSC performance under varying traffic conditions. The data processing and analysis procedures were implemented using customized scripts, which are available at https://github.com/mdaslamcee/RL_robust_data.

4.3 Overview

This chapter presented the data collection and preprocessing procedures used to evaluate traffic efficiency, traffic safety, and the performance of the RL-based ATSC. It described the key evaluation metrics, including travel time rate, number of conflicts, average travel time, and recovery time, and explained how these data were obtained from simulation runs using SUMO and the modified LibSignal framework. The chapter also outlined the data processing methods employed to prepare the simulation outputs for analysis. The results and findings derived from these data are presented in the following chapter.

CHAPTER 5

RESULTS AND ANALYSIS

5.1 Introduction

Following the data collection procedures described in Chapter 4 and the methodology presented in Chapter 3, this chapter presents the results and findings of the study. The chapter is organized into three main sections. First, the traffic efficiency results are presented and analyzed. Second, the traffic safety performance is evaluated based on the collected safety data. Finally, the performance of the RL-based ATSC system is assessed using average travel time and recovery time as the primary evaluation measures.

5.2 Traffic Efficiency

In this section, the impact of CAVs on traffic efficiency is evaluated using the Travel Time Rate (TTR) metric. The results are first presented and analyzed for both the Grid 4x4 and DCC network scenarios under three traffic conditions: HDV condition (100 % HDVs), mix condition (50 % HDVs and 50 % CAVs), and CAV condition (100 % CAVs). For each scenario, the effects of different vehicle compositions on traffic efficiency are examined. Subsequently, a comprehensive comparative analysis is conducted to assess the overall influence of increasing CAV penetration levels on network-wide traffic efficiency.

5.2.1 Grid 4x4 Scenario

5.2.1.1 Case 1 (Morning Peak-Hour Traffic)

This case represents morning peak-hour traffic and is generated by combining morning traffic demand with random traffic demand. In Case 1, the HDV case shown in Figure 5.1(a) exhibits high Travel Time Rate (TTR) values throughout the grid network. Nearly all horizontal and vertical links are colored deep red, indicating severe congestion across the network.

The mix case shown in Figure 5.1(b) demonstrates a noticeable improvement. Several horizontal links change from deep red to orange, indicating that the introduction of 50% CAVs helps reduce congestion on these road sections. However, some vertical links still remain heavily congested.

In the CAV case shown in Figure 5.1(c), the improvement becomes much more evident. Most of the horizontal links are colored light blue, indicating travel times close to free-flow conditions. Although a few vertical links still show relatively high TTR values, the overall level of congestion is significantly reduced compared to the HDV and mixed traffic cases. Overall, Case 1 shows that increasing the penetration of CAVs improves traffic efficiency and reduces congestion across the network.

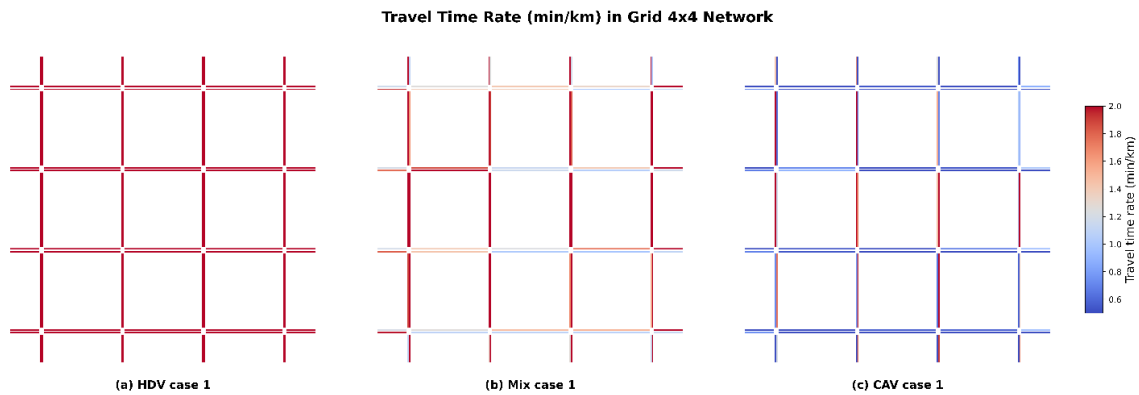


Figure 5.1: Spatial distribution of Travel Time Rate (min/km) in the Grid 4x4 network under Case 1: (a) HDV case 1, (b) mix case 1, and (c) CAV case 1

5.2.1.2 Case 2 (Morning Peak-Hour Traffic with Road Blockages)

This case represents morning rush-hour traffic and is generated by combining morning traffic demand with random traffic demand, along with the addition of two blocked roads.

In Case 2, the HDV case shown in Figure 5.2(a) exhibits high Travel Time Rate (TTR) values across the entire grid network. Both horizontal and vertical links are predominantly colored deep red, indicating severe congestion throughout the network. The presence of the blocked roads further increases congestion by forcing vehicles to use alternative routes.

The mix case shown in Figure 5.2(b) shows a noticeable improvement. Congestion is reduced mainly on the horizontal links, which change from deep red to lighter colors. This indicates that the introduction of 50% CAVs helps improve traffic flow despite the road blockages. However, several vertical links remain red, indicating that congestion still persists in those areas.

In the CAV case shown in Figure 5.2(c), a significant improvement in traffic efficiency can be observed. Most horizontal links are colored light blue, indicating low TTR values and traffic conditions close to free-flow operation. However, compared to Case 1, a slightly larger number of vertical links, particularly in the lower portion of the grid, continue to show relatively high TTR values. This suggests that the combination of road blockages and traffic demand places greater pressure on certain routes, resulting in localized congestion even under full CAV penetration.

Overall, Case 2 shows that increasing the penetration of CAVs improves traffic efficiency and reduces congestion across the network. However, the effects of the blocked roads can still be observed on several links, even in the CAV case.

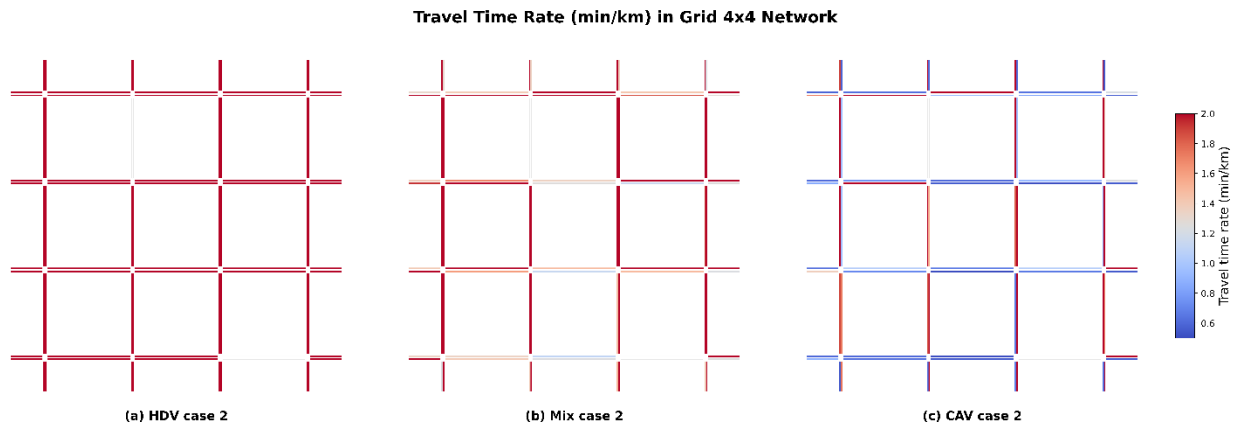


Figure 5.2: Spatial distribution of Travel Time Rate (min/km) in the Grid 4x4 network under Case 2: (a) HDV case 2, (b) mix case 2, and (c) CAV case 2

5.2.1.3 Case 3 (Evening Peak-Hour Traffic)

This case represents evening peak-hour traffic and is generated by combining evening traffic demand with random traffic demand.

In Case 3, the HDV case shown in Figure 5.3(a) exhibits high TTR values throughout the grid network. Both horizontal and vertical links are predominantly colored deep red, indicating severe congestion across the entire network.

The mix case shown in Figure 5.3(b) shows a noticeable improvement in traffic efficiency. Most horizontal links change from deep red to orange, indicating a reduction in congestion with the introduction of 50% CAVs. However, several vertical links, particularly in the central part of the grid, remain heavily congested and continue to exhibit high TTR values.

In the CAV case shown in Figure 5.3(c), traffic conditions improve further. Most horizontal links are colored light blue, indicating low TTR values and traffic flow close to free-flow conditions. However, a small group of vertical links in the central-left area of the grid still shows relatively high TTR values. This indicates that traffic demand remains concentrated on certain routes, causing localized congestion even under full CAV penetration.

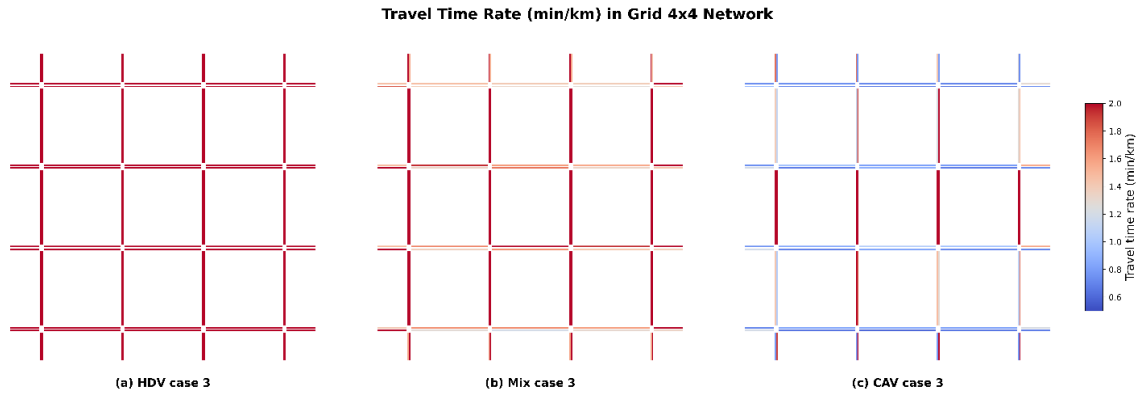


Figure 5.3: Spatial distribution of Travel Time Rate (min/km) in the Grid 4x4 network under Case 3: (a) HDV case 3, (b) mix case 3, and (c) CAV case 3

Overall, Case 3 shows that increasing the penetration of CAVs significantly improves traffic efficiency and reduces congestion across the network. However, some localized congestion remains on a few vertical links due to the uneven distribution of traffic demand.

5.2.1.4 Case 4 (Evening Peak-Hour Traffic with Special Event Traffic)

This case represents evening rush-hour traffic with special event traffic and is generated by combining evening traffic demand, random traffic demand, and special event traffic demand. Similar to the previous cases, increasing the penetration of CAVs improves traffic efficiency across the network. However, compared to Case 1, a greater level of residual congestion can be observed in both the mix case and CAV case.

In Case 4, the HDV case shown in Figure 5.4(a) exhibits high TTR values throughout the grid network. Both horizontal and vertical links are predominantly colored deep red, indicating severe congestion across the entire network.

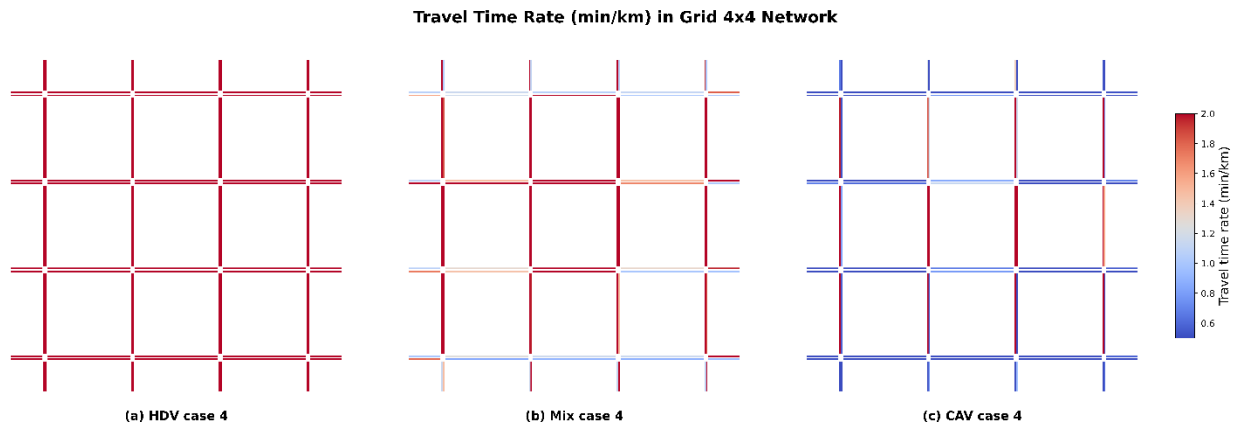


Figure 5.4: Spatial distribution of Travel Time Rate (min/km) in the Grid 4x4 network under Case 4: (a) HDV case 4, (b) mix case 4, and (c) CAV case 4

The mix case shown in Figure 5.4(b) shows a noticeable improvement in traffic efficiency. Several horizontal links change from deep red to orange, indicating a reduction in congestion with the introduction of 50% CAVs. However, the improvement is not uniform across the network, and several vertical links remain heavily congested, particularly in the right portion of the grid.

In the CAV case shown in Figure 5.4(c), traffic conditions improve further. Most horizontal links are colored light blue, indicating low TTR values and traffic flow close to free-flow conditions. However, compared to the CAV case in Case 1, several vertical links,

particularly in the right portion of the grid, continue to show relatively high TTR values. This suggests that the combination of evening traffic and special event traffic places greater pressure on these routes, resulting in localized congestion even under full CAV penetration.

Overall, Case 4 shows that increasing the penetration of CAVs improves traffic efficiency and reduces congestion across the network. However, compared to Case 1, a larger number of links continue to experience localized congestion due to the more challenging traffic demand pattern.

5.2.1.5 Case 5 (Evening Peak-Hour Traffic under Bad Weather Conditions)

This case represents evening rush-hour traffic under bad weather conditions and is generated by combining evening traffic demand with random traffic demand. Similar to the previous cases, increasing the penetration of CAVs improves traffic efficiency across the network. However, a greater level of residual congestion can still be observed in both the mix case and CAV case due to the adverse weather conditions.

In Case 5, the HDV case shown in Figure 5.5(a) exhibits high Travel Time Rate (TTR) values throughout the grid network. Both horizontal and vertical links are predominantly colored deep red, indicating severe congestion across the entire network.

The mix case shown in Figure 5.5(b) shows a noticeable improvement in traffic efficiency. Several horizontal links change from deep red to orange, indicating a reduction in congestion with the introduction of 50% CAVs. However, the improvement is not uniform across the network, and some links continue to exhibit relatively high TTR values. Several vertical links also remain heavily congested, indicating that traffic flow is still restricted in these areas.

In the CAV case shown in Figure 5.5(c), traffic conditions improve considerably. Most horizontal links are colored light blue, indicating low TTR values and traffic flow close to free-flow conditions. However, several vertical links, continue to show relatively high TTR values compared to the corresponding CAV cases in the earlier scenarios. This suggests that the combination of heavy traffic demand and adverse weather conditions places greater pressure on these routes, resulting in localized congestion even under full CAV penetration.

Overall, Case 5 shows that increasing the penetration of CAVs improves traffic efficiency and reduces congestion across the network. However, the efficiency gains are somewhat limited by the bad weather conditions, and several links continue to experience localized congestion.

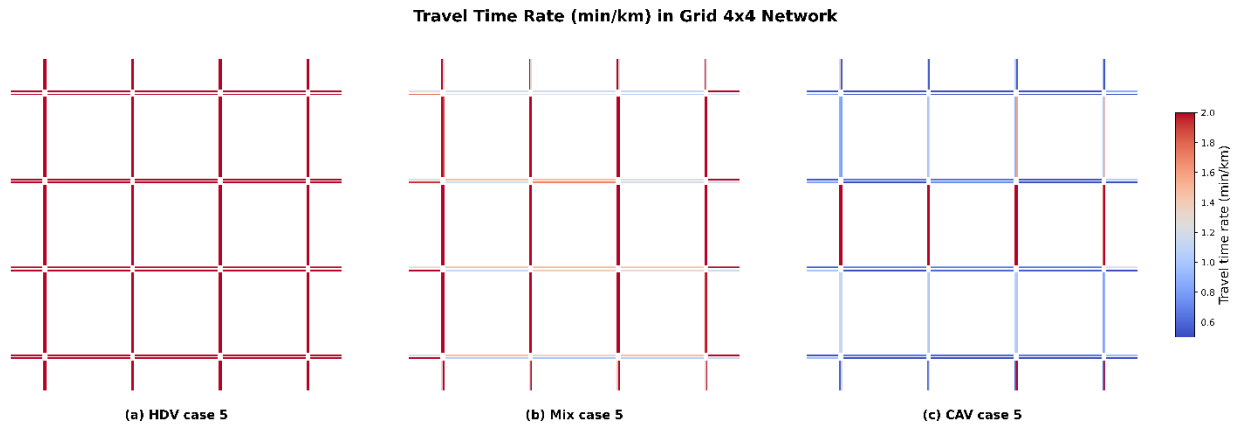


Figure 5.5: Spatial distribution of Travel Time Rate (min/km) in the Grid 4x4 network under Case 5: (a) HDV case 5, (b) mix case 5, and (c) CAV case 5

5.2.1.6 Case 6 (Combined Traffic Scenario)

This case combines all four traffic types: morning peak-hour traffic, evening peak-hour traffic, special event traffic, and random traffic. It is categorized as a long-duration case, with the simulation running for 14 hours. Due to the combination of multiple traffic demand patterns over an extended period, Case 6 represents the most complex traffic condition among all six cases.

In Case 6, the HDV case shown in Figure 5.6(a) exhibits high Travel Time Rate (TTR) values throughout the grid network. Both horizontal and vertical links are predominantly colored deep red, indicating severe congestion across the entire network.

The mix case shown in Figure 5.6(b) shows a noticeable improvement in traffic efficiency. Similar to the previous cases, several horizontal links change from deep red to orange,

indicating a reduction in congestion with the introduction of 50% CAVs. However, the level of improvement varies across the network. Some horizontal links experience greater reductions in TTR values than others.

In the CAV case shown in Figure 5.6(c), traffic conditions improve significantly across most of the network. The majority of horizontal and vertical links are colored light blue, indicating low TTR values and traffic flow close to free-flow conditions. However, a vertical link in the lower- two links exhibit higher TTR values than the rest of the network. This indicates that certain traffic demand patterns continue to create localized congestion even under full CAV penetration.

Overall, Case 6 shows that increasing the penetration of CAVs substantially improves traffic efficiency and reduces congestion across the network. However, due to the complexity of the traffic demand pattern and the long simulation duration, a small number of links continue to experience localized congestion.

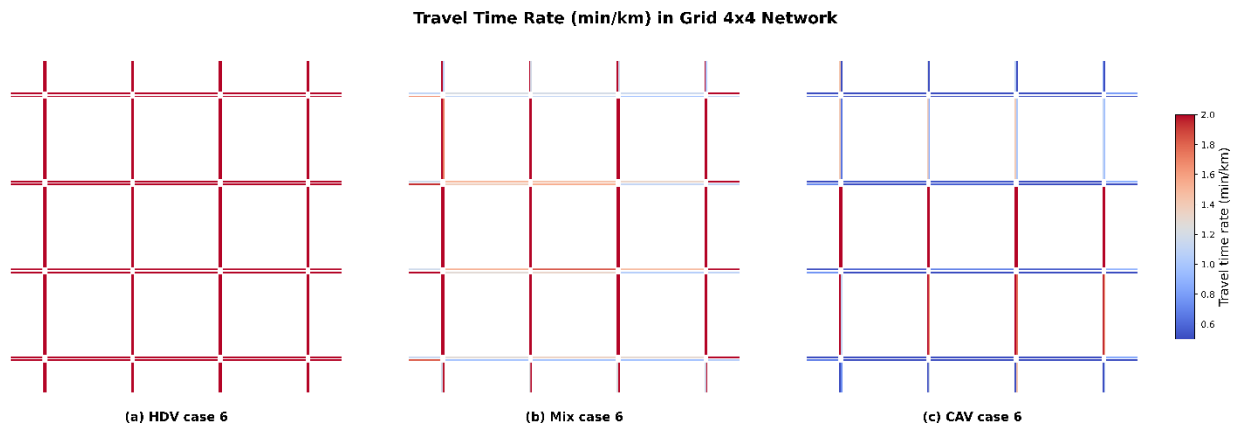


Figure 5.6: Spatial distribution of Travel Time Rate (min/km) in the Grid 4x4 network under Case 6: (a) HDV case 6, (b) mix case 6, and (c) CAV case 6

5.2.2 DCC Scenario

5.2.2.1 Case 1 (Free-Flow Traffic Conditions)

This case represents the free-flow condition of a real-world urban traffic network in Dublin city centre.

In Case 1, the HDV case shown in Figure 5.7(a) exhibits high Travel Time Rate (TTR) values across most parts of the network. The majority of streets are colored deep red, indicating significant congestion throughout the city centre. Despite the relatively low traffic demand, the dense road network and large number of intersections contribute to increased travel times and reduced traffic efficiency.

The mix case shown in Figure 5.7(b) demonstrates a noticeable improvement in traffic efficiency. Many roads, particularly the major arterial roads and routes along the outer parts of the network, change from deep red to orange, indicating a reduction in congestion with the introduction of 50% CAVs. In addition, several roads near the boundary of the network are colored light blue, suggesting traffic conditions close to free-flow operation. However, parts of the city centre continue to exhibit relatively high TTR values, indicating that congestion still remains in the most densely connected areas.

In the CAV case shown in Figure 5.7(c), traffic conditions improve further across the network. Most roads are colored light blue, indicating low TTR values and traffic flow close to free-flow conditions. The greatest improvements can be observed on the peripheral routes. However, a group of streets in the central parts of network continues to show orange and red colors, indicating the presence of localized congestion. This suggests that certain bottlenecks associated with complex intersections remain even under full CAV penetration.

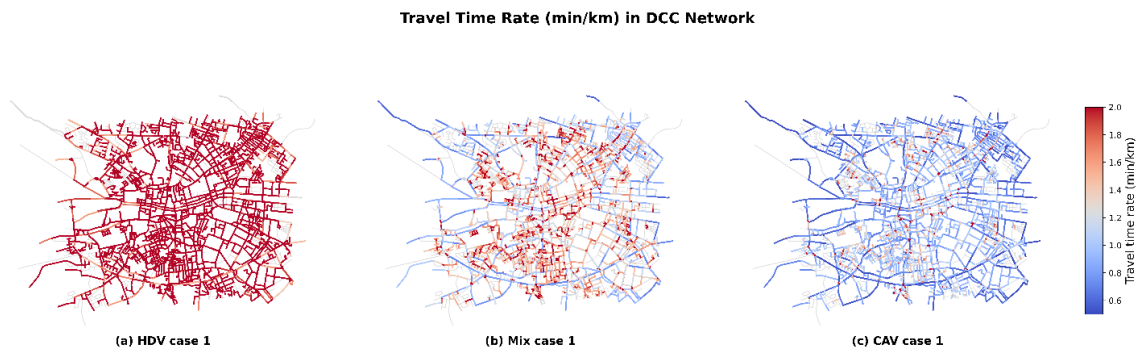


Figure 5.7: Spatial distribution of Travel Time Rate (min/km) in the DCC network under Case 1:
(a) HDV case 1, (b) mix case 1, and (c) CAV case 1

Overall, Case 1 shows that increasing the penetration of CAVs significantly improves traffic efficiency throughout the Dublin city centre network. However, some localized

congestion remains in highly connected urban areas where intersection complexity continues to affect traffic performance.

5.2.2.2 Case 2 (Saturated Traffic Conditions)

This case represents the saturated condition of a real-world urban traffic network in Dublin city centre, where traffic volumes are high and congestion begins to form across the network. Compared to Case 1, a higher level of congestion can be observed due to the increased traffic demand.

In Case 2, the HDV case shown in Figure 5.8(a) exhibits high Travel Time Rate (TTR) values throughout the network. Nearly all streets are colored deep red, indicating severe congestion across the city centre. Compared to the HDV case in Case 1, congestion is more widespread and uniformly distributed, reflecting the effects of increased traffic volume under saturated conditions.

The mix case shown in Figure 5.8(b) demonstrates a noticeable improvement in traffic efficiency. Many arterial roads, peripheral roads, and major radial corridors change from deep red to orange, indicating a reduction in congestion with the introduction of 50% CAVs. However, compared to Case 1, a larger portion of the city centre remains heavily congested, suggesting that the higher traffic demand limits the overall improvement. The most visible improvements are observed on roads located outside the dense urban core.

In the CAV case shown in Figure 5.8(c), traffic conditions improve further across the network. More roads transition to lighter colors compared to the mix case, indicating lower TTR values and improved traffic flow. However, several streets in the central and eastern parts of the network continue to exhibit relatively high TTR values. This indicates that although full CAV penetration significantly improves traffic efficiency, congestion is not completely eliminated under saturated traffic conditions.

Overall, Case 2 shows that increasing the penetration of CAVs improves traffic efficiency throughout the Dublin city centre network. However, compared to Case 1, a greater level of congestion remains within the urban core, indicating that high traffic demand and the

complexity of the city centre road network continue to influence traffic performance even under full CAV penetration.

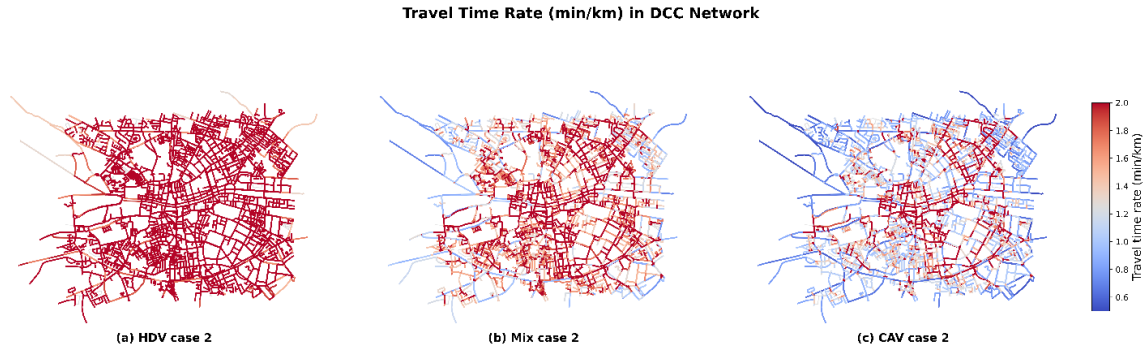


Figure 5.8: Spatial distribution of Travel Time Rate (min/km) in the DCC network under Case 2:
(a) HDV case 2, (b) mix case 2, and (c) CAV case 2

5.2.2.3 Case 3 (Congested Traffic Conditions)

This case represents the congested condition of a real-world urban traffic network in Dublin city centre. Among the three traffic demand scenarios, Case 3 represents the most demanding condition, with traffic volumes high enough to create severe congestion across the network.

In Case 3, the HDV case shown in Figure 5.9(a) exhibits extremely high Travel Time Rate (TTR) values throughout the network. Almost all streets are colored deep red, indicating severe congestion across the city centre. Unlike the previous cases, even many peripheral roads experience high TTR values, demonstrating the impact of the heavy traffic demand on the entire network.

The mix case shown in Figure 5.9(b) demonstrates an improvement in traffic efficiency compared to the HDV case. Several peripheral roads and major arterial routes change from deep red to orange and light pink, indicating a reduction in congestion with the introduction of 50% CAVs. However, compared to Cases 1 and 2, the improvement is less pronounced, and the overall reduction in TTR values is more limited. A large portion of the city centre remains heavily congested, reflecting the challenging traffic conditions of this scenario.

In the CAV case shown in Figure 5.9(c), traffic conditions improve. Several peripheral roads and some arterial routes exhibit lighter colors, indicating lower TTR values and improved traffic flow. However, compared to the CAV case in Cases 1 and 2, the improvement is noticeably smaller. Much of the central city area continues to display red and orange colors, indicating persistent congestion even under full CAV penetration.

Overall, Case 3 shows that increasing the penetration of CAVs improves traffic efficiency throughout the Dublin city centre network. However, the magnitude of the improvement is smaller than that observed in Cases 1 and 2. Under highly congested traffic conditions, the benefits of CAVs are reduced because high traffic demand, dense intersections, and the complex urban road network continue to create significant congestion across the city centre.

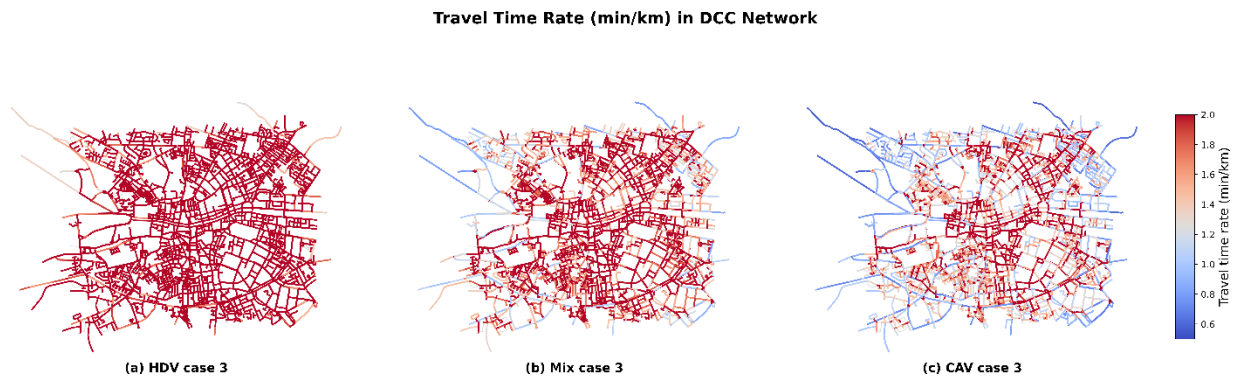


Figure 5.9: Spatial distribution of Travel Time Rate (min/km) in the DCC network under Case 3:
(a) HDV case 3, (b) mix case 3, and (c) CAV case

Note: For a clearer view of the graphs discussed in Section 5.2, please refer to the following link: https://github.com/mdaslamcee/graph_plot_data

5.2.3 Results Analysis

The results obtained from both the Grid 4×4 and DCC network scenarios show a consistent trend of improving traffic efficiency with increasing CAV penetration. Across all cases, the

HDV case exhibited the highest Travel Time Rate (TTR) values and the most severe congestion, while the mix case showed noticeable improvements in traffic flow. The CAV case consistently achieved the lowest TTR values and the highest level of traffic efficiency. Overall, the findings indicate that CAVs can significantly improve traffic efficiency and reduce congestion in both synthetic and real-world urban networks. However, the magnitude of the improvement depends on traffic demand, network geometry, and the spatial distribution of traffic flow. While full CAV penetration provides the greatest benefits, it does not completely eliminate congestion in locations affected by high demand concentrations and complex intersection layouts.

5.3 Traffic Safety

In this section, the impact of CAVs on traffic safety is evaluated using the total number of conflicts in terms of Surrogate Safety Measures (SSMs) based on Post-Encroachment Time (PET). The results are presented and analyzed for both the Grid 4x4 and DCC network scenarios under three traffic conditions: HDV condition (100% HDVs), mix condition (50% HDVs and 50% CAVs), and CAV condition (100% CAVs). For each scenario, the effects of different vehicle compositions on traffic safety are examined. Subsequently, a comprehensive comparative analysis is conducted to assess the overall influence of increasing CAV penetration levels on network-wide traffic safety.

5.3.1 Grid 4x4 Scenario

5.3.1.1 Case 1 (Morning Peak-Hour Traffic)

This case represents morning peak-hour traffic and is generated by combining morning traffic demand with random traffic demand.

In Case 1, the HDV case shown in Figure 5.10(a) recorded a total of 38 conflicts distributed across multiple intersections within the network, indicating a relatively high level of traffic conflict under conventional driving conditions.

The mix case shown in Figure 5.10(b) reduced the total number of conflicts to 23, representing a 39.5% reduction compared to the HDV case. Although conflicts were still

observed at several intersections, their overall frequency was noticeably lower, indicating an improvement in traffic safety with the introduction of 50% CAVs.

In the CAV case shown in Figure 5.10(c), the total number of conflicts was further reduced to 10, corresponding to a 73.7% reduction compared to the HDV case. In addition, conflict locations were limited to only a small number of intersections, demonstrating a substantial improvement in network safety under full CAV penetration.

Overall, Case 1 shows that increasing the penetration of CAVs significantly improves traffic safety by reducing both the total number of conflicts and the number of intersections experiencing conflicts. The progressive reduction in conflicts from the HDV case to the mix case and finally to the CAV case highlights the positive impact of CAVs on network-wide safety performance.

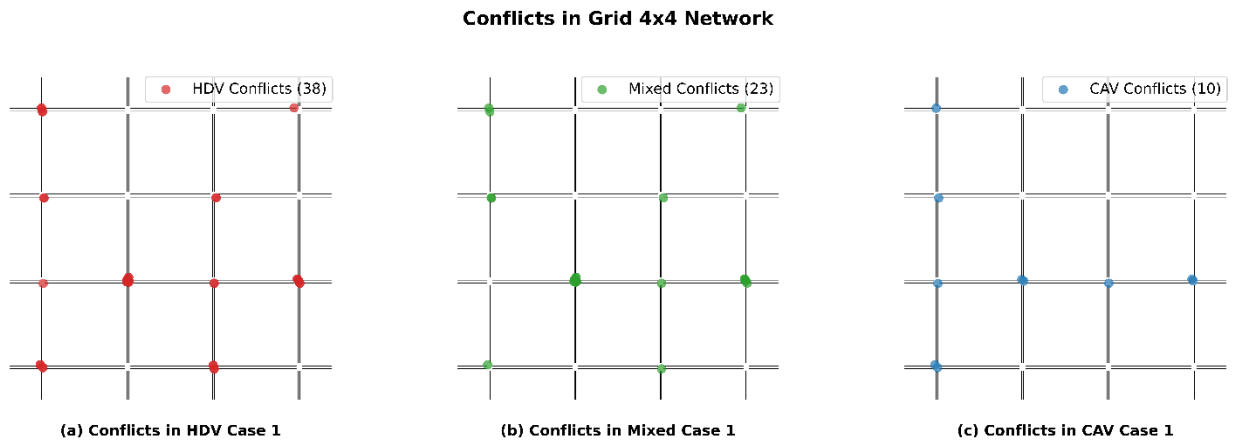


Figure 5.10: Spatial distribution of the total number of conflicts at intersections in the Grid 4x4 network under Case 1: (a) HDV Case 1, (b) Mix Case 1, and (c) CAV Case 1

5.3.1.2 Case 2 (Morning Peak-Hour Traffic with Road Blockages)

This case represents morning rush-hour traffic and is generated by combining morning traffic demand with random traffic demand, along with the addition of two blocked roads.

In Case 2, the HDV case shown in Figure 5.11(a) recorded a total of 16 conflicts distributed across several intersections within the network. Compared to Case 1, the total number of conflicts is lower, with most conflicts occurring at scattered intersections.

The mix case shown in Figure 5.11(b) reduced the total number of conflicts to 10, representing a 37.5% reduction compared to the HDV case. The number of intersections experiencing conflicts also decreased in the network. This indicates an improvement in traffic safety with the introduction of 50% CAVs.

In the CAV case shown in Figure 5.11(c), a total of 11 conflicts were recorded. Although this represents a 31.3% reduction compared to the HDV case, the conflict count is slightly higher than that observed in the mix case. Unlike the trend observed in Case 1, the transition from the mix case to the CAV case does not result in an additional reduction in conflicts. This suggests that the specific traffic demand pattern and road blockages in Case 2 may create interaction patterns at certain intersections that continue to generate conflicts even under full CAV penetration.

Overall, Case 2 shows that increasing CAV penetration generally improves traffic safety compared to the HDV case. However, the slight increase in conflicts from the mix case to the CAV case indicates that higher CAV penetration does not always guarantee further safety improvements under every traffic condition and network configuration.

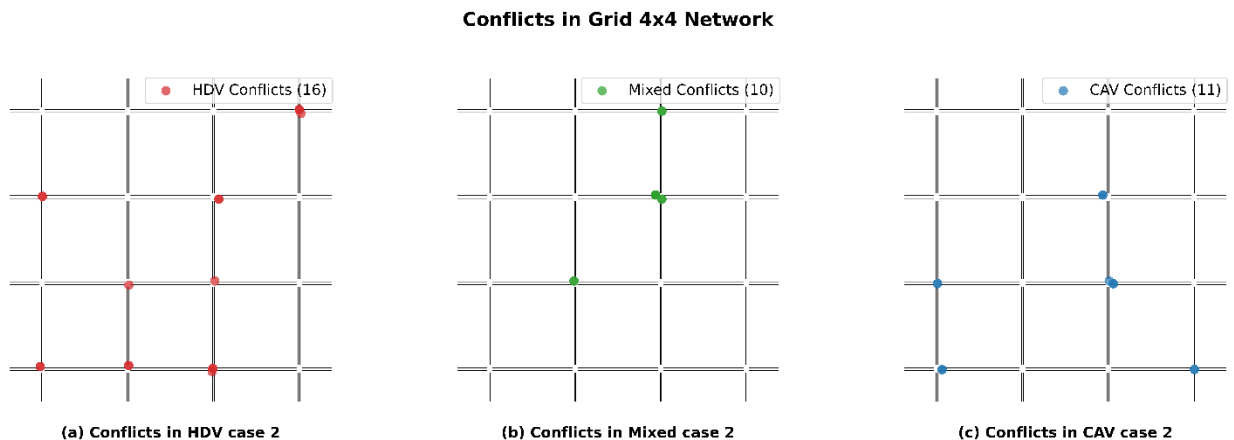


Figure 5.11: Spatial distribution of the total number of conflicts at intersections in the Grid 4x4 network under Case 2: (a) HDV Case 2, (b) Mix Case 2, and (c) CAV Case 2

5.3.1.3 Case 3 (Evening Peak-Hour Traffic)

This case represents evening peak-hour traffic and is generated by combining evening traffic demand with random traffic demand.

In Case 3, the HDV case shown in Figure 5.12(a) recorded a total of 27 conflicts across the network, indicating a relatively high level of traffic conflict under conventional driving conditions.

The mix case shown in Figure 5.12(b) reduced the total number of conflicts to 18, representing a 33.3% reduction compared to the HDV case. Although conflicts were still observed at several intersections, the overall number of conflicts decreased, demonstrating an improvement in traffic safety with the introduction of 50% CAVs.

In the CAV case shown in Figure 5.12(c), the total number of conflicts was further reduced to 9. This corresponds to a 66.7% reduction compared to the HDV case and a 50% reduction compared to the mix case. The lower number of conflicts indicates a substantial improvement in safety under full CAV penetration.

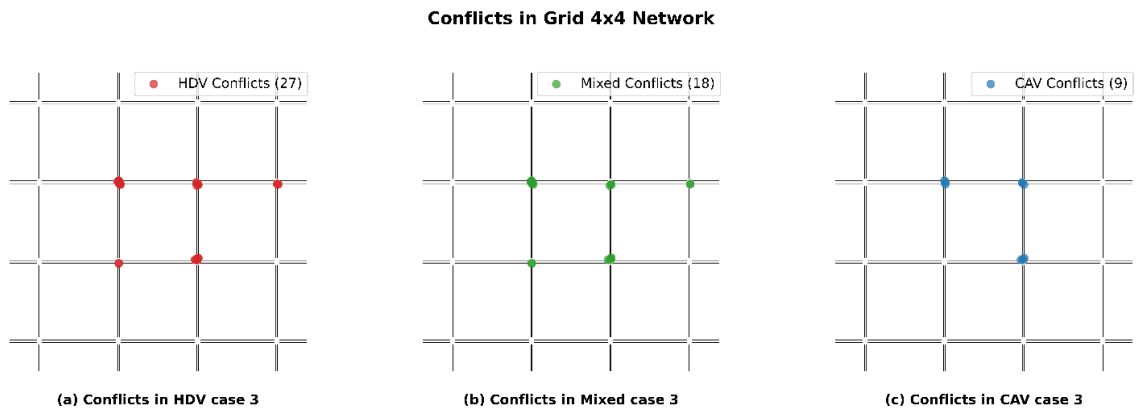


Figure 5.12: Spatial distribution of the total number of conflicts at intersections in the Grid 4x4 network under Case 3: (a) HDV Case 3, (b) Mix Case 3, and (c) CAV Case 3

Overall, Case 3 shows that increasing the penetration of CAVs significantly improves traffic safety by reducing the total number of conflicts throughout the network. The consistent reduction in conflicts from the HDV case to the mix case and finally to the CAV case demonstrates the positive impact of CAVs on network-wide safety performance.

5.3.1.4 Case 4 (Evening Peak-Hour Traffic with Special Event Traffic)

This case represents evening rush-hour traffic with special event traffic and is generated by combining evening traffic demand, random traffic demand, and special event traffic demand.

In Case 4, the HDV case shown in Figure 5.13(a) recorded a total of 37 conflicts across the network. This is the highest number of conflicts observed among Cases 1–5, indicating that the combination of evening traffic and special event traffic creates a highly challenging traffic environment.

The mix case shown in Figure 5.13(b) reduced the total number of conflicts to 17, representing a 54.1% reduction compared to the HDV case. The substantial decrease in conflicts demonstrates a significant improvement in traffic safety with the introduction of 50% CAVs.

In the CAV case shown in Figure 5.13(c), the total number of conflicts was further reduced to 5. This corresponds to an 86.5% reduction compared to the HDV case and represents the lowest conflict count recorded among all cases. The sharp decline in conflicts indicates that full CAV penetration is highly effective in reducing conflicting vehicle interactions under this traffic demand condition.

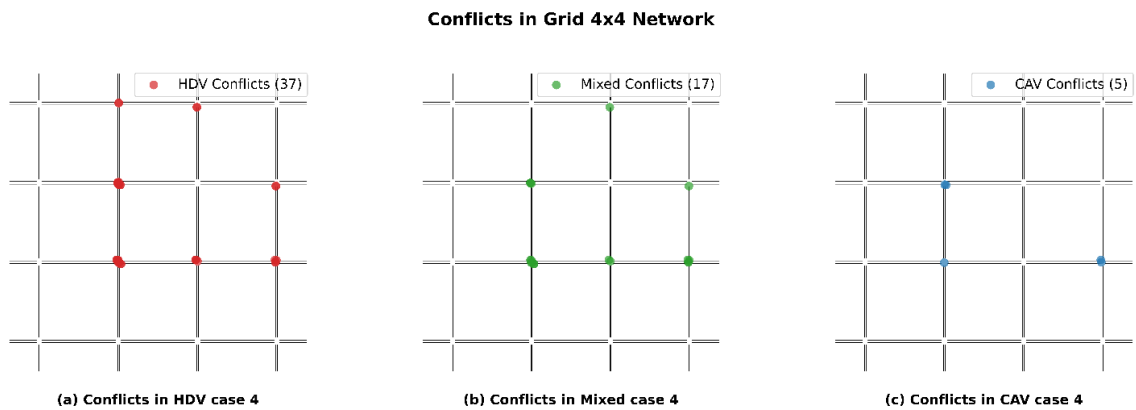


Figure 5.13: Spatial distribution of the total number of conflicts at intersections in the Grid 4x4 network under Case 4: (a) HDV Case 4, (b) Mix Case 4, and (c) CAV Case 4

Overall, Case 4 shows that increasing the penetration of CAVs significantly improves traffic safety throughout the network. The large reduction in conflicts from the HDV case to the CAV case indicates that this traffic demand pattern benefits greatly from CAV deployment, resulting in the highest relative safety improvement among all evaluated cases.

5.3.1.5 Case 5 (Evening Peak-Hour Traffic under Bad Weather Conditions)

This case represents evening rush-hour traffic under bad weather conditions and is generated by combining evening traffic demand with random traffic demand.

In Case 5, the HDV case shown in Figure 5.14(a) recorded a total of 24 conflicts across the network, indicating a relatively high level of traffic conflict under adverse weather conditions.

The mix case shown in Figure 5.14(b) reduced the total number of conflicts to 13, representing a 45.8% reduction compared to the HDV case. The decrease in conflicts demonstrates an improvement in traffic safety with the introduction of 50% CAVs, despite the challenging weather conditions.

In the CAV case shown in Figure 5.14(c), the total number of conflicts was further reduced to 6. This corresponds to a 75.0% reduction compared to the HDV case, indicating a substantial improvement in network safety under full CAV penetration. The lower conflict count suggests that CAVs are effective in reducing conflicting vehicle interactions even in adverse weather conditions.

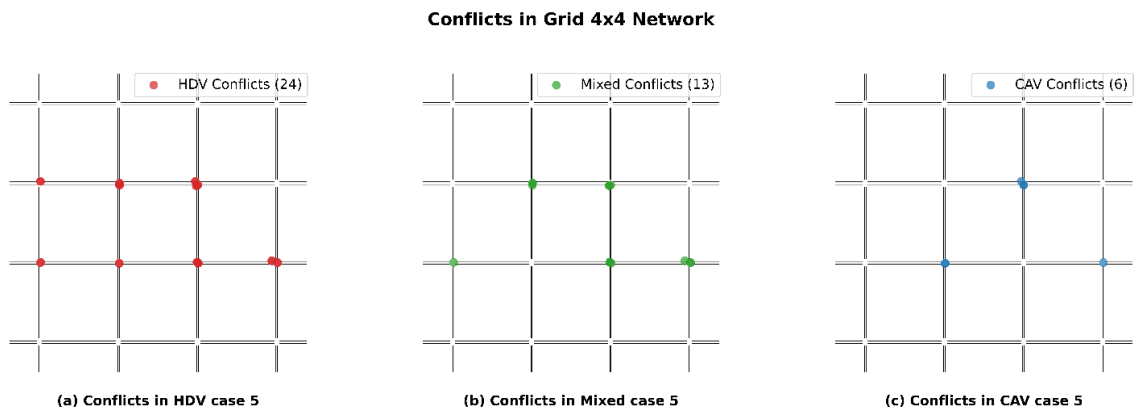


Figure 5.14: Spatial distribution of the total number of conflicts at intersections in the Grid 4x4 network under Case 5: (a) HDV Case 5, (b) Mix Case 5, and (c) CAV Case 5

Overall, Case 5 shows a strong and consistent improvement in traffic safety as CAV penetration increases. The progressive reduction in conflicts from the HDV case to the mix case and finally to the CAV case demonstrates that CAV deployment can significantly enhance safety performance under challenging traffic and weather conditions.

5.3.1.6 Case 6 (Combined Traffic Scenario)

This case combines all four traffic types: morning peak-hour traffic, evening peak-hour traffic, special event traffic, and random traffic. It is also categorized as a long-duration case, with the simulation running for 14 hours. Due to the combination of multiple traffic demand patterns over an extended period, Case 6 represents the most demanding traffic scenario in the Grid 4×4 network.

In Case 6, the HDV case shown in Figure 5.15(a) recorded a total of 260 conflicts across the network. This is by far the highest conflict count observed among all six cases, indicating a substantial deterioration in traffic safety under extremely high traffic demand.

The mix case shown in Figure 5.15(b) reduced the total number of conflicts to 166, representing a 36.2% reduction compared to the HDV case. Although the introduction of 50% CAVs improves safety performance, the number of conflicts remains considerably higher than in the previous cases, reflecting the challenging nature of this traffic scenario.

In the CAV case shown in Figure 5.15(c), the total number of conflicts was further reduced to 80. This corresponds to a 69.2% reduction compared to the HDV case and demonstrates the safety benefits of full CAV penetration. However, despite the significant reduction, the conflict count remains much higher than those observed in Cases 1–5, indicating that high traffic demand continues to generate a large number of potential conflict situations.

Overall, Case 6 shows that increasing the penetration of CAVs significantly improves traffic safety by reducing the total number of conflicts throughout the network. However, compared to all other cases, the absolute conflict count remains high even in the CAV case,

highlighting the strong influence of traffic demand on safety performance. These results suggest that while CAV deployment can substantially improve safety, additional traffic demand management strategies may be required to achieve safer operating conditions under extreme traffic loads.

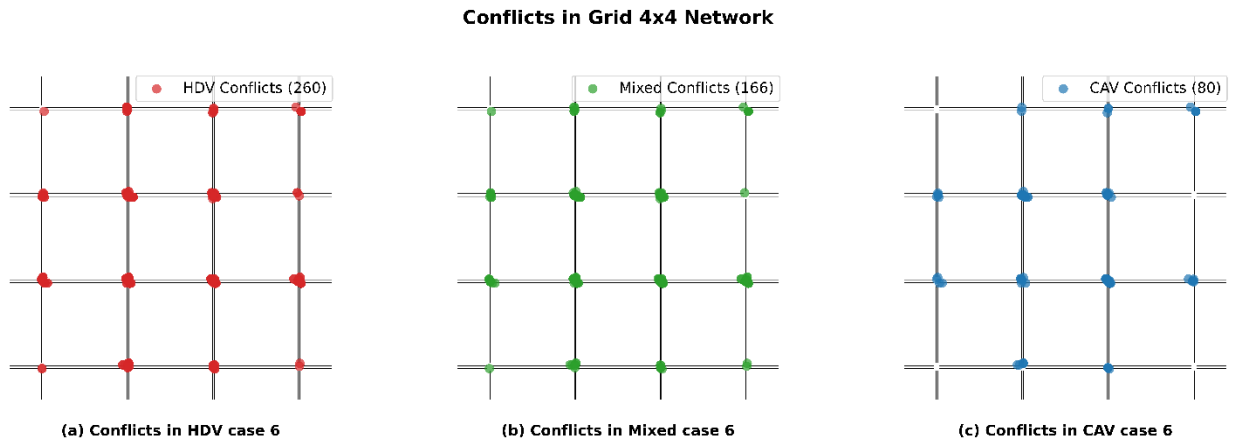


Figure 5.15: Spatial distribution of the total number of conflicts at intersections in the Grid 4x4 network under Case 6: (a) HDV Case 6, (b) Mix Case 6, and (c) CAV Case 6

Table 5.1 summarizes the traffic safety analysis for the six cases of the Grid 4x4 network scenario under three traffic conditions: the HDV condition (100% HDVs), mix condition (50% HDVs and 50% CAVs), and CAV condition (100% CAVs), as discussed previously.

Table 5.1: Traffic safety analysis summary for the six cases of the Grid 4x4 scenarios

Case	HDV Case	Mix Case	CAV Case	HDV to CAV reduction
1	38	23	10	73.7%
2	16	10	11	31.3%
3	27	18	9	66.7%
4	37	17	5	86.5%
5	24	13	6	75.0%
6	260	166	80	69.2%

5.3.2 DCC Scenario

5.3.2.1 Case 1 (Free-Flow Traffic Conditions)

This case represents the free-flow condition of a real-world urban traffic network in Dublin city centre.

In Case 1, the HDV case shown in Figure 5.16(a) recorded a total of 20 conflicts across the network. Under free-flow traffic conditions, the number of conflicts is relatively low, indicating limited vehicle interactions and a lower likelihood of conflicts at intersections.

The mix case shown in Figure 5.16(b) reduced the total number of conflicts to 12, representing a 40.0% reduction compared to the HDV case. The decrease in conflicts demonstrates an improvement in traffic safety with the introduction of 50% CAVs.

In the CAV case shown in Figure 5.16(c), the total number of conflicts was further reduced to 5. This corresponds to a 75.0% reduction compared to the HDV case and indicates a substantial improvement in network safety under full CAV penetration. The low conflict count suggests that CAVs are highly effective in reducing conflicting vehicle interactions under free-flow traffic conditions.

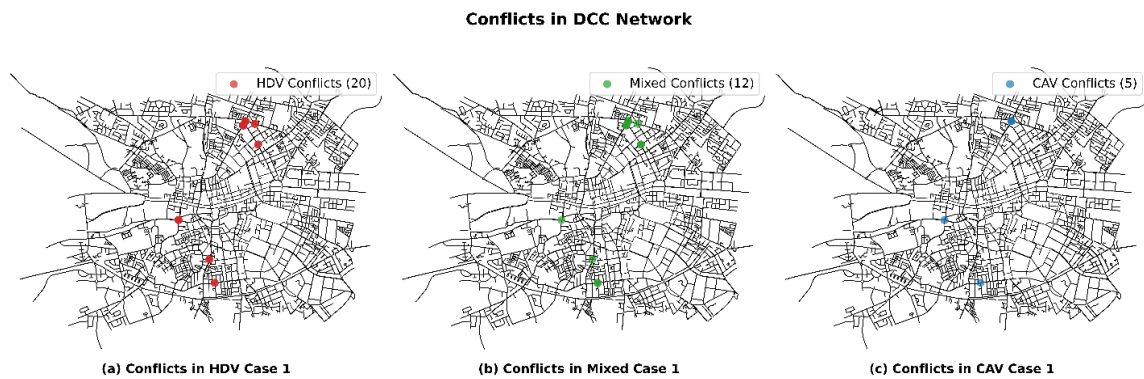


Figure 5.16: Spatial distribution of the total number of conflicts at intersections in the DCC network under Case 1: (a) HDV Case 1, (b) Mix Case 1, and (c) CAV Case 1

Overall, Case 1 shows that increasing the penetration of CAVs significantly improves traffic safety by reducing the total number of conflicts throughout the network. The progressive reduction in conflicts from the HDV case to the mix case and finally to the

CAV case demonstrates the positive impact of CAVs on safety performance, even under relatively low traffic demand conditions.

5.3.2.2 Case 2 (Saturated Traffic Conditions)

This case represents the saturated condition of a real-world urban traffic network in Dublin city centre, where traffic volumes are high and congestion begins to form.

In Case 2, the HDV case shown in Figure 5.17(a) recorded a total of 538 conflicts across the network. Compared to Case 1, the number of conflicts increased dramatically, indicating that higher traffic demand significantly increases the frequency of vehicle interactions and potential conflicts throughout the network.

The mix case shown in Figure 5.17(b) reduced the total number of conflicts to 285, representing a 47.0% reduction compared to the HDV case. Although the conflict count remains high, the substantial decrease demonstrates that the introduction of 50% CAVs can significantly improve traffic safety under saturated traffic conditions.

In the CAV case shown in Figure 5.17(c), the total number of conflicts was further reduced to 112. This corresponds to a 79.2% reduction compared to the HDV case and represents the largest relative safety improvement among the three DCC cases. The considerable reduction in conflicts indicates that full CAV penetration is highly effective in improving safety under saturated traffic conditions.

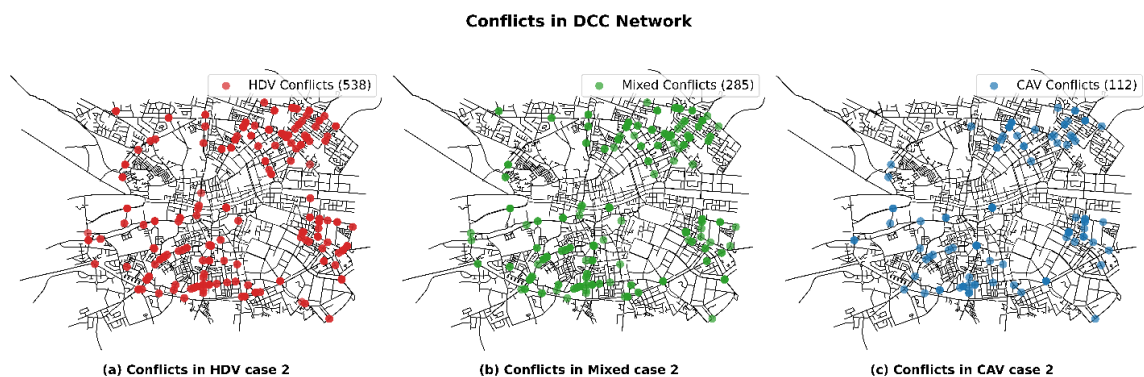


Figure 5.17: Spatial distribution of the total number of conflicts at intersections in the DCC network under Case 2: (a) HDV Case 2, (b) Mix Case 2, and (c) CAV Case 2

Overall, Case 2 shows that increasing the penetration of CAVs significantly improves traffic safety by reducing the total number of conflicts throughout the network. Although the absolute conflict count remains relatively high due to the increased traffic demand, the large reduction achieved in the CAV case demonstrates the strong potential of CAVs to enhance safety in congested urban traffic environments.

5.3.2.2 Case 3 (Congested Traffic Conditions)

This case represents the congested condition of a real-world urban traffic network in Dublin city centre.

In Case 3, the HDV case shown in Figure 5.18(a) recorded a total of 488 conflicts across the network. This represents a very high level of traffic conflict, reflecting the challenging traffic conditions associated with severe congestion. Interestingly, the conflict count is slightly lower than that observed in Case 2, which recorded 538 conflicts under saturated conditions. A possible explanation for this observation is the speed-suppression effect. Under severe congestion, vehicle speeds become extremely low, reducing the likelihood of certain rear-end and crossing conflicts despite the high density of vehicle interactions throughout the network.

The mix case shown in Figure 5.18(b) reduced the total number of conflicts to 322, representing a 34.0% reduction compared to the HDV case. This is the smallest proportional reduction among the three DCC cases, suggesting that the safety benefits of 50% CAV penetration are more limited under highly congested traffic conditions. Despite the reduction, the overall conflict count remains high.

In the CAV case shown in Figure 5.18(c), the total number of conflicts was further reduced to 161. This corresponds to a 67.0% reduction compared to the HDV case and demonstrates a significant improvement in traffic safety under full CAV penetration. However, compared to Cases 1 and 2, the absolute number of conflicts remains considerably higher, indicating that severe congestion continues to generate a large number of potential conflict situations even when all vehicles are CAVs.

Overall, Case 3 shows that increasing the penetration of CAVs improves traffic safety by reducing the total number of conflicts throughout the network. However, the magnitude of the improvement is lower than that observed under saturated traffic conditions. These results suggest that while CAVs can significantly enhance safety, the benefits become more limited under severe congestion, where high traffic demand and complex urban traffic interactions continue to influence conflict occurrence.

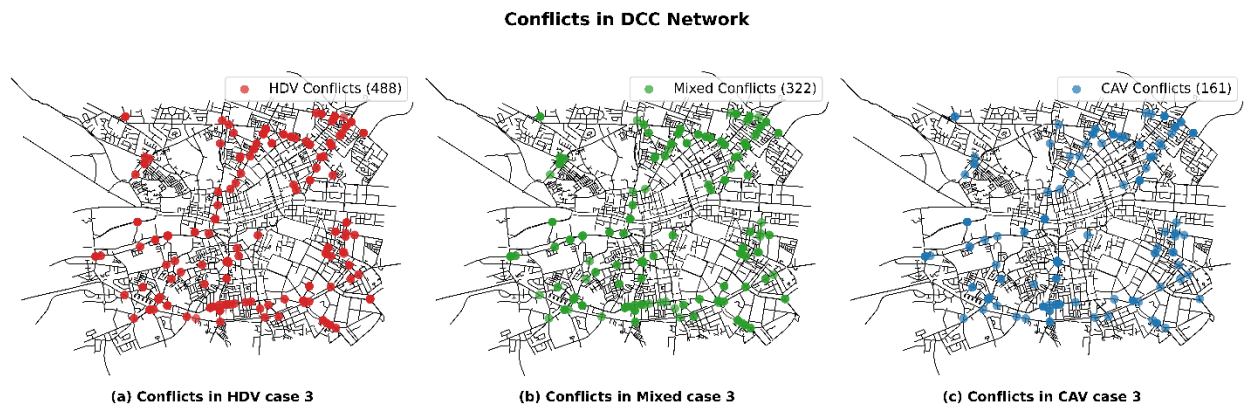


Figure 5.18: Spatial distribution of the total number of conflicts at intersections in the DCC network under Case 3: (a) HDV Case 3, (b) Mix Case 3, and (c) CAV Case 3

Table 5.2 summarizes the traffic safety analysis for the three cases of the DCC network scenario under three traffic conditions: the HDV condition (100% HDVs), mix condition (50% HDVs and 50% CAVs), and CAV condition (100% CAVs), as discussed previously.

Table 5.2: Traffic safety analysis summary for the three cases of the DCC scenarios

Case	HDV Case	Mix Case	CAV Case	HDV to CAV reduction
1(free flow)	20	12	5	75.0%
2(saturated)	538	285	112	79.2%
3(congested)	488	322	161	67.0%

Note: For a clearer view of the graphs discussed in Section 5.3, please refer to the following link: https://github.com/mdaslamcee/graph_plot_data

5.3.3 Results Analysis

The results obtained from both the Grid 4×4 and DCC network scenarios show a consistent trend of improving traffic safety with increasing CAV penetration. Across almost all cases, the HDV case recorded the highest number of conflicts, indicating the lowest level of safety, while the mix case showed a noticeable reduction in conflicts. The CAV case consistently achieved the lowest conflict counts and the highest level of traffic safety.

For the Grid 4×4 network, increasing CAV penetration resulted in significant reductions in the total number of conflicts across all six traffic demand scenarios. The largest safety improvements were observed in Cases 4 and 5, where the CAV case reduced conflicts by more than 75% compared to the HDV case. Although Case 2 showed a slight increase in conflicts from the mix case to the CAV case, the overall conflict count in the CAV case remained lower than that of the HDV case, indicating an overall safety benefit from CAV deployment. Case 6 recorded the highest conflict counts due to the combination of multiple traffic demand patterns and the long simulation duration, highlighting the strong influence of traffic demand on safety performance.

Similarly, the DCC network demonstrated substantial safety improvements as CAV penetration increased. The HDV case consistently produced the highest conflict counts, while the CAV case achieved the greatest reductions. The most significant improvement was observed under saturated traffic conditions, where the CAV case reduced conflicts by approximately 79% compared to the HDV case. Under congested conditions, the safety benefits of CAVs remained substantial but were comparatively lower, indicating that severe congestion can limit the extent of conflict reduction. A notable observation was that the HDV case in the congested condition recorded slightly fewer conflicts than in the saturated condition, which may be explained by speed-suppression effect where extremely low vehicle speeds reduce the occurrence of certain conflict types despite high traffic density.

Overall, the findings indicate that CAVs can significantly improve traffic safety and reduce conflict occurrences in both synthetic and real-world urban networks. However, the magnitude of the improvement depends on traffic demand, network characteristics, and traffic conditions. While full CAV penetration provides the greatest safety benefits, it does not completely eliminate conflicts under high-demand conditions, suggesting that traffic demand management and network-level improvements may be required to achieve further safety enhancements.

5.4 Performance Evaluation of RL-Based ATSC

To analyze the performance of the RL-based ATSC under varying traffic conditions and patterns at different penetration levels of CAVs, only the Grid 4x4 network scenario is considered in this study. The results are presented and analyzed under three traffic conditions: HDV condition (100% HDVs), the mix condition (50% HDVs and 50% CAVs), and the CAV condition (100% CAVs). Each traffic condition is evaluated using six different traffic demand cases to assess the performance of the RL-based ATSC.

5.4.1 Initial Training

For the initial training process, the RL-based traffic control model IDQN was trained using Case 1 of the HDV condition in the Grid 4×4 network for 200 episodes. After training, the model weights were saved and used to evaluate the performance of the model under all six cases of each traffic condition, including the HDV condition, mix condition, and CAV condition.

The initial training process required approximately 23 hours to reach stabilization, which is considered the recovery time of the model. At convergence, the model achieved an average travel time of 135.6 s. Figure 5.19 presents the training results of the IDQN model. In contrast, the non-RL-based FixedTime model does not require any training process and was directly evaluated under the same traffic conditions.

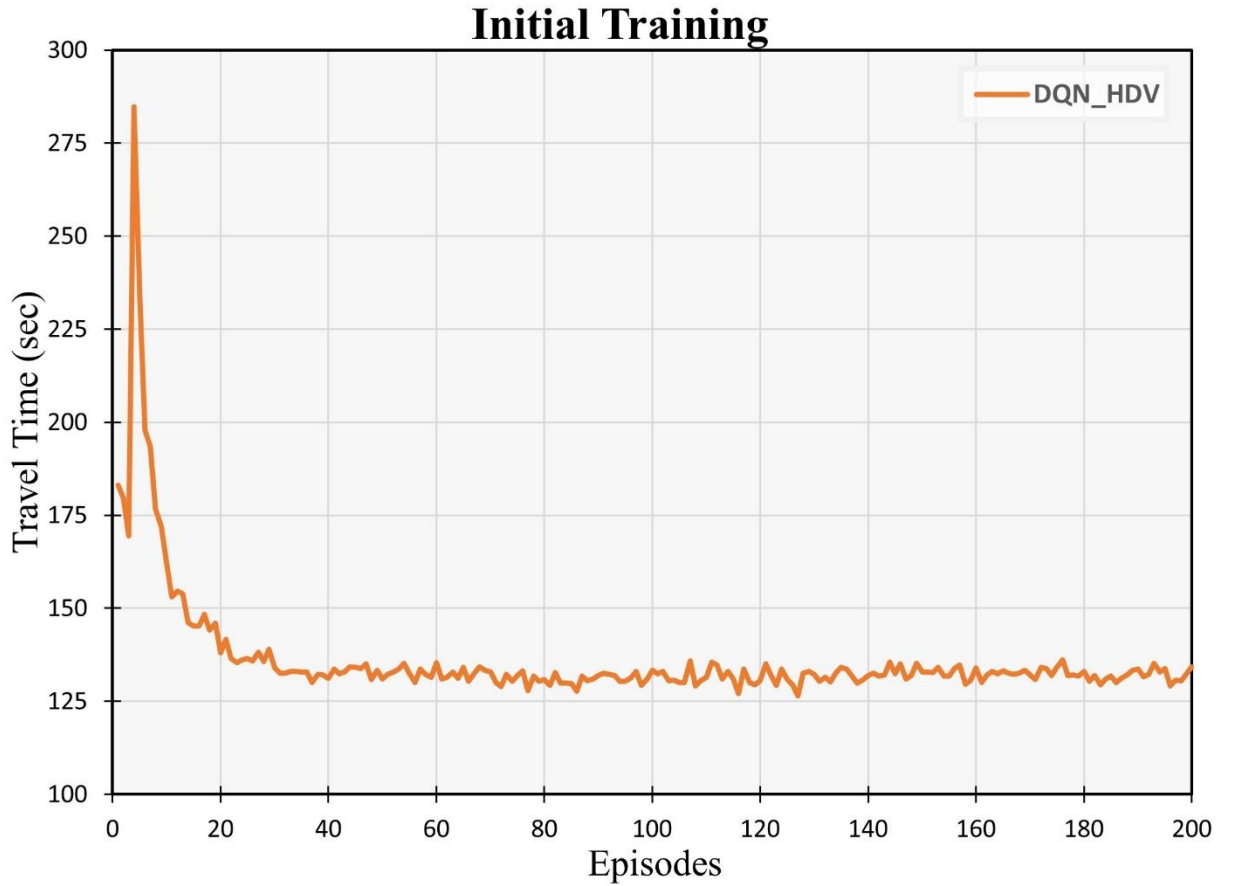


Figure 5.19: Initial training results of the IDQN model for HDV case 1, showing Average Travel Time (sec) versus Episodes. Each episode represents one hour of simulation

5.4.2 Recovery Time

This section presents the recovery time analysis of the IDQN model in the Grid 4x4 network scenario. In addition to evaluating the recovery time of the model, the training curves for all six cases under the HDV condition, mix condition, and CAV condition are also presented and analyzed. A case-by-case analysis is provided to better understand the model's learning performance and adaptation under different traffic conditions and demand patterns.

The Grid 4×4 network consists of 16 signalized intersections arranged in a 4x4 grid, where traffic flow is synthetically generated. In this scenario, the RL-based traffic control model

IDQN was evaluated and compared with the non-RL-based FixedTime model. To assess the recovery time performance of the IDQN model under varying traffic conditions and demand patterns, Cases 1 to 6 were evaluated for each traffic condition, namely the HDV condition, mix condition, and CAV condition. The recovery time results are summarized in Table 5.3.

Table 5.3: Recovery Time (Hours) of IDQN RL Models Across Different Cases for the Grid 4x4 Scenario

Cases	HDV Condition	Mix Condition	CAV Condition
case1	5	6	6
case2	7	8	8
case3	5	6	5
case4	8	6	6
case5	5	6	6
case6	1	1	1

For better visualization and comparison of the recovery time results presented in Table 5.3, the data are illustrated in Figure 5.20.

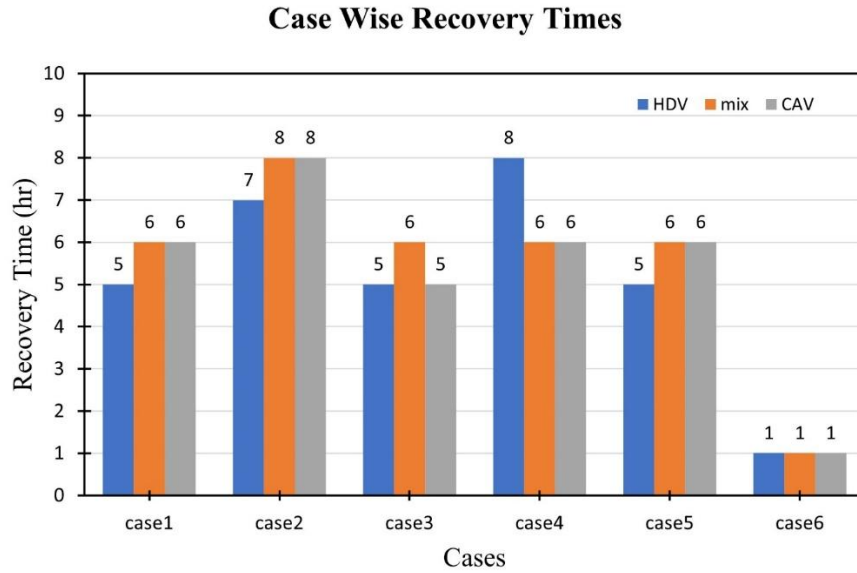


Figure 5.20: Case-wise recovery times of the IDQN model under different traffic conditions in the Grid 4×4 network scenario

To better understand the results in Table 5.3, it is useful to analyze the training curves for individual cases to explain the recovery time of the models. Therefore, the training curves for each case are presented in Figure 5.21, Figure 5.22, Figure 5.23, Figure 5.24, Figure 5.25, Figure 5.26. The results for each case are discussed in the following subsections.

5.4.2.1 Case 1 (Morning Peak-Hour Traffic)

This case represents morning peak-hour traffic and is created by combining morning traffic demand with random traffic. Figure 5.21 presents the training curves for Case 1 under the HDV condition, mix condition, and CAV condition.

In the HDV condition, the IDQN model begins with an initial average travel time of 180.1 s and gradually converges to an optimum travel time of 130.8 s. This represents a reduction of 27.4% from the initial value. The converged travel time is substantially lower than the FixedTime model travel time of 208.25 s, indicating that the IDQN model effectively adapts to the HDV traffic condition and achieves better traffic performance. The recovery time for the HDV condition is 5 hours.

In the mix condition, the IDQN model starts with an initial average travel time of 147.4 s and converges to an optimum travel time of 99.5 s, corresponding to a reduction of 32.5%. The final travel time is considerably lower than the FixedTime model travel time of 146.53 s, demonstrating the effectiveness of the IDQN model under mixed traffic conditions. The recovery time for the mix condition is 6 hours.

In the CAV condition, the IDQN model begins with an initial average travel time of 96.9 s and converges to an optimum travel time of 68.3 s. This represents a reduction of 29.5% from the initial value. The converged travel time is significantly lower than the FixedTime model travel time of 101.24 s, highlighting the ability of the IDQN model to take advantage of the improved operational characteristics of CAVs. The recovery time for the CAV condition is 6 hours.

Overall, Case 1 shows that the IDQN model successfully adapts to all three traffic conditions and consistently outperforms the FixedTime model. Among the three

conditions, the CAV condition achieves the lowest optimum travel time, while the mix condition records the largest percentage reduction in travel time during the learning process.

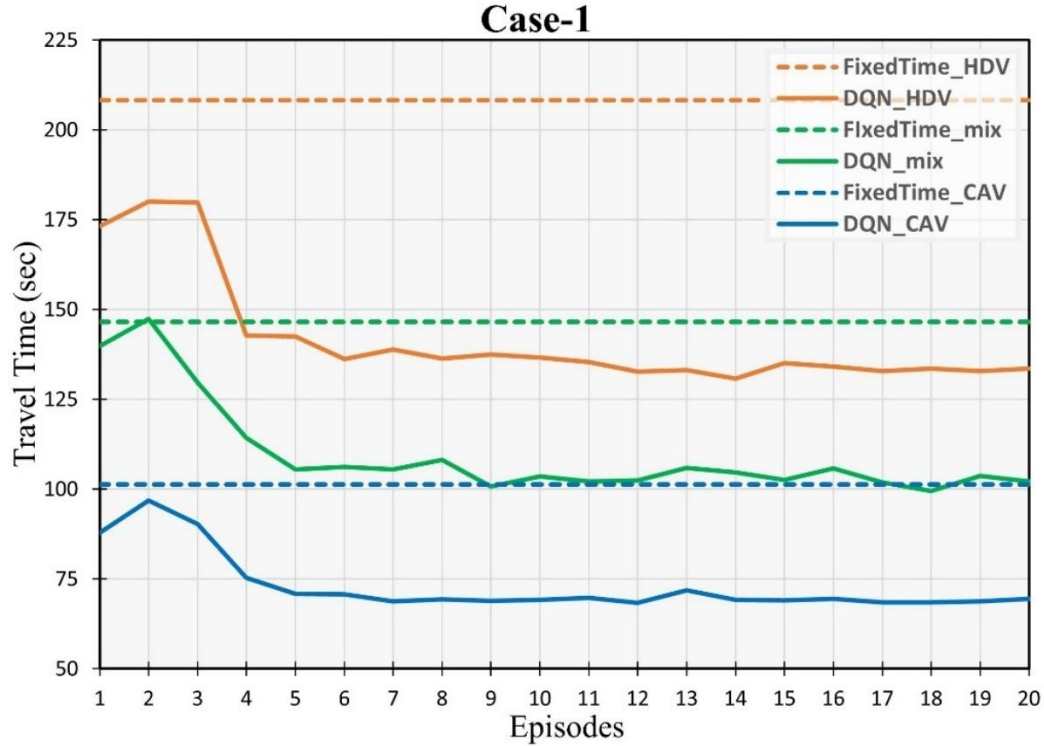


Figure 5.21: Average travel time versus episodes for the IDQN model and FixedTime model under the HDV condition, mix condition, and CAV condition in Case 1 of the Grid 4x4 network

5.4.2.2 Case 2 (Morning Peak-Hour Traffic with Road Blockages)

This case also represents morning rush-hour traffic and is created by combining morning traffic demand with random traffic demand, along with the addition of two blocked roads. Figure 5.22 presents the training curves for Case 2 under the HDV condition, mix condition, and CAV condition.

In the HDV condition, the IDQN model starts with an average travel time of 209.4 s and gradually converges to 175.5 s. This corresponds to a reduction of 16.2% from the initial value. Although the improvement is smaller than that observed in Case 1, the converged

travel time remains considerably lower than the FixedTime model travel time of 239.81 s, demonstrating the ability of the IDQN model to adapt to this more challenging traffic condition. The recovery time for the HDV condition is 7 hours.

In the mix condition, the IDQN model begins with an average travel time of 172.5 s and reaches an optimum travel time of 139.4 s, representing a reduction of 19.2%. The final travel time is noticeably lower than the FixedTime model travel time of 183.31 s, indicating that the RL-based controller is able to improve traffic performance under mixed traffic operations. The recovery time for the mix condition is 8 hours.

In the CAV condition, the IDQN model initially records an average travel time of 141.6 s and converges to 110.4 s. This represents a reduction of 22.0% during the learning process. The optimum travel time is substantially lower than the FixedTime model travel time of 140.42 s, showing that the IDQN model can effectively utilize the operational advantages offered by CAVs. The recovery time for the CAV condition is 8 hours.

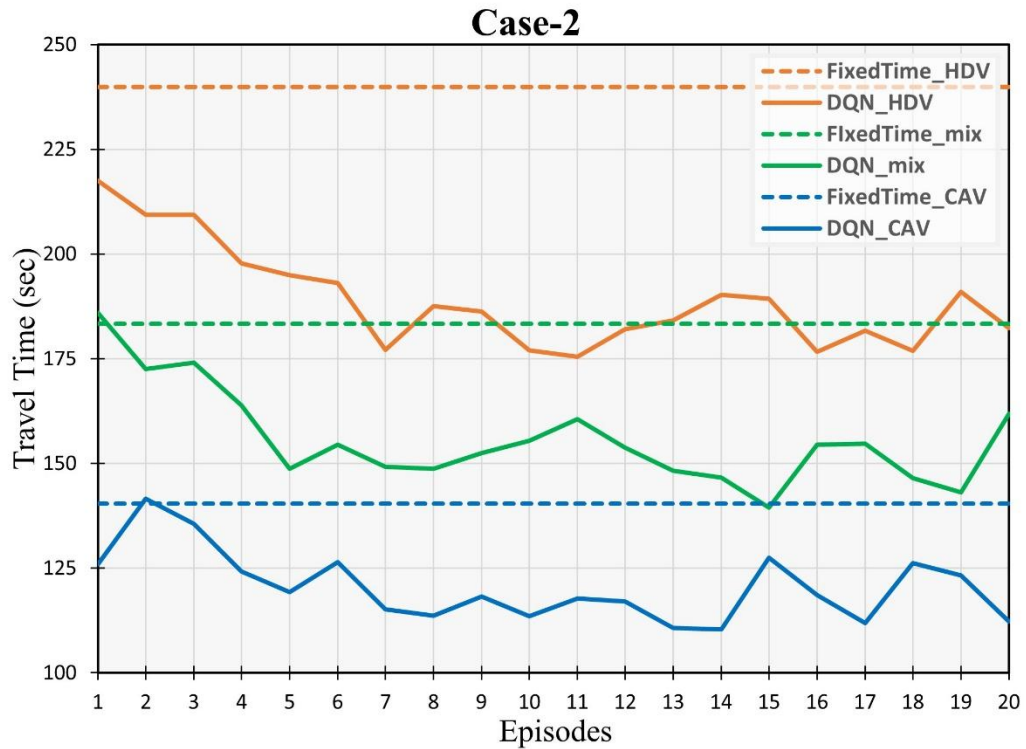


Figure 5.22: Average travel time versus episodes for the IDQN model and FixedTime model under the HDV condition, mix condition, and CAV condition in Case 2 of the Grid 4x4 network

Overall, Case 2 demonstrates that the IDQN model remains effective under increased traffic complexity and consistently outperforms the FixedTime model across all traffic conditions. However, compared to Case 1, the longer recovery times and smaller travel time reductions suggest that the presence of blocked roads and higher traffic demand increases the difficulty of the learning task.

5.4.2.3 Case 3 (Evening Peak-Hour Traffic)

This case represents evening peak-hour traffic and is created by combining evening traffic demand with random traffic demand. Figure 5.23 presents the training curves for Case 3 under the HDV condition, mix condition, and CAV condition.

In the HDV condition, the IDQN model begins with an initial average travel time of 183.5 s and gradually converges to an optimum travel time of 130.1 s. This represents a reduction of 29.1% from the initial value and is the largest absolute reduction achieved among all six cases under the HDV condition. The converged travel time is substantially lower than the FixedTime model travel time of 194.62 s, indicating that the IDQN model effectively adapts to the traffic demand pattern and achieves better traffic performance. The recovery time for the HDV condition is 5 hours.

In the mix condition, the IDQN model starts with an initial average travel time of 134.2 s and converges to an optimum travel time of 98.4 s, corresponding to a reduction of 26.7%. The final travel time is considerably lower than the FixedTime model travel time of 137.78 s, demonstrating the effectiveness of the IDQN model under mixed traffic conditions. The recovery time for the mix condition is 6 hours.

In the CAV condition, the IDQN model begins with an initial average travel time of 89.9 s and converges to an optimum travel time of 68.6 s. This represents a reduction of 23.7% from the initial value. The converged travel time is significantly lower than the FixedTime model travel time of 97.66 s, highlighting the ability of the IDQN model to take advantage of the improved operational characteristics of CAVs. The recovery time for the CAV condition is 5 hours, which is one hour shorter than that observed in Case 1, indicating a faster adaptation to the traffic demand pattern.

Overall, Case 3 shows smooth and consistent convergence behavior across all three traffic conditions, with the IDQN model achieving reliable reductions in average travel time and consistently outperforming the FixedTime model. Among the three conditions, the CAV condition achieves the lowest optimum travel time, while the HDV condition records the largest absolute reduction in travel time during the learning process.

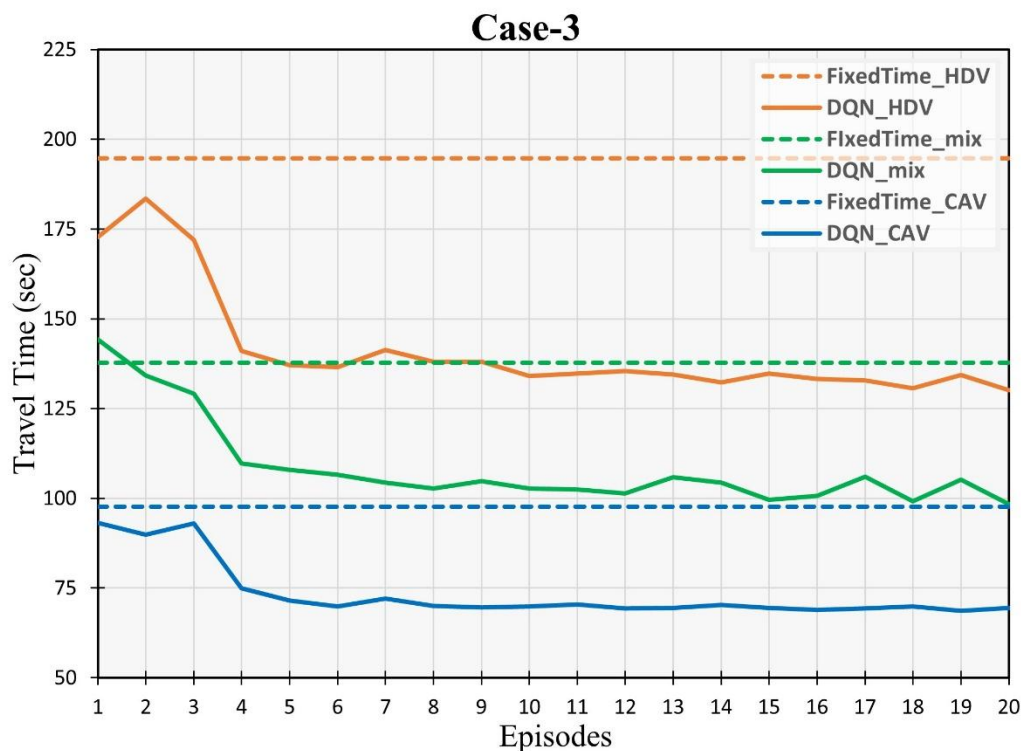


Figure 5.23: Average travel time versus episodes for the IDQN model and FixedTime model under the HDV condition, mix condition, and CAV condition in Case 3 of the Grid 4x4 network

5.4.2.4 Case 4 (Evening Peak-Hour Traffic with Special Event Traffic)

This case represents evening rush-hour traffic with special event traffic and is created by combining evening traffic demand, random traffic demand, and special event traffic demand. Figure 5.24 presents the training curves for Case 4 under the HDV condition, mix condition, and CAV condition.

In the HDV condition, the IDQN model begins with an initial average travel time of 207.6 s and gradually converges to an optimum travel time of 144.0 s. This represents a reduction of 30.6% from the initial value and is the largest proportional reduction achieved among all six cases under the HDV condition. The converged travel time is substantially lower than the FixedTime model travel time of 208.43 s, indicating that the IDQN model effectively adapts to this traffic demand pattern and achieves significant improvements in traffic performance. The recovery time for the HDV condition is 8 hours, which is the longest among all HDV conditions, suggesting that the model requires additional learning time to reach stable convergence.

In the mix condition, the IDQN model starts with an initial average travel time of 166.6 s and converges to an optimum travel time of 110.4 s, corresponding to a reduction of 33.7%. The final travel time is considerably lower than the FixedTime model travel time of 164.48 s, demonstrating the strong effectiveness of the IDQN model under mixed traffic conditions. The recovery time for the mix condition is 6 hours.

In the CAV condition, the IDQN model begins with an initial average travel time of 111.1 s and converges to an optimum travel time of 73.0 s. This represents a reduction of 34.3% from the initial value, which is the largest proportional improvement achieved among all six cases under the CAV condition. The converged travel time is significantly lower than the FixedTime model travel time of 112.47 s, highlighting the substantial benefits achieved by the IDQN model after convergence. The recovery time for the CAV condition is 6 hours.

Overall, Case 4 demonstrates excellent learning performance across all three traffic conditions, with the IDQN model consistently outperforming the FixedTime model. This case is particularly notable because it achieves the largest proportional reductions in travel time across the mix condition and CAV condition, while also producing the strongest improvement under the HDV condition. Although the HDV condition requires a longer recovery time, the final performance gains are among the highest observed across all cases.

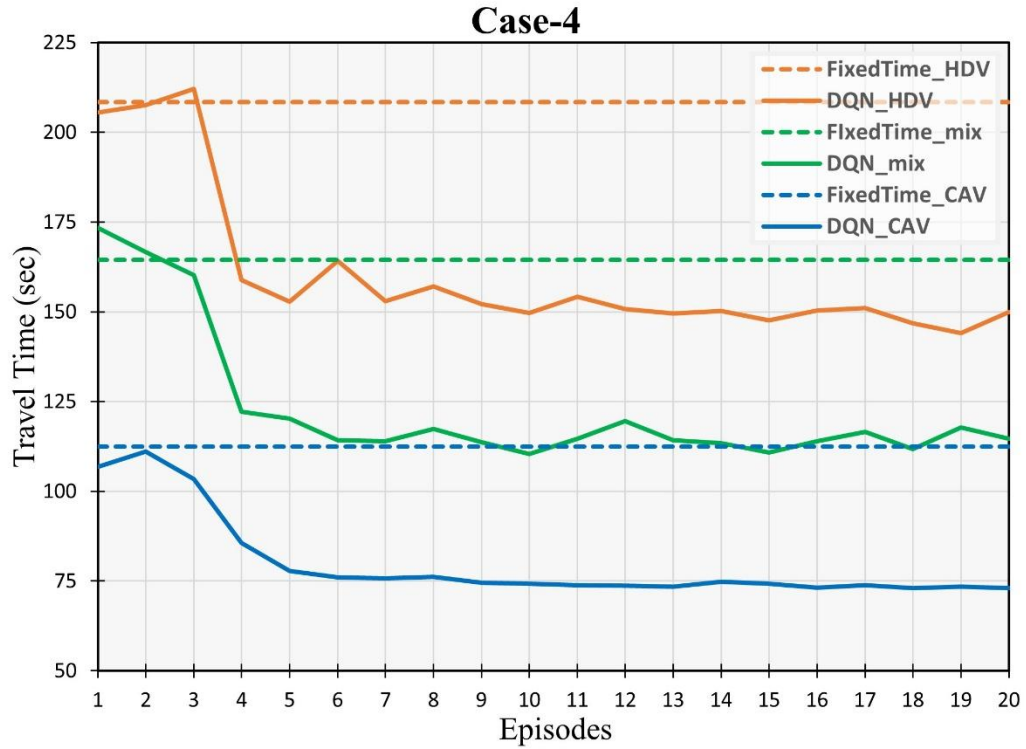


Figure 5.24: Average travel time versus episodes for the IDQN model and FixedTime model under the HDV condition, mix condition, and CAV condition in Case 4 of the Grid 4x4 network

5.4.2.5 Case 5 (Evening Peak-Hour Traffic under Bad Weather Conditions)

This case represents evening rush-hour traffic and is created by combining evening traffic demand with random traffic demand under bad weather conditions. Figure 5.25 presents the training curves for Case 5 under the HDV condition, mix condition, and CAV condition.

In the HDV condition, the IDQN model begins with an initial average travel time of 176.6 s and gradually converges to an optimum travel time of 131.6 s. This represents a reduction of 25.5% from the initial value. The converged travel time is substantially lower than the FixedTime model travel time of 196.19 s, indicating that the IDQN model effectively adapts to the traffic demand pattern and achieves improved traffic performance despite the adverse weather conditions. The recovery time for the HDV condition is 5 hours.

In the mix condition, the IDQN model starts with an initial average travel time of 143.0 s and converges to an optimum travel time of 103.6 s, corresponding to a reduction of 27.6%. The final travel time is considerably lower than the FixedTime model travel time of 154.77 s, demonstrating the effectiveness of the IDQN model under mixed traffic conditions. The recovery time for the mix condition is 6 hours.

In the CAV condition, the IDQN model begins with an initial average travel time of 97.7 s and converges to an optimum travel time of 75.4 s. This represents a reduction of 22.8% from the initial value. The converged travel time is significantly lower than the FixedTime model travel time of 102.18 s, highlighting the ability of the IDQN model to maintain efficient traffic operations under full CAV penetration even in unfavorable weather conditions. The recovery time for the CAV condition is 6 hours.

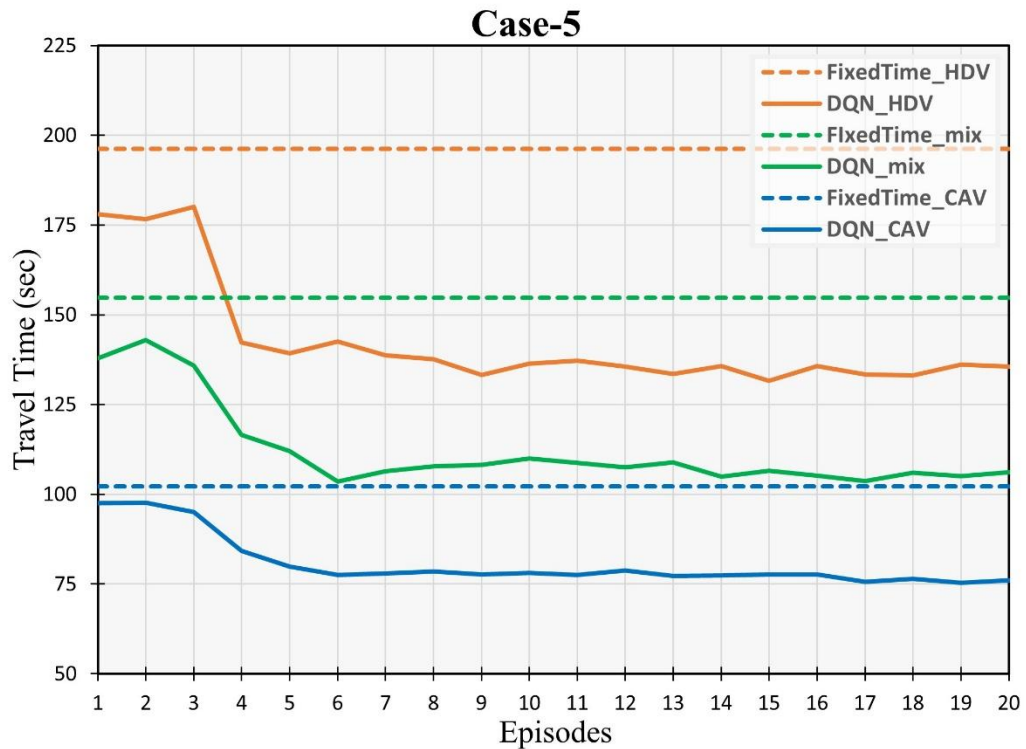


Figure 5.25: Average travel time versus episodes for the IDQN model and FixedTime model under the HDV condition, mix condition, and CAV condition in Case 5 of the Grid 4x4 network

Overall, Case 5 shows smooth and stable convergence behavior across all three traffic conditions, with the IDQN model consistently outperforming the FixedTime model.

Similar to Cases 1 and 3, the model adapts efficiently to the traffic demand pattern and achieves reliable reductions in average travel time while maintaining relatively short recovery times.

5.4.2.6 Case 6 (Combined Traffic Scenario)

This case combines all four traffic types: morning peak-hour traffic, evening peak-hour traffic, special event traffic, and random traffic. It is also categorized as a long-duration case, as the simulation runs for 14 hours. Figure 5.26 presents the training curves for Case 6 under the HDV condition, mix condition, and CAV condition.

In the HDV condition, the IDQN model starts with an average travel time of 136.0 s and converges to 133.9 s. This corresponds to a modest reduction of only 1.5%, which is the smallest improvement observed among all six cases. However, despite the limited reduction during adaptation, the final travel time remains substantially lower than the FixedTime model travel time of 250.91 s. This indicates that the IDQN model is already performing near its optimum level from the beginning of the evaluation. As a result, the model requires only 1 hour to stabilize, giving the shortest recovery time among all HDV conditions.

In the mix condition, the average travel time decreases from 105.1 s to 103.1 s, representing a reduction of 1.9%. Similar to the HDV condition, only a small adjustment is required before the model reaches a stable state. The converged travel time is considerably lower than the FixedTime model travel time of 149.80 s, confirming the effectiveness of the learned policy. The recovery time for the mix condition is also 1 hour.

A similar trend can be observed in the CAV condition. The IDQN model begins with an average travel time of 68.3 s and converges to 68.1 s, resulting in a negligible reduction of 0.3%. The almost identical initial and final travel times suggest that the model is operating very close to its optimum performance from the first episode. Even so, the converged value remains significantly lower than the FixedTime model travel time of 99.06 s. The recovery time is again only 1 hour.

Overall, Case 6 stands out from the other cases due to its exceptionally fast convergence. While the reductions from initial to optimum travel time are very small, this is not a sign of poor performance. Rather, it indicates that the IDQN model is already well suited to the traffic conditions encountered in this case and therefore requires very little additional learning. The consistently low travel times and shortest recovery times across all three traffic conditions demonstrate the strong generalization capability of the trained model.

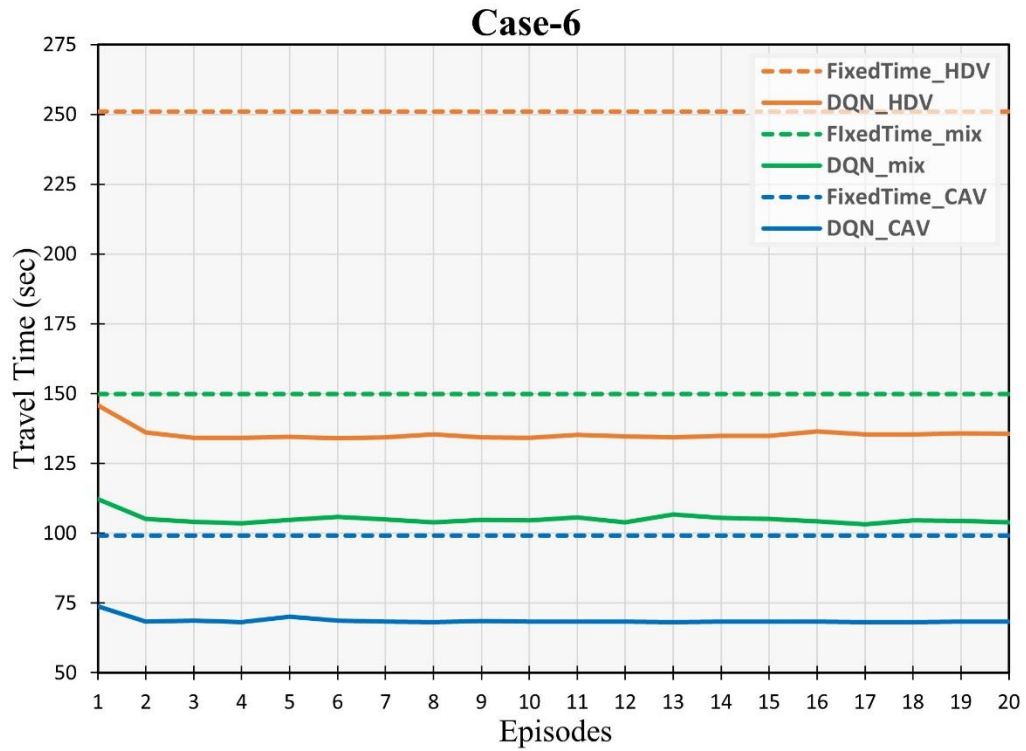


Figure 5.26: Average travel time versus episodes for the IDQN model and FixedTime model under the HDV condition, mix condition, and CAV condition in Case 6 of the Grid 4x4 network.

5.4.2.7: Relevant Figures and Plots for Case-wise Analysis

Table 5.4 presents the case-wise initial and optimum average travel times for the six cases of the Grid 4×4 network scenario under three traffic conditions: the HDV condition (100% HDVs), mix condition (50% HDVs and 50% CAVs), and CAV condition (100% CAVs), as

discussed previously. These values are used to evaluate the adaptation performance of the IDQN model and its ability to reduce travel time under different traffic demand patterns.

Table 5.4: Case-wise Initial and Optimum Average Travel Times (s) of the IDQN Model for the Grid 4x4 Network under HDV, Mix, and CAV Conditions

Cases	Traffic Condition	Initial Travel Time (sec)	Optimum Travel Time (sec)
Case1	HDV	180.1	130.8
	mix	147.4	99.5
	CAV	96.9	68.3
Case2	HDV	209.4	175.5
	mix	172.5	139.4
	CAV	141.6	110.4
Case3	HDV	183.5	130.1
	mix	134.2	98.4
	CAV	89.9	68.6
Case4	HDV	207.6	144
	mix	166.6	110.4
	CAV	111.1	73
Case5	HDV	176.6	131.6
	mix	143	103.6
	CAV	97.7	75.4
Case6	HDV	136	133.9
	mix	105.1	103.1
	CAV	68.3	68.1

Table 5.5 presents the case-wise percent decrease in travel times for the six cases of the Grid 4x4 network scenario under three traffic conditions: HDV condition (100% HDVs), mix condition (50% HDVs and 50% CAVs), and CAV condition (100% CAVs), as discussed previously.

Table 5.5: Case-Wise Percent Decrease in Travel Times (%) of the IDQN Model for the Grid 4x4 Network under HDV, Mix, And CAV Conditions

Cases	HDV Condition	Mix Condition	CAV Condition
case1	27.4	32.5	29.5
case2	16.2	19.2	22.0
case3	29.1	26.7	23.7
case4	30.6	33.7	34.3
case5	25.5	27.6	22.8
case6	1.5	1.9	0.3

For better visualization and comparison Case-Wise Percent Decrease in Travel Times (%) of the IDQN Model for the Grid 4x4 Network under HDV, Mix, And CAV Conditions, the data are illustrated in Figure 5.27.

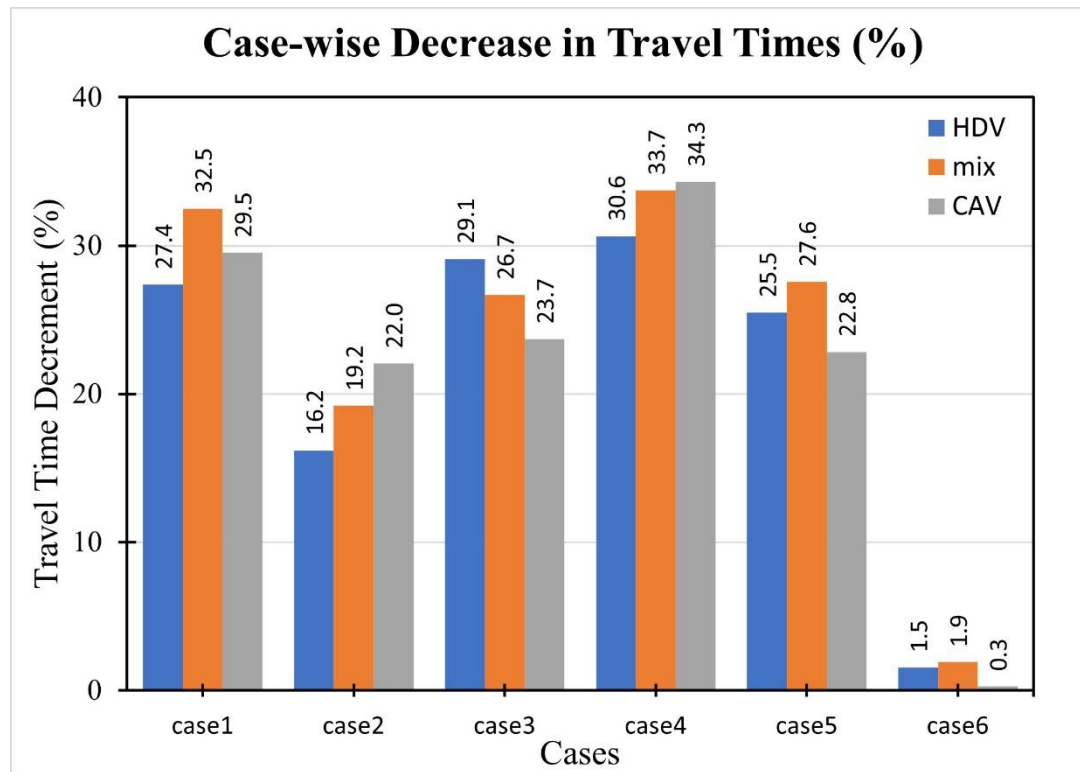


Figure 5.27: Case-Wise Percent Decrease in Travel Times (%) of the IDQN Model for the Grid 4x4 Network under HDV, Mix, And CAV Conditions

Table 5.6 presents the case-wise Fixed Travel Times (sec) for the six cases of the Grid 4x4 network scenario under three traffic conditions: HDV condition (100% HDVs), mix condition (50% HDVs and 50% CAVs), and CAV condition (100% CAVs), as discussed previously.

Table 5.6: Case-Wise Fixed Travel Times (sec) of the FixedTime Model for the Grid 4x4 Network under HDV, Mix, And CAV Conditions

	Fixed Travel Times (sec)		
Cases	HDV Condition	Mix Condition	CAV Condition
case1	208.25	146.53	101.24
case2	239.81	183.31	140.42
case3	194.62	137.78	97.66
case4	208.43	164.48	112.47
case5	196.19	154.77	102.18
case6	250.91	149.80	99.06

5.4.3 Results Analysis

The results obtained from the six cases of the Grid 4×4 network demonstrate that the IDQN-based ATSC is capable of adapting to a wide range of traffic conditions and demand patterns. Across all cases and traffic conditions, the IDQN model consistently achieved lower average travel times than the FixedTime controller, indicating the effectiveness of reinforcement learning in optimizing traffic signal operations.

For Cases 1, 3, and 5, the model exhibited smooth convergence behavior with relatively short recovery times ranging from 5 to 6 hours. These cases showed substantial reductions in average travel time from the initial to the optimum state, indicating that the learned policy was able to adapt efficiently to the corresponding traffic demand patterns. Among these cases, the CAV

condition consistently achieved the lowest optimum travel times, highlighting the benefits of higher CAV penetration when combined with adaptive signal control.

Case 2 required longer recovery times, reaching 7 hours in the HDV condition and 8 hours in both the mix condition and CAV condition. The smaller reductions in travel time and slower convergence suggest that the presence of blocked roads created a more challenging traffic environment for the learning process. Nevertheless, the IDQN model still outperformed the FixedTime controller under all traffic conditions.

Case 4 produced some of the largest reductions in travel time across all traffic conditions. Although the HDV condition required the longest recovery time of 8 hours, the model ultimately achieved significant performance improvements after convergence. This indicates that while certain traffic demand patterns may require additional adaptation time, the RL-based controller can still learn highly effective control strategies.

Case 6 exhibited distinctly different behavior from the other cases. The recovery time was only 1 hour for all three traffic conditions, and the reductions between initial and optimum travel times were minimal. This suggests that the pre-trained model was already operating close to its optimum performance from the beginning of the evaluation. The rapid stabilization observed in Case 6 demonstrates the strong generalization capability of the learned policy and its ability to transfer effectively to similar traffic conditions without requiring extensive re-training.

A clear trend can also be observed across all cases with increasing CAV penetration. The CAV condition consistently achieved the lowest optimum travel times, followed by the mix condition and the HDV condition. This indicates that the combination of connected and automated vehicles with RL-based adaptive traffic signal control can further enhance network performance by improving traffic flow and reducing delays.

Overall, the findings confirm that the IDQN-based ATSC is robust under varying traffic conditions and demand patterns. The model not only achieves lower travel times than the FixedTime controller but also demonstrates effective adaptation and recovery characteristics across different traffic scenarios. While recovery time and performance

gains vary depending on the complexity of the traffic demand pattern, the RL-based controller consistently provides superior operational performance, particularly under higher CAV penetration levels.

5.5 Overview

This chapter presents the results of the study based on the methodology outlined in Chapter 3 and the data collection procedures described in Chapter 4. The analysis is divided into three main sections. First, the impact of different CAV penetration levels on traffic efficiency is evaluated using the Travel Time Rate (TTR) metric in both the Grid 4×4 and DCC network scenarios. Next, traffic safety is assessed by analyzing the total number of conflicts under different traffic conditions and demand patterns. Finally, the performance of the RL-based Adaptive Traffic Signal Control (ATSC) system is evaluated using average travel time and recovery time to examine the adaptability and effectiveness of the IDQN model under varying traffic conditions and CAV penetration levels.

CHAPTER 6

CONCLUSIONS

6.1 Background

The main objective of this study was to evaluate the impact of CAVs on traffic efficiency and safety and to analyze the performance of a RL-based Adaptive Traffic Signal Control (ATSC) system under different traffic conditions and demand patterns. Two network scenarios were considered: a synthetic Grid 4x4 network and a real-world DCC network.

Traffic efficiency was evaluated using the Travel Time Rate (TTR), while traffic safety was assessed using the total number of conflicts obtained through PET based Surrogate Safety Measure (SSM) analysis. Three traffic conditions were examined: the HDV condition (100% HDVs), mix condition (50% HDVs and 50% CAVs), and CAV condition (100% CAVs).

The performance of the RL-based ATSC was evaluated in the Grid 4x4 network using the IDQN model. The model was assessed based on average travel time and recovery time and compared with a FixedTime controller. Based on the results obtained, this chapter summarizes the key findings of the study, presents the overall conclusions, and provides recommendations for future research.

6.2 Summary of the Findings

The findings of this study provide insights into the effects of CAVs on traffic efficiency and safety, as well as the performance of a RL-based ATSC system under varying traffic conditions and demand patterns.

Based on the experimental results obtained from the Grid 4x4 and DCC network scenarios, the key findings are summarized as follows:

1. Traffic efficiency consistently improved with increasing CAV penetration. Across both the Grid 4x4 and DCC networks, the HDV condition exhibited the highest

Travel Time Rate (TTR) values and the most severe congestion, while the CAV condition achieved the lowest TTR values and the highest level of traffic efficiency.

2. The magnitude of traffic efficiency improvement depended on traffic demand and network characteristics. Greater improvements were observed under low and moderate traffic demand conditions, whereas some localized congestion remained under high-demand conditions even with full CAV penetration.
3. Traffic safety improved significantly as the proportion of CAVs increased. In almost all cases, the total number of conflicts decreased progressively from the HDV condition to the mix condition and further to the CAV condition, demonstrating the positive impact of vehicle automation on road safety.
4. The DCC network recorded substantially higher conflict counts than the Grid 4x4 network due to its larger size, higher traffic volumes, and more complex road geometry. However, the overall trend of conflict reduction with increasing CAV penetration remained consistent across both network scenarios.
5. The IDQN-based ATSC model successfully adapted to different traffic demand patterns and consistently achieved lower average travel times than the FixedTime controller in all evaluated cases.
6. Recovery time varied depending on the traffic demand pattern. Most cases required between 5 and 8 hours to reach stable performance, while Case 6 exhibited the shortest recovery time of only 1 hour, indicating strong transferability of the learned control policy.
7. The CAV condition consistently achieved the lowest optimum travel times among the three traffic conditions, demonstrating that the combination of CAV technology and RL-based adaptive signal control can significantly improve overall network performance.

8. Although the RL-based controller performed effectively across all cases, the level of improvement was influenced by traffic demand, network conditions, and the complexity of traffic interactions. More challenging traffic scenarios generally required longer adaptation periods before stable performance was achieved.

Overall, the results demonstrate that increasing CAV penetration can significantly improve both traffic efficiency and traffic safety. Furthermore, the IDQN-based ATSC model proved capable of adapting to a wide range of traffic conditions while consistently outperforming the conventional FixedTime controller. These findings highlight the potential benefits of integrating CAV technology with RL-based traffic signal control systems for future intelligent transportation networks.

6.3 Recommendations of the Study

Based on the findings of this study, several recommendations are proposed to further improve traffic efficiency, traffic safety, and the performance of RL-based Adaptive Traffic Signal Control (ATSC) systems:

1. Increasing the penetration of CAVs is recommended, as the results consistently showed improvements in both traffic efficiency and traffic safety with higher CAV penetration levels. The greatest benefits were observed under the CAV condition (100% CAVs), which achieved the lowest Travel Time Rate (TTR) values and the fewest traffic conflicts in most cases.
2. Future traffic management strategies should consider the combined implementation of CAV technology and RL-based ATSC. The results demonstrated that the integration of vehicle automation and adaptive signal control can significantly reduce travel times and improve overall network performance compared to conventional fixed-time traffic control systems.
3. To improve the adaptability of RL-based traffic signal controllers, future training datasets should include a wider variety of traffic conditions and demand patterns. Exposure to different traffic situations during training can enhance the models

ability to generalize and maintain stable performance when encountering previously unseen traffic conditions.

4. The present study considered only one RL-based traffic signal control model (IDQN). Future studies should compare multiple RL algorithms to identify the most effective approaches for improving traffic efficiency, traffic safety, and adaptability under different traffic conditions.
5. Additional factors such as communication delays, sensor errors, vehicle-to-infrastructure communication reliability, and varying driver behaviour should be incorporated into future simulations to better represent real-world operating conditions and assess the performance of RL-based traffic control systems.

By implementing these recommendations, future intelligent transportation systems can better utilize the benefits of CAV technology and RL-based traffic signal control to improve traffic operations, reduce congestion, and enhance road safety.

6.4 Limitations of the Study

This study evaluated the impact of CAVs on traffic efficiency and safety and analyzed the performance of an RL-based ATSC system under different traffic conditions, demand patterns, and CAV penetration levels. However, several limitations should be acknowledged.

1. All simulation experiments were conducted using the SUMO traffic simulation environment. Although SUMO provides a realistic representation of traffic operations, simulation results may not fully capture the complexity and uncertainty of real-world traffic conditions.
2. The study considered only two network scenarios: a synthetic Grid 4x4 network and a real-world DCC network. Additional network types with different geometries, traffic characteristics, and operational conditions should be investigated to improve the generalizability of the findings.

3. Traffic efficiency was evaluated using the Travel Time Rate (TTR), while traffic safety was assessed using conflict counts derived from PET-based surrogate safety measures. Other performance indicators, such as delay, throughput, fuel consumption, emissions, and crash risk, were not considered and may provide additional insights into network performance.
4. The RL-based ATSC analysis was limited to a single reinforcement learning model, namely IDQN. Although the model demonstrated promising performance, the results cannot be generalized to other RL algorithms without further comparative studies.
5. The RL model was trained using only Case 1 of the HDV condition before being evaluated under the remaining traffic conditions and demand patterns. Different training strategies, datasets, and training durations may influence the models adaptation capability and overall performance.
6. The study assumed ideal communication and sensing conditions for CAV operations. Factors such as communication delays, packet loss, sensor inaccuracies, and equipment failures were not considered, although they may affect the real-world performance of both CAVs and RL-based traffic signal control systems.
7. Human driver behavior was represented using predefined vehicle parameters in the simulation environment. Real-world variations in driver behavior, compliance, and decision-making were not explicitly modeled and may influence traffic efficiency and safety outcomes.

Despite these limitations, the study provides valuable insights into the potential benefits of CAV deployment and RL-based traffic signal control. The findings establish a foundation for future research aimed at developing more realistic, scalable, and robust intelligent transportation systems.

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APPENDIX

APPENDIX A

CODE AND DATASETS

This chapter presents the links to the source code and datasets used in the implementation and analysis of this thesis.

A.1 Codes

Relevant codes for traffic efficiency evaluation are available at:

https://github.com/mdaslamcee/graph_plot_data

Relevant codes for traffic safety evaluation are available at:

https://github.com/mdaslamcee/graph_plot_data

The modified LibSignal framework is available at:

<https://github.com/mdaslamcee/LibSignal>

The main source link for the LibSignal framework is available at:

<https://github.com/DaRL-LibSignal/LibSignal>

Data post-processing code is available at:

https://github.com/mdaslamcee/RL_robust_data

A.2 Datasets

The Grid 4×4 scenario files can be accessed here:

https://github.com/mdaslamcee/row_data_SUMO

The DCC scenario files can be accessed here:

https://github.com/mdaslamcee/row_data_SUMO

APPENDIX B

Table B-1: DQN Summary Output for All Cases in the Grid 4x4 Network Scenario

Model name	case	Initial Travel Time	Optimum Travel Time	Recovery Time	Initial Drop
dqn	init_HDV_case1	179.5	126.4	23	29.58
dqn	HDV_case1	180.1	130.8	5	27.37
dqn	HDV_case2	209.4	175.5	7	16.19
dqn	HDV_case3	183.5	130.1	5	29.1
dqn	HDV_case4	207.6	144	8	30.64
dqn	HDV_case5	176.6	131.6	5	25.48
dqn	HDV_case6	136	133.9	1	1.54
dqn	CAV_case1	96.9	68.3	6	29.51
dqn	CAV_case2	141.6	110.4	8	22.03
dqn	CAV_case3	89.9	68.6	5	23.69
dqn	CAV_case4	111.1	73	6	34.29
dqn	CAV_case5	97.7	75.4	6	22.82
dqn	CAV_case6	68.3	68.1	1	0.29
dqn	mix_case1	147.4	99.5	6	32.5
dqn	mix_case2	172.5	139.4	8	19.19
dqn	mix_case3	134.2	98.4	6	26.68
dqn	mix_case4	166.6	110.4	6	33.73
dqn	mix_case5	143	103.6	6	27.55
dqn	mix_case6	105.1	103.1	1	1.9

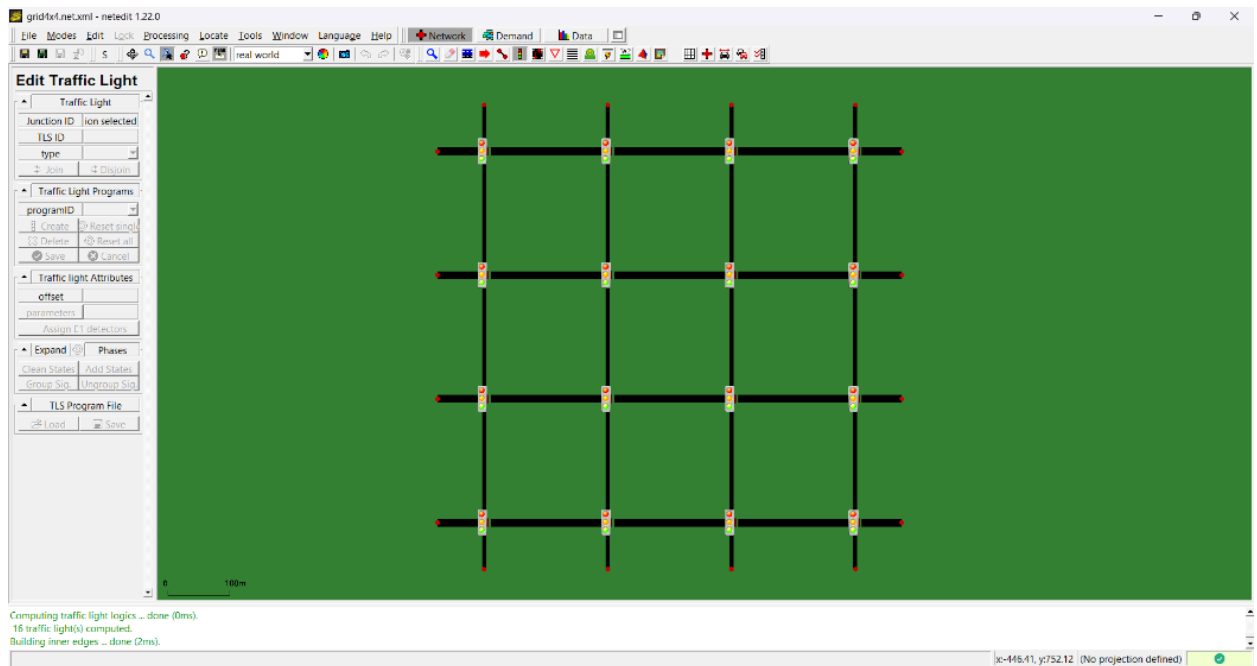


Figure B-1: Grid 4x4 Network Scenario Created in NetEdit, the Graphical Network Editor of SUMO

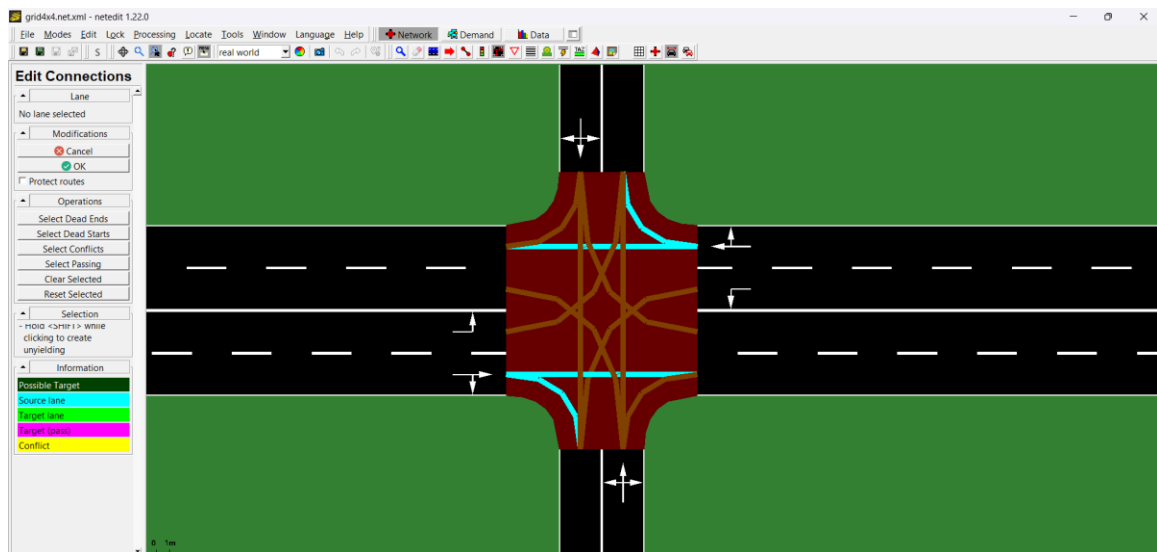


Figure B-2: Single Section of the Grid 4x4 Network Scenario in NetEdit

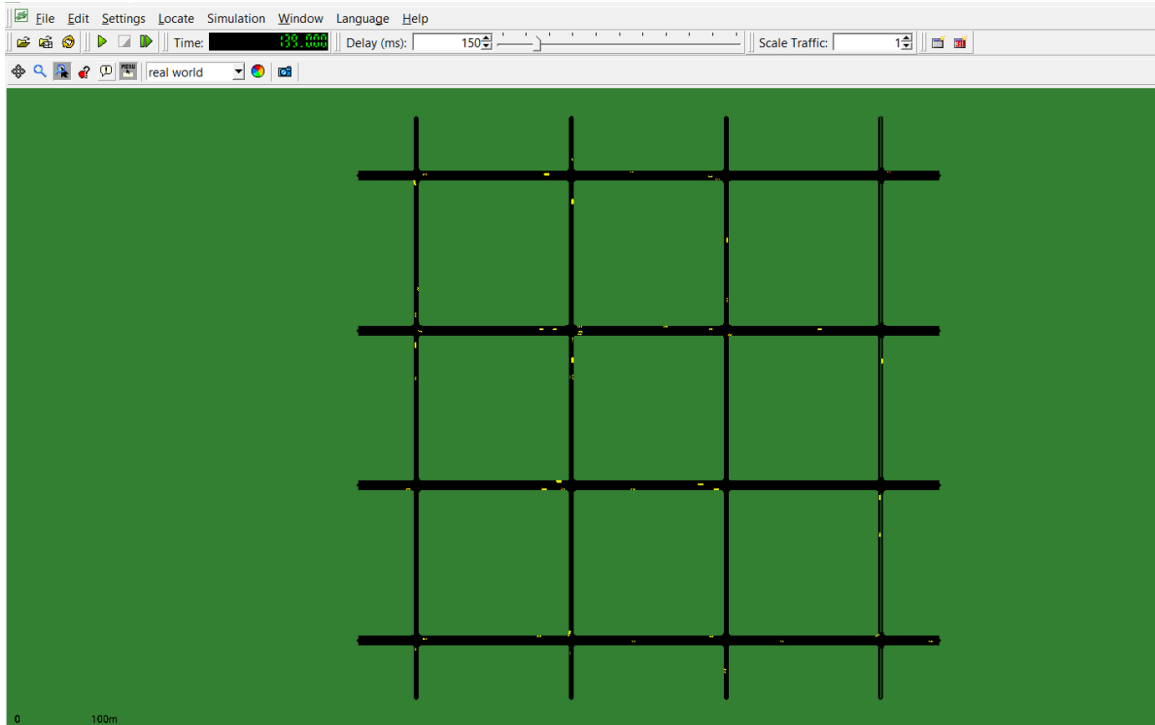


Figure B-3: Running Simulation in SUMO for Case 1 in the Grid 4×4 Scenario

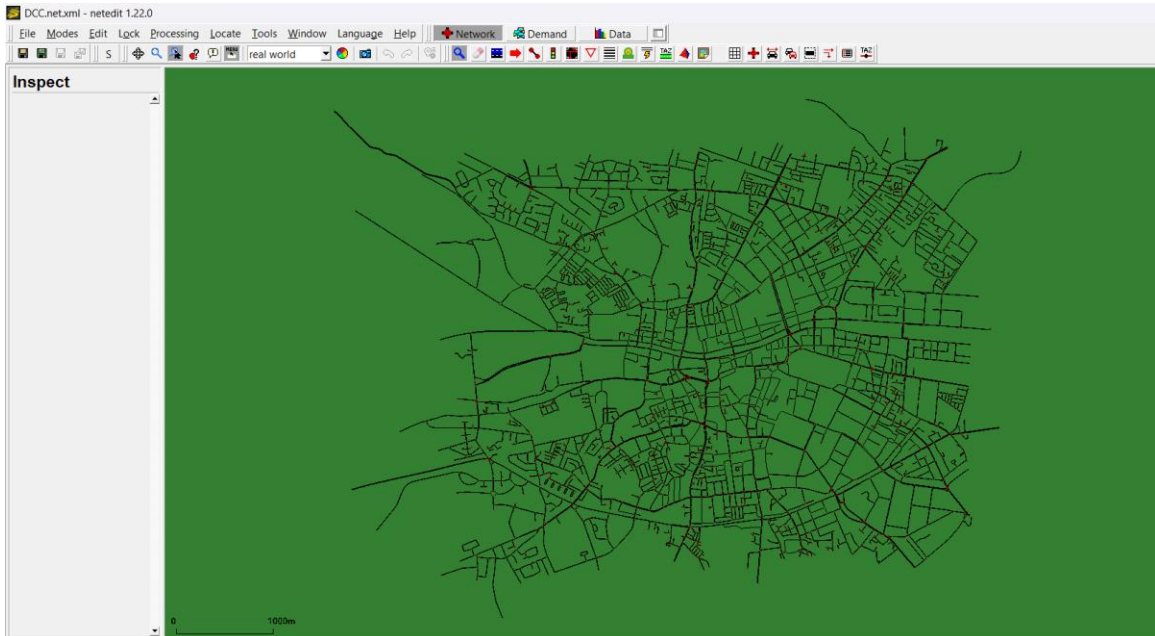


Figure B-4: DCC Network Scenario in NetEdit



Figure B-5: Running Simulation in SUMO for Case 1 in the DCC Scenario (Partial Segment)